MODULE 1

Managing in the Digital World

An **information system** is the combination of people and information technology that creates, collects, processes, stores, and distributes useful data. That one sentence is doing a lot of work, and this module unpacks all of it: the world these systems now run, what the parts actually are, why the same technology can make one company and break another, what responsible use of it looks like, and how a manager turns any of that into a decision worth defending.

1 OBJECTIVE 1.1
Describe the digital world, the societal issues it raises, and how rising digital density is shaping what comes next

2 OBJECTIVE 1.2
Explain what an information system is, contrasting its data, technology, people, and organizational parts

3 OBJECTIVE 1.3
Describe the dual nature of information systems in the success and failure of modern organizations

4 OBJECTIVE 1.4
Describe how computer ethics affect the use of information systems, including privacy and intellectual property

5 APPLICATION 1.5
Use supplemental strategy frameworks to turn a business problem into a defensible technology recommendation

How to use this page

1. **Read the short setup first.** Every term is explained in ordinary language before any activity uses it, so nothing here assumes a background you have not been given yet.
2. **Answer as you go.** The activities sit inside the reading, not at the end. When you answer a question, **every option explains itself** — including the three you did not pick, because knowing why a wrong answer is wrong is the part that transfers.

3. **Use the diagrams.** The tabs redraw one idea from several angles. On a phone, a wide diagram scrolls sideways rather than shrinking to nothing.
4. **Retry anything.** Restart one activity, one section, or the whole page. Your completion record stays in this browser until you clear it; the answers themselves reset on reload so you always get a fresh attempt.
5. **Finish with the final challenge.** It unlocks once you have answered all twenty-five situations, then reports the four chapter objectives and the strategy supplement separately so a weak area shows up instead of being averaged away.

WORLD

The setting

Ubiquitous computing, knowledge work, globalization, and the widening reach of connected devices.

SYSTEM

The parts

Data becomes information becomes knowledge — carried by hardware, software, networks, and people.

STAKES

The double edge

The same investment that carries one company past its rivals can sink another. Both happen for reasons.

JUDGMENT

The responsibility

Privacy, accuracy, property, access — and a method for recommending technology you can defend.

ONE IDEA CONNECTS THE WHOLE MODULE

Technology by itself is not the story. An information system is technology *plus* the people who use it and the organization it sits inside, and it is those last two parts that decide whether the first one is worth anything. That is why a company can buy exactly what its competitor bought and get a completely different result.

SECTION 1–1A

The digital world we already live in

You already know computers are everywhere. What you have probably not done is look at that fact the way a manager must: as forces deciding who gets paid well, who gets locked out, where work happens, and which enormous problems businesses get asked to help solve.

Information systems are ubiquitous

An **information system** is the combination of people and information technology that creates, collects, processes, stores, and distributes useful data. Something **ubiquitous** is present everywhere at once, so present you stop noticing it — which is what information systems now are, whether you see them or not. They already run in places you would not think to look:

> **Shipping** — FedEx and UPS use them to route trucks and track packages, so the estimate on your phone comes from a system.

> **Retail** — Walgreens and Walmart use them to optimize supply chains and analyze customer tastes and preferences.

> **Public infrastructure** — cities use them for adaptive traffic control systems and variable speed limits, so a light that changes with traffic is deciding something.

> **Vehicles** — your car uses them for ignition control, airbags, park assist, and even self-driving.

Some companies do not just use one; they are one. A **business model** is a summary of a business's strategic direction that outlines how the objectives will be achieved. Airbnb and Uber are innovative business models built on or around information systems; delete the software matching riders to drivers and no company remains, only strangers with cars.

Then come the systems you use on purpose:

- Registering for classes online rather than in person.
- A paycheck generated by computer and deposited automatically.
- One phone for photos, music, games, and navigation.
- Submitting work through Blackboard, Canvas, or Moodle.
- Algorithms built to explore your browsing and shopping habits so platforms such as Facebook or TikTok can steer you toward new interests.

The post-PC era

Computing used to mean walking to a machine and sitting down at it. Now connected devices, smartphones, and tablets are replacing traditional desktop and laptop computers — the **post-PC era**. Tablets began as consumer devices and are now commonplace in warehouses, showrooms, airplane cockpits, and hospitals.

Newer form factors do not kill older ones; they work in tandem, giving different devices to different users for different tasks, because what matters is not the device but the services and data it reaches. The same shift produced wearable computers, augmented reality devices, and smart home devices.

Two ordinary constraints fell away, and the activity changed with them:

> **Place** — people once bound to a stationary PC are now bound to no particular location.

> **Time** — a workday once ran from powering the computer on to turning it off at night, and today casual tasks such as email happen at any hour in small chunks, like waiting in line at a cashier.

> **Activity** — computing was once focused primarily on automating work, and now it also provides new benefits and services and carries social and casual activity.

Mobile broadband makes computing **instant-on**, and **cloud computing** — Gmail, Microsoft 365, Dropbox — keeps files reachable from whichever device you hold.

The cycle runs in both directions

It is tempting to assume technology arrives and society adjusts. It runs as a loop instead: changes in technology lead to societal changes, and societal changes shape technological changes. The same loop earns two names, depending on the side you describe:

> **Virtuous cycle** — the benefit side, because communication, social networking, and online investing almost necessitate mobility and connectivity, and people have grown accustomed to checking email, posting status updates, or checking real-time stock quotes while on the go.

> **Vicious cycle** — the cost side, because work life creeps into leisure time and the fixation on being permanently "on call" grows.

Society then pushes back on the technology: with the boundary between work and leisure blurred, employees increasingly demand devices that support both and often bring their own devices into the workplace.

Knowledge workers and the knowledge society

In 1959, more than 65 years ago, **Peter Drucker** predicted that information and information systems would become increasingly important, and coined the term **knowledge worker**. Knowledge workers are typically professionals who are relatively well educated and who create, modify, and/or synthesize knowledge as a fundamental part of their jobs — an analyst building a forecast, a nurse practitioner reading a chart.

He was right, and the advantage shows up in four places:

> **Pay** — they are generally paid better than their agricultural and industrial counterparts.

Education — they are empowered by formal education while often also possessing valuable real-world skills.

Learning — they are continually learning to do their jobs better.

Bargaining power — they hold much better career opportunities and far more bargaining power than workers ever had before.

They make up about a quarter of the workforce in the United States and other developed nations, and their numbers are rising quickly.

Drucker also predicted the society they would build. Because education matters so much to knowledge workers and the firms that need them, education would become the cornerstone of a **knowledge society**, in which possessing knowledge is as important as possessing land, labor, or capital, if not more so.

Others call it the knowledge economy, the new economy, the digital society, the network era, or the internet era; this book simply calls it the **digital world**.

What the earnings data show. The most recent data from the U.S. Census Bureau's American Community Survey (2023 data) give median earnings for workers 25 and over:

HIGHEST LEVEL OF EDUCATION	MEDIAN EARNINGS PER YEAR
Graduate or professional degree	US$86,524
Bachelor's degree	US$67,256
High school diploma	US$39,428
Without a high school diploma	US$31,660

That gap is increasing, not closing, and a degree is often required to qualify for advancement at all.

Knowledge work is also leaking into occupations nobody used to file under that heading:

The delivery driver — the UPS delivery person uses global positioning system (GPS) technology to take the best route, so a system picks the route, not memory of the neighborhood.

> **The farmer** — the farmer in Iowa flies a drone to monitor soil conditions and crop yield, turning a field into data somebody must read.

So the lines between "knowledge workers" and "manual workers" are blurring, to the point that almost every worker can now be considered a knowledge worker — and workers of the future need to become **learning workers**, because not the knowledge itself but the knowledge of how to learn will be of primary importance.

CHECK YOURSELF **The post-PC era, the cycle, and the knowledge society**

Pick the option that matches how the chapter actually describes each idea, then read every explanation.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. The chapter says we are living in the **post-PC era**. What does that mean?

- **A.** The PC is finished as a work machine, so an organization can issue every employee one mobile device and cover every task with it. — This is the one-device version of the idea, and it is what people usually mean when they say the PC is dead. The chapter says devices with newer form factors work in tandem with older ones, providing different devices for different users and different tasks, which is why a warehouse runs on tablets while a design studio still runs on workstations. It would be right if it said mobile devices are becoming the first thing people reach for.
- **B.** Connected devices, smartphones, and tablets are replacing traditional desktops and laptops, and new and old form factors work in tandem. **— correct.** The chapter defines the post-PC era as connected devices, smartphones, and tablets replacing traditional desktop and laptop computers, with newer and older form factors complementing each other because not the device but the services and data provided are of primary importance.
- **C.** It is another name for the move to cloud computing, so what changed is where files and applications live rather than which devices people use. — Cloud computing is real, related, and named separately in the chapter (Gmail, Microsoft 365, Dropbox) as what lets you reach email, files, and notes from different devices. But it answers where the data and applications sit, while the post-PC era answers which devices people carry and use, so the chapter treats cloud computing as an enabler of the shift rather than the shift itself.
- **D.** Computing shifted toward social and casual activity, which is why tablets remain consumer devices rather than professional tools. — The first half is right: computing changed from an activity primarily focused on automating work to one that also carries social and casual activity. The second half contradicts the chapter, which says tablets began as consumer devices and are now commonplace in warehouses, showrooms, airplane cockpits, and hospitals.

2. What does the chapter mean by calling the relationship between technology and society a **virtuous cycle** — or a **vicious cycle**?

- **A.** Technology drives society in one direction: new devices change how people live, but how people live has no real effect on what gets built next. — This is the one-way version of the story, and it is the most common mistake here. The chapter makes the causation run in both directions: changes in technology enabled new ways of working and socializing, and then the blurring of work and leisure led employees to demand devices that support both and to bring their own devices into the workplace. It would be right only if what people wanted never changed what gets built next.
- **B.** The virtuous cycle is the helpful technology and the vicious cycle is the harmful technology, so they refer to two different groups of devices. — It is one cycle viewed two ways, not two sets of devices. Mobility and connectivity are simultaneously the convenience and the reason people feel permanently on call; the label depends on whether you are describing the benefit or the cost of the same loop.
- **C.** Technological change produces societal change and societal change reshapes technology; the same loop turns vicious when work invades leisure time. **— correct.** The chapter describes a loop in which technological change produces societal change, which in turn shapes further technological change, and it offers 'vicious' as the same loop judged by the creep of work into people's leisure time.
- **D.** Companies must keep upgrading their information systems as competitors upgrade theirs, so falling behind for even one cycle costs market share. — This describes competitive pressure to adopt technology, which is a real idea in this course, but it is about firms racing each other rather than technology and society reshaping one another. It would answer a question about competitive advantage instead.

3. Drucker argued that in a **knowledge society**, possessing knowledge would be as important as possessing land, labor, or capital. Which evidence does the chapter offer for that claim?

- **A.** Knowledge workers now make up roughly three quarters of the workforce in the United States and other developed nations. — The direction is right but the size is wrong: the chapter says knowledge workers make up about a quarter of the workforce in the United States and other developed nations, with their numbers rising quickly. The separate claim that almost every worker can now be considered a knowledge worker is about how the work has spread, not about that percentage.
- **B.** 2023 Census survey data show median earnings of US$67,256 for bachelor's degree holders versus US$39,428 for high school graduates. **— correct.** The chapter cites the U.S. Census Bureau's American Community Survey (2023 data): US$86,524 for a graduate or professional degree, US$67,256 for a bachelor's degree, US$39,428 for a high school diploma, and US$31,660 without a high school diploma, and it notes the gap is increasing.
- **C.** The UPS delivery person uses GPS to take the best route, and the farmer in Iowa flies a drone to monitor soil conditions and crop yield. — Every word of this is in the chapter, which is what makes it tempting, but it is evidence for a different claim. Those two examples show that traditional occupations now use information systems too, which is why the line between knowledge workers and manual workers is blurring. They say nothing about knowledge being worth as much as land, labor, or capital; the earnings comparison is what carries that argument.

- **D.** A college degree raises an individual worker's pay, which is exactly what a median earnings figure reports. — This misreads what a median is. A median is the midpoint of a whole group, so it compares two populations rather than predicting what any one person would earn. It would be right if it said degree holders as a group earn more at the midpoint than non-degree holders as a group.

The digital divide

All of that assumes you are on the inside; not everyone is. The **digital divide** is the gap between those with access to information systems, who hold great advantages, and those without it. The chapter calls it one of the major ethical challenges facing society today, because of the strong linkage between computer literacy and a person’s ability to compete in the digital world.

As the economist John Kenneth Galbraith put it, “in the informational society, the fuel, the power, is knowledge,” splitting society between those who have information and those who function out of ignorance.

The divide is not the same everywhere, and neither is the fix:

> **Inside America** — the divide is rapidly shrinking, but people in rural communities, the elderly, people with disabilities, and minorities lag behind national averages for internet access and computer literacy.

> **Outside developed countries** — the gap gets even wider and the obstacles much harder, particularly where infrastructure and financial resources are lacking, and most developing countries still lack affordable internet access or efficient electronic payment methods.

Two people can both be called “offline” when the fix for one is a cheaper data plan and for the other a power grid.

Some argue there is a downside to being a knowledge worker at all, and that knowledge workers will be the first replaced by automation with information systems. There is a downside to overreliance on information systems.

What is not in dispute is that knowledge workers and information systems are now critical to the success of modern organizations, economies, and societies.

COMPLETE IT

The vocabulary of the knowledge society

Choose the word or phrase the chapter uses in each sentence; the explanation appears once you pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- Professionals who are relatively well educated and who create, modify, and/or synthesize knowledge as a fundamental part of their jobs are **knowledge workers**, a term Peter Drucker coined in 1959. Drucker coined knowledge worker; manual worker is the category the chapter contrasts it with, and learning worker is the future role the chapter says everyone must grow into.
- Drucker predicted that as those workers grew in number and influence, a **knowledge society** would emerge, with education as its cornerstone and knowledge as important as land, labor, or capital. Knowledge society is Drucker's own term; the chapter notes others call the same thing the knowledge economy, the new economy, the digital society, the network era, or the internet era, and that this book calls it the digital world.
- Knowledge workers currently make up about **a quarter** of the workforce in the United States and other developed nations, and their numbers are rising quickly. The chapter's figure is about a quarter, which is easy to overestimate because the chapter separately says the lines between knowledge workers and manual workers are blurring.
- The gap between those who have access to information systems and those who do not is the **digital divide**, which the chapter calls one of the major ethical challenges facing society today. It counts as an ethical challenge because of the strong linkage between computer literacy and a person's ability to compete in the digital world, not merely because it is inconvenient.
- Inside America that gap is rapidly shrinking, but people in rural communities, the elderly, people with disabilities, and minorities still lag behind national averages for internet access and **computer literacy**. The chapter pairs internet access with computer literacy because having a connection and knowing how to use it are two separate barriers, and closing only one leaves the divide open.
- In most developing countries the obstacles are much harder to overcome, because infrastructure and financial resources are lacking and modern informational resources such as affordable internet access or **efficient electronic payment methods** are missing. The chapter names affordable internet access and efficient electronic payment methods specifically, which is why the divide abroad is an economic and infrastructure problem rather than a device problem.
- Because what will matter most is not the knowledge itself but the knowledge of how to learn, the chapter says workers of the future need to become **learning workers**. Learning worker is the chapter's forward-looking term; systems analyst and data entry clerk are examples of how narrowly people once pictured computer-related occupations.

Globalization: what it opens and what it costs

Beneath these trends sits **globalization** — the integration of economies throughout the world, enabled by innovation and technological progress, visible in the greater international movement of commodities, money, information, and labor. Falling transportation and telecommunication costs opened many opportunities for organizations; three of them drive the rest of this section:

- **Cultures moved closer together**, fueled by movies, television, and other media — Netflix reaches many countries and major movies are increasingly international — creating shared understanding about norms of behavior and desirable goods. The counterweight: the same technology has facilitated an increase in authoritarianism and restrictions on the free flow of communication and content, such as on WeChat and TikTok.
- **New markets appeared**, because the rapid rise of a new middle class in many developing countries lets established companies sell to millions of new customers.
- **The talent pool went global**, because with communication costs falling a company can draw on skilled professionals anywhere. Russia, China, and India offer high-quality education, and some countries built industries around one competency — software development and tax preparation in India, call centers in Ireland.

That third opportunity has a name. **Outsourcing** is the moving of business processes or tasks — such as accounting, manufacturing, or security — to another company or another country. Companies do it for eight reasons, and cost is only one of the eight:

- To reduce or control costs.
- To gain access to world-class capabilities.
- To reduce time to market.
- To focus on core activities.
- To free up internal resources.
- To increase revenue potential.
- To increase process efficiencies.
- To compensate for a lack of specific skills.

India's Wipro and Infosys lead in IT services, and even the reading of X-rays is outsourced by U.S. hospitals to skilled radiologists abroad, often while doctors in the United States are sleeping.

Then the bill arrives, in three parts:

> **Governmental challenges** — differences in political systems, regulatory environments, laws, standards, or individual freedoms, so what is routine at home may be forbidden there.

> **Geoeconomic challenges** — differences in infrastructure, demographics, welfare, or workers' expertise.

> **Cultural challenges** — differences in languages, beliefs, attitudes, religions, or life focus, including differing viewpoints regarding intellectual property.

So locations must be chosen carefully, weighing English proficiency, salaries, and geopolitical risk. Singapore, Canada, and Ireland are declining because of rising salaries, pushing outsourcers toward emerging countries such as Bulgaria, Egypt, Ghana, Bangladesh, or Vietnam.

> Time zones cut both ways: the reason a hospital can have X-rays read overnight is the reason a project team loses a day waiting on an answer — the deciders are asleep while you work.

SORT

The digital divide and globalization: accurate or a common misconception?

Drop each statement into the bucket where it belongs, then read why the chapter puts it there.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Accurate — matches what the chapter actually says

- The digital divide in America is rapidly shrinking, yet rural communities, the elderly, people with disabilities, and minorities still lag national averages. — The chapter reports both halves at once: real national progress, and specific groups still behind on internet access and computer literacy.
- Outside developed countries the gap widens, because many developing countries lack affordable internet access and efficient electronic payment methods. — The chapter attributes the wider gap to missing infrastructure and financial resources rather than to individual choices.
- Outsourcing means moving business processes or tasks to another company, which may or may not be in another country. — The chapter's definition covers both: moving processes such as accounting, manufacturing, or security to another company or another country.
- U.S. hospitals outsource the reading of X-rays to skilled radiologists abroad, often while doctors in the United States are sleeping. — The chapter uses this to show that even highly specialized professional services, not just routine tasks, can move across borders.
- Singapore, Canada, and Ireland are declining as outsourcing destinations because salaries there are rising, pushing outsourcers toward Bulgaria, Egypt, Ghana, Bangladesh, or Vietnam. — The chapter treats the outsourcing map as something that keeps moving, which is why an outsourcing decision has to be revisited rather than made once.

Common misconception — sounds reasonable, but the chapter says otherwise

- The digital divide is essentially about who can afford the newest smartphone. — It is about access to information systems and the computer literacy that lets a person compete at all; a new phone with no affordable connection, no payment system, and no skills does not close it.

- Companies outsource mainly to pay lower wages, so cost is the reason that really matters. — Reducing or controlling costs is one of eight listed reasons, alongside freeing internal resources, gaining world-class capabilities, increasing revenue potential, reducing time to market, increasing process efficiencies, focusing on core activities, and compensating for missing skills.
- Once a company has compared labor costs across countries, it has the information it needs to pick an outsourcing location. — Location choice also involves governmental challenges (political systems, regulation, laws, individual freedoms), geoeconomic challenges (infrastructure, demographics, welfare, expertise), and cultural challenges (language, beliefs, religion, views on intellectual property), plus English proficiency and geopolitical risk.
- Globalization pushes cultures apart, since every country now produces media for its own audience. — The chapter says the reverse: movies, television, and other media have moved cultures closer together, creating shared understanding about norms, desirable goods and services, and even forms of government.
- Spreading internet access reliably expands the free flow of communication in the countries it reaches. — The chapter notes the same technology has facilitated an increase in authoritarianism and restrictions on the free flow of communication and content, naming WeChat and TikTok.

The societal issues information systems collide with

The rapid development of transportation and telecommunication technologies, national and global infrastructures, and information systems has created a number of pressing societal issues. Three very different groups discuss them as interrelated societal "megatrends" tied to ever-increasing globalization:

- Consulting firms such as PricewaterhouseCoopers (PwC) and Ernst & Young (EY), who sell advice about them.
- Political and business leaders at the World Economic Forum.
- Local and national governments, who must legislate around them.

They interact with, affect, and fuel one another, so none of them can be managed on its own; the chain is easy to trace:

1. An aging population in one region and a population boom in another change migration.
2. Migration feeds the growth of cities, and about half the world's population now lives in one.
3. Cities consume resources, pressing on limited supplies of fossil fuels and other natural resources.

EXPLORE **The societal issues of the digital world**

Open each issue and read all four panels; the last panel is the part managers underestimate.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Demographic changes · Population structure

- **What it is:** Changes in the structure of populations related to factors such as age, birth rates, and migration.
- **Where you can see it:** Many countries in the developed world see rapidly aging populations, while developing regions such as Africa are expected to rise rapidly in population, fueling massive global population growth.
- **Why it matters now:** These differences shift the balance of demand and supply of labor, so the workers an organization needs and the customers it sells to are increasingly in different places.
- **What makes it hard:** Differences in welfare are likely to keep increasing, and many countries are already experiencing both positive and negative effects of mass migrations, making this a political problem as much as an economic one.

Urbanization · Rural to urban

- **What it is:** The movement of rural populations to urban areas.
- **Where you can see it:** 50 percent of the world's population is now living in cities, and cities are where the chapter's own public-infrastructure examples sit, such as adaptive traffic control systems and variable speed limits.
- **Why it matters now:** Every additional city dweller needs housing, transport, water, and power inside the same square miles, so density turns ordinary services into engineering problems.
- **What makes it hard:** Sustaining that growth while providing livable environments for the inhabitants will pose major challenges, and the growth does not pause while the solutions are being built.

Shifts in economic power · Where the leverage sits

- **What it is:** Changes in countries' purchasing power and control over natural resources.
- **Where you can see it:** Established economies are losing their dominating positions in the world's economy, while a new middle class in developing countries becomes a market of millions of new customers.
- **Why it matters now:** Where purchasing power sits determines which markets a company must serve and which countries set the terms of trade it has to accept.
- **What makes it hard:** The chapter notes the shift results in the need to resolve political struggles, and political struggle is exactly the governmental challenge that makes operating across borders risky.

Resource scarcity · Finite supply

- **What it is:** Pressure created by the limited availability of fossil fuels and other natural resources.
- **Where you can see it:** Population growth, global trade, consumerism, and other factors contribute to increasing waste and pollution as well as a growing need for resources.
- **Why it matters now:** Humans already live beyond the finite natural resources the planet can provide, so demand and supply are moving in opposite directions rather than converging.
- **What makes it hard:** Scarcity is entangled with every other issue on this list: more people, in more cities, with more purchasing power, all need more of the same finite materials.

Climate change · Long-term shifts

- **What it is:** Large-scale, long-term shifts in regional and global temperatures and weather patterns.
- **Where you can see it:** Changing weather patterns, a rise in sea levels, and an increase in the severity of storms.
- **Why it matters now:** These pose many challenges for individuals, societies, and the world, including for any organization whose stores, factories, or supply routes sit in the path of a storm.
- **What makes it hard:** The chapter is careful about scope here: the consequences arrive regardless of what one concludes about the causes, so an organization still has to plan around them.

Sustainable development · Present versus future

- **What it is:** Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.
- **Where you can see it:** The wording is the World Commission on Environment and Development's, from 1987, and the chapter attaches it to every issue above it: waste, pollution, and the need for resources keep rising while the planet's supply is finite.
- **Why it matters now:** The chapter says it will become an increasingly important aspect, which turns a moral argument into a test a manager can apply to a proposal: does this option leave the next generation the same range of choices?
- **What makes it hard:** The needs of the present are concrete and loud while the needs of future generations are abstract and silent, which is why the trade-off usually gets decided in favor of the present.

Technology stands on both sides of these problems. Breakthroughs and transformations enabled by technology disrupt traditional business models — Uber wreaked havoc on the taxi industry — but can also help address pressing societal issues.

Medicine shows how far the systems now reach: professionals in the medical industry use **healthcare IS**, information systems that support healthcare processes ranging from patient diagnosis and treatment to analyzing patient and disease data to running doctors' offices and hospitals.

> Put each new technology through three questions: who gains access, what work crosses a border because of it, and which societal issue it eases or worsens. That turns a gadget into a management decision.

CHECK YOURSELF

Globalization, societal issues, and healthcare IS

Answer each question, then read all four explanations before moving on.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A U.S. hospital sends overnight X-ray images to skilled radiologists in another country to be read while American doctors sleep. Which term does the chapter use for this specific arrangement?

- **A.** Globalization, which the chapter defines as the practice of hiring workers in lower-wage countries. — Globalization is the right family but the wrong definition. The chapter defines globalization as the integration of economies throughout the world, enabled by innovation and technological progress; outsourcing is one activity that integration makes possible. This would be right if it defined globalization properly and called the X-ray example an instance of it.
- **B.** Outsourcing, which shows that even highly specialized professional services can move to another company or country. **— correct.** Outsourcing is the moving of business processes or tasks to another company or another country, and the chapter uses radiology precisely to show that even highly specialized services are candidates.
- **C.** The digital divide, because the country receiving the work has less access to information systems. — The digital divide is the gap between those with and those without access to information systems. It is a separate idea, and it does not fit here: the radiologists reading these images clearly have access to advanced systems.
- **D.** A geoeconomic challenge, because the arrangement rests on differences in infrastructure, welfare, and workers' expertise between the two countries. — Geoeconomic challenges are genuinely differences in infrastructure, demographics, welfare, or workers' expertise, so the words fit the situation. But the chapter uses that term for the difficulties an organization has to manage once it operates across borders, not as a name for the practice of moving the work. It would be right if the question asked what makes choosing this location risky.

2. The chapter quotes a 1987 definition of **sustainable development**. Which statement matches it?

- **A.** Growth a company can maintain year after year without losing market share to competitors. — This is the everyday business sense of 'sustainable' meaning repeatable, and it is the most common mix-up. The chapter's definition is about generations rather than quarters, and this option would be right in a discussion of business strategy instead of societal issues.
- **B.** Reducing waste through recycling programs so that materials are reused rather than discarded. — Recycling is one practice that can serve sustainable development, but the definition is broader: it covers every kind of development, including energy, housing, and agriculture, and it states a constraint on the future rather than a technique in the present.
- **C.** Development that meets the needs of the present without compromising the ability of future generations to meet their own needs. **— correct.** That is the World Commission on Environment and Development's 1987 wording, quoted in the chapter, which presents it as an increasingly important standard as resource scarcity and climate change intensify.
- **D.** Economic activity that avoids fossil fuels, since fossil fuels are the resource the chapter says is running short. — Fossil fuels appear in the chapter under resource scarcity, and using fewer of them may serve sustainable development, but the definition never names any specific resource. It would be right if it were offered as one example of sustainable development rather than as its definition.

3. How does the chapter define **healthcare IS**?

- **A.** Information systems supporting healthcare processes, from diagnosis and treatment to analyzing patient data to running offices and hospitals. **— correct.** The chapter's definition deliberately spans clinical work, analysis, and administration, which is why healthcare IS shows information systems reaching all the way into a traditional profession rather than being a single product.
- **B.** Software that lets patients look up symptoms, view test results, and book appointments online. — This is the patient-facing slice only. Patient portals are part of healthcare IS, but the definition also covers diagnosis and treatment, the analysis of patient and disease data, and the running of doctors' offices and hospitals, so this option is too narrow.
- **C.** The medical equipment itself, such as cardiac monitors and imaging machines, once it is connected to a network. — Devices are hardware. An information system includes the technology, the data, and the people and processes that use it; a cardiac monitor generates data that a healthcare IS then stores, analyzes, and routes to a physician.
- **D.** A specialization within computer science that prepares programmers to build software for hospitals. — This describes a field of study and the people trained in it, not a system. Healthcare IS refers to the systems themselves, which are used by professionals in the medical industry rather than only by programmers.

SECTION 1–1B

Digital density and the digital future

Twenty years ago, driving to work left no record at all. The same drive today throws off a stream of readings — tire pressure, engine temperature, position, how hard you braked — each of which can be stored and pooled with readings from a million other drivers. The activity did not change; the connected data it produces changed enormously, and that change has a name.

Two conditions set this up.

> **Information technologies have become pervasive** — they are used throughout society rather than locked in a computer room, so the technology sits inside ordinary activity.

> **Innovation keeps accelerating** — radical innovations displace existing products and whole industries, which is why in just a few years drones went from mostly military use to farmers, aerial photographers, filmmakers, and hobbyists, why self-parking systems ship in many vehicles, why self-driving cars and trucks are being actively tested, and why autonomous Caterpillar mining trucks are already at work.

The result is an exponential increase in **digital density**: the amount of connected data per unit of activity, where every unit of activity generates ever more connected data and enables new value-added interactions and business models. The chapter builds that in three moves: **connections**, **data**, and the **interactions** they make possible.

> **Read that as a ratio, not a total.** Digital density is not how much data exists in the world; it is how much connected data **one unit of activity** produces. A cash sale of a hammer in 2004 recorded one thing: a hammer left. The same sale today can record who bought it, which ad they saw first, and whether it came back.

INTERACTIVE DIAGRAM **One idea, four angles**

Pick a driver to redraw the diagram; each model is the same digital-density story told from a different starting point.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Connections — Anything physical can now be wired into the digital world.

Mobile devices (Within reach 24/7) → Sensors (Detect, record, report) → IoT and IIoT (Things share data on their own) → Wearables (The body reports in)

- The old picture connected people, organizations, or computers; the new one connects **things** too, from a tire with a pressure sensor to a cow with an injectable ID chip.
- Cheaper **sensors**, better chips, and wireless radios are what made connecting objects affordable instead of theoretical.
- Phones stay within reach around the clock, so customer contact arrives in **micro-moments** rather than long desktop sessions.
- On a factory floor the same idea is the **Industrial Internet of Things**, where information technology and operations technology converge.

Data — Every connection leaves a record behind, and records pile up fast.

Volume (Extremely large datasets) → Variety (Many different types) → Velocity (Collected and analyzed faster) → Data quality (Fit to decide on?)

- **Big Data** is the chapter's name for extremely large and complex datasets high in volume, variety, and velocity.
- Social media poured in **unstructured** data, such as opinions about a product posted on blogs and networks.
- Storage got cheap and moved to the **cloud**, so keeping the data is rarely the constraint; making sense of it is.
- More data is not better data: **data quality** means completeness, accuracy, timeliness, validity, and consistency.

Interactions — Connected data becomes a different way of doing business.

Network effect (More users, more value) → APIs (Use someone else's system) → Servitization (Sensors sell it by usage) → Value from data (Small headcount, large value)

- The **network effect** means a network gets more valuable as more people use it, which is how Uber and Airbnb disrupted established industries.
- **APIs** let a company borrow functionality it never built; Lyft takes payments through Stripe and maps through Google.
- Sensors monitoring performance, temperature, or mileage turn a product into a service, which is how Bridgestone and Michelin can sell tires by usage while staying responsible for how the tires perform.
- Value now comes from data rather than headcount: digital-world firms reached the top with 2,500 to 35,000 employees.

Workforce — The same shift, arriving in your job description.

BYOD (Your device, real work) → Consumerization of IT (Consumer first, work second) → Computer literacy (Use the tool you know) → Computer fluency (Learn the next tool)

- **BYOD** means employees use personal devices for work that now reaches into customer relationship management and other enterprise systems.
- In the **consumerization of IT**, innovations reach shoppers first and organizations second, so managers must keep evaluating consumer technology.
- **Computer literacy**, knowing how to use a computer and certain applications, is now assumed rather than impressive.
- **Computer fluency** is independently learning new technologies as they emerge and assessing their impact on your work and life.

Connections: anything physical can join the digital world

Connections once ran between people, organizations, or computers; today just about any element of the physical world — people, organizations, or **things** — can be joined to the digital realm. The first enabler is the mobile phone, which changes four things at once.

> **Constant reach** — most adults in developed countries keep one within reach 24/7, and in the developing world mobile devices frequently leapfrog PCs, because without stable power or landlines the phone is often the primary means of reaching the internet.

> **Real-time organizations** — for the employer that means more collaboration and the ability to run a business in real time, at any time, from anywhere.

> **Mobile apps** — companies must build these software programs, each designed to perform a particular, well-defined function, to market their products or services, because the customer visit has moved onto the phone.

> **Micro-moments** — interaction happens less in long desktop sessions and more in the moments when a person almost instinctively picks up a mobile device to buy something, know something, do something, or go somewhere.

Checking a store's closing time while walking to your car is a micro-moment, and it leaves a record behind.

The second enabler is the **Internet of Things (IoT)**: a network of a broad range of physical objects that can automatically share data over the internet, such as a tire fitted with a pressure sensor, a smart meter a utility reads remotely, or a cow with an injectable ID chip.

Already in 2008, more devices were connected to the internet than there were people on earth, driven by better chips and wireless radios plus decreasing costs of **sensors** — devices that can detect, record, and report changes in the physical environment. The base technology arrives under several names.

Smart home technologies, or **home automation** — they allow remote monitoring and control of lighting, heating, or appliances such as the Nest Learning Thermostat, a market expected to exceed US$250 billion by 2029.

Wearable technologies — clothing or accessories that incorporate electronic technologies such as the Apple Watch, whose sensors record body movements or heart rate as well as ambient light, orientation, or altitude.

The quantified self — trackers such as the Fitbit are worn passively all day, supporting the logging of all aspects of one's daily life, from activities to moods, to improve health and performance.

Industrial Internet of Things (IIoT) — in manufacturing the same idea converges information technology with operations technology and brings improvements in efficiency, product quality, agility, and flexibility.

Internet of Everything (IoE) — as the number of connected sensors and devices grows, the Internet of Things is expected to become this, where just about any device's functionality is enhanced through connectivity and intelligence.

3D printing — the link also runs the other way, building physical objects from digital models by adding thin layers, instead of milling a part out of a slab and scrapping up to 90 percent of it.

EXPLORE

The technologies doing the connecting

Open each card and read all four facets; the fourth one is the part students usually skip.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Internet of Things · Connections

- **What it is:** A network of a broad range of physical objects that can automatically share data over the internet. The word automatically is the point: no person types the reading in.
- **Where you already see it:** A tire that reports its own pressure, a smart meter your utility reads remotely, an injectable ID chip in a cow, a thermostat you adjust from a parking lot.
- **Why it matters:** It is the reason a unit of activity generates data at all. Already in 2008, more devices were connected to the internet than there were people living on earth.

- **The catch:** Connecting an object is now the easy part. Analyzing the enormous, often unstructured data that results poses tremendous challenges for organizations.

Sensors · Connections

- **What it is:** Devices that can detect, record, and report changes in the physical environment, such as temperature, movement, light, moisture, or heart rate.
- **Where you already see it:** Road surfaces that trigger dynamic speed limits when ice is likely, sensors reporting open parking spaces or traffic flow, soil-moisture sensors on golf courses.
- **Why it matters:** Decreasing sensor costs, plus advances in chips and wireless radios, are what turned the Internet of Things from an idea into an affordable build.
- **The catch:** A sensor reports; it does not judge. A miscalibrated sensor sends confident wrong numbers downstream, which is the data-quality problem in physical form.

Wearable technologies · Connections

- **What it is:** Clothing or accessories that incorporate electronic technologies, such as the Apple Watch, Samsung's Galaxy Watch, or the Google Pixel Watch.
- **Where you already see it:** A smartwatch extending your phone by showing notifications and giving quick access to phone functions, or a Fitbit worn passively all day.
- **Why it matters:** Their sensors record physiological data such as body movements or heart rate and environmental data such as ambient light, orientation, or altitude.
- **The catch:** They enable the quantified self, the logging of all aspects of daily life including moods and physiological states, which is the most personal data a system can hold.

Artificial intelligence · Interactions

- **What it is:** Using information technologies to simulate human intelligence, typically working alongside machine learning algorithms.
- **Where you already see it:** The software layer inside connected products: continuous sensor input paired with machine learning is what the chapter credits for advances in robotics, from self-parking cars to autonomous mining trucks.
- **Why it matters:** Without it the streams from connected things are just volume. AI is what lets organizations anticipate changes, coordinate resources, and personalize offerings.
- **The catch:** Its output inherits the quality of its input. Garbage in, garbage out applies here at enormous scale and without any obvious warning sign.

Robotics · Interactions

- **What it is:** The use of robots to perform manual tasks, which the chapter treats as a direct result of sensors and AI advancing together.
- **Where you already see it:** Autonomous Caterpillar mining trucks already in use, self-parking systems sold in many vehicles, self-driving cars and trucks in active testing.

- **Why it matters:** It shows how connections and data become physical action: continuous sensor input paired with machine learning is what enables the advances.
- **The catch:** Machines taking over manual tasks is part of why the chapter argues that computer fluency, not one fixed skill, is what protects a career.

3D printing · Digital to physical

- **What it is:** Creating physical three-dimensional objects from digital models by successively adding thin layers of material.
- **Where you already see it:** Prototypes and manufactured parts that traditionally were drilled, cut, or milled out of a solid piece of material.
- **Why it matters:** Traditional machining can leave up to 90 percent of a slab as scrap, while additive printing builds only the object itself.
- **The catch:** Better, faster, cheaper printers also make counterfeit goods quick and inexpensive to produce, deceiving buyers and causing intellectual property losses.

Data: what the connections leave behind

Connecting the physical world to the digital one generates tremendous amounts of **Big Data**: extremely large and complex datasets. The chapter defines them by three characteristics.

> **Volume** — the datasets run high in volume, extremely large and complex, because connected things report constantly.

> **Variety** — many different types of data arrive at once, from a sensor reading to a paragraph of opinion.

> **Velocity** — the data are collected and analyzed at ever-increasing rates, so the window for acting keeps shrinking.

Social media added a flood of **unstructured** data, since people voice thoughts about products on blogs and networks — text nobody designed to fit a spreadsheet column. Storage keeps getting cheaper, so organizations use cloud computing to hold the data and to run advanced analytics on what mobile devices, sensors, and social networks generate.

Volume is not virtue. Garbage in, garbage out applies to data, so a key consideration in judging whether data are reliable enough to decide on is **data quality**, which the chapter breaks into five tests.

- **Completeness** — whether a half-filled record is passing as complete.
- **Timeliness** — whether the reading is current, because an old number still reads like a fact.
- **Accuracy** — whether the values match what really happened.
- **Validity** — whether a value is of the kind the field should hold.

- **Consistency** — whether one fact agrees with itself across systems.

A report built on stale readings does not fail loudly, it quietly recommends the wrong thing.

Technology today and tomorrow · Memory crystals. University of Southampton researchers write data into fused quartz with laser pulses, encoding it in five dimensions — height, length, width, position, and orientation. A glass disc the size of a large coin holds 360 terabytes, several hundred times a standard desktop computer (1–4 TB), and the quartz is stable up to 13.8 billion years at room temperature, so data can be archived essentially forever.

CHECK YOURSELF **Connections and data**

Choose the best answer, then read every explanation, including the ones for options you did not pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A ride-hailing trip today produces route data, driver and rider ratings, payment records, and map corrections. The same trip by taxi in 2004 produced a paper receipt. Which chapter term names that change?

- **A.** The digital divide, because the two trips rely on different levels of technology — The digital divide is the gap in which those with access to information systems hold great advantages over those without. It would be right for a question comparing this rider to someone with no computer access or skills at all.
- **B.** Digital density, because each unit of activity now generates far more connected data **— correct.** Digital density is the amount of connected data per unit of activity, and here the unit of activity is identical while the connected data around it multiplied.
- **C.** The network effect, because many more riders joined the service over those years — The network effect is the rise in a network's value as more users join. That is true of ride-hailing, but it describes value from many users, not how much data a single trip creates.
- **D.** Big Data, because a trip record is an extremely large and complex dataset — Big Data names the datasets that result, defined by high volume, variety, and velocity. One trip's records are not Big Data; the pooled records of millions of trips are.

2. A dairy farm gives each cow an injectable ID chip that automatically uploads the animal's location and health readings. In the chapter's vocabulary, this is an example of:

- **A.** The Internet of Things, because a physical object automatically shares data over the internet **— correct.** The chapter uses the cow with an injectable ID chip as one of its own examples of a thing on the Internet of Things: a physical object automatically sharing data over the internet.
- **B.** The Industrial Internet of Things, because the animals are part of a production operation — The Industrial Internet of Things is the chapter's term for IoT technologies used in manufacturing, where information technology converges with operations technology. Sensors on an assembly line would fit; livestock in a field does not.
- **C.** Wearable technology, because the chip travels on the animal all day — Wearable technologies are clothing or accessories that incorporate electronic technologies, such as an Apple Watch or a Fitbit. This option would be right if the device were worn as an accessory rather than implanted.
- **D.** The quantified self, because everything about the cow's day is being logged — The quantified self is a person logging all aspects of their own daily life to improve their health and performance. The defining feature is self-tracking by a person, not a farm tracking an animal.

3. Which set of characteristics does the chapter use to define Big Data?

- **A.** Completeness, accuracy, timeliness, validity, and consistency — Those five are the chapter's definition of data quality, which is how you judge whether data are reliable enough to decide on. They describe the fitness of data, not its scale.
- **B.** Volume, veracity, and value — Veracity and value appear in other popular lists of data attributes and veracity is a genuine concern, but the chapter's three characteristics are volume, variety, and velocity.
- **C.** High volume, variety, and velocity **— correct.** Big Data is described as extremely large and complex datasets of high volume, variety (many different types of data), and velocity (collected and analyzed at ever-increasing rates).
- **D.** Cloud storage, machine learning, and unstructured social media posts — Those three are things the chapter associates with Big Data — cloud computing stores it, machine learning helps make sense of it, social posts supply unstructured input — but they are sources and tools, not the characteristics that define it.

Interactions: connected data becomes a business model

Together, connections and data enable new value-added interactions and business models, because value now comes from data rather than from headcount.

ERA	EXAMPLES THE CHAPTER NAMES	EMPLOYEES	VALUE COMES FROM
Old economy	GE, Dow, Ford	100,000–300,000	Value from people

ERA	EXAMPLES THE CHAPTER NAMES	EMPLOYEES	VALUE COMES FROM
New economy	Microsoft, HP, Oracle	Comparable headcounts	Value from people
Digital world	Airbnb, Nvidia, X	2,500–35,000	Value from data

Continuous sensor input, paired with machine learning and **artificial intelligence (AI)** — using information technologies to simulate human intelligence — makes sense of those Big Data streams and enables advances in **robotics**, the use of robots to perform manual tasks. Big Data also drives research from genomics to climate change, though analyzing enormous and often unstructured data poses tremendous challenges.

Two patterns recur here.

> **The network effect** — the value of a network, or of a tool built on one, increases with the number of other users, which is how Uber and Airbnb disrupted traditional industries, since riders come for drivers and drivers come for riders.

> **Servitization** — a company shifts from selling a physical product to providing it as a service, so sensors monitoring performance, temperature, or mileage let Bridgestone and Michelin sell tires by usage while staying responsible for how they perform.

Where data once mainly improved efficiency, connected data now lets firms anticipate changes, coordinate resources, and personalize offerings.

The API economy

APIs (application programming interfaces) are intermediaries that let different software components exchange data or functionality over common web communication protocols, so a provider can open part of its functionality to others without their needing intimate knowledge of its inner workings. Think of a power socket: a standardized interface delivering electricity from a utility, used without knowing how the power was made.

The value runs both ways.

> **Providers gain revenue and reach** — Stripe processes payments for companies from Target to Lyft.

Users build on functionality they never wrote — Lyft pulls maps from Google Maps' API, and Uber built almost its entire app on other companies' APIs.

The interface holds still — when Stripe changes its internals the API stays the same and users notice nothing, which is why some argue we live in an **API economy**.

MATCH

The vocabulary of digital density

Pair each term with the chapter's definition, then read the note to see what separates it from its nearest neighbor.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Digital density
: The amount of connected data per unit of activity It is a ratio, not a total: the same trip or sale now generates far more connected data than it once did.

Internet of Things (IoT)
: A network of a broad range of physical objects that can automatically share data over the internet Automatically is the key word; a person emailing a photo is not the Internet of Things.

Sensors
: Devices that can detect, record, and report changes in the physical environment Falling sensor costs plus better chips and wireless radios are what made the Internet of Things practical.

Quantified self
: The logging of all aspects of one's daily life to improve health and performance Activity trackers such as the Fitbit are designed to be worn passively so the logging never has to be started.

Micro-moments
: The instinctive moments when a person picks up a mobile device to buy, know, do, or go They replaced the long, well-defined desktop session as the typical shape of a customer interaction.

Network effect
: The value of a network increases with the number of other users It explains why Uber and Airbnb could disrupt established industries so quickly once they had critical mass.

APIs
: Intermediaries that let different software components exchange data or functionality over common web protocols Like a power socket, an API delivers a service in a standard format without exposing how it is produced.

Big Data

Extremely large and complex datasets of high volume, variety, and velocity Volume alone is not the definition; variety means many different types of data and velocity means data collected and analyzed at ever-increasing rates.

Digital density and today's workforce

Fueled by consumer mobile devices and cloud access, employees increasingly use their own devices and familiar software for work — no longer just email, but customer relationship management and other enterprise systems. Two named patterns follow from that.

> **BYOD** (bring your own device) — managing it worries business and IT managers because of security, compliance, and support costs, though it can also raise productivity, retention, and customer satisfaction.

> **The consumerization of IT** — many innovations reach the consumer marketplace first and organizations second.

Knowing how to use a computer is **computer literacy** (or information literacy), which can mean the difference between being employed and being unemployed; some fear the Information Age will not treat information haves, who have almost unlimited access to information, the same as information have-nots, who have limited or no computer access or skills.

	COMPUTER LITERACY	COMPUTER FLUENCY
Definition	Knowing how to use a computer and certain applications	Independently learning new technologies as they emerge and assessing their impact
What it buys	Access to information, and to most jobs at all	What sets you apart in the future

Computer work also stopped being a separate category, because information systems sit inside ordinary jobs.

- Information systems manage air traffic, perform medical tests, and control construction machinery, so their operators are computer users too.
- Engineers and architects use computer-aided design (CAD) to draw and test what they build.
- Medical staff use **healthcare IS**, systems supporting everything from patient diagnosis and treatment to running a hospital.

Sustainability · Alphabet renewables. Running the Google empire takes enormous electricity, so Alphabet aims for 100 percent clean energy by 2030: renewable capacity reached nearly 5.5 gigawatts by late 2019 and contracted agreements topped 14 GW by 2023. Long-term contracts and investments in clean energy companies lock in prices and improve reliability, so this is good business, not only good citizenship.

CHECK YOURSELF

Interactions, APIs, and the workforce

Choose the best answer, then read all four explanations so the near-miss options stop looking attractive.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Airbnb becomes more useful to travelers as more hosts list on it, and more useful to hosts as more travelers search it. The chapter's name for this is:

- **A.** Economies of scale, because Airbnb's cost per booking falls as the company grows — Economies of scale is about a company's own costs per unit falling as volume rises. That may well be true here, but it says nothing about the value each user gets from other users.
- **B.** The consumerization of IT, because ordinary consumers supply the rooms — The consumerization of IT is the pattern of technologies appearing in the consumer marketplace before organizations adopt them. It describes where a technology shows up first, not why a platform gains value as it fills up.
- **C.** The network effect, because a network's value rises with the number of other users **— correct.** The network effect is the notion that the value of a network, or of a tool or application based on a network, increases with the number of other users, and the chapter names Uber and Airbnb as examples.
- **D.** Servitization, because Airbnb provides a service rather than selling a product — Servitization is the shift from selling physical products to providing them as services, as when Bridgestone and Michelin sell tires by usage. Airbnb never sold the room as a product to begin with.

2. Lyft handles payments by connecting to Stripe's API and shows maps by connecting to Google Maps' API. Using the chapter's power-socket analogy, what does an API give the company that uses it?

- **A.** A copy of the provider's software that Lyft can host and modify on its own servers — That describes licensing software to run yourself, which is exactly what an API lets a company avoid. If Stripe handed over its code, Lyft would inherit the job of running payments reliably and securely.

- **B.** A direct connection into the provider's database so Lyft can query the provider's tables — APIs are intermediaries that exchange data or functionality over common web communication protocols, and they deliberately hide the provider's internal structure. Direct table access would break that, since Stripe rewrites its internals without API users noticing.
- **C.** Ownership of the data the provider processes on Lyft's behalf — Using an API is consuming someone else's service, not acquiring their data. The chapter frames the value as two-sided: the provider gains revenue and reach, the user gains functionality it did not have to build.
- **D.** Access to the provider's functionality through a standard interface, without needing to know how it works inside **— correct.** Like a power socket, the API supplies the service in a standardized format while the user stays ignorant of how it is produced, which is why Stripe can change its systems and Lyft notices nothing.

3. An employee is fast with the payroll application she has used for six years, but when the company replaces it she waits to be trained rather than exploring it on her own. In the chapter's terms:

- **A.** She is computer fluent but not computer literate — This reverses the two terms. Fluency is the broader ability and is defined as learning new technologies independently, so it cannot exist without the basic skills that literacy names.
- **B.** She is computer literate but not computer fluent **— correct.** Computer literacy is knowing how to use a computer and certain applications, which she clearly has; computer fluency is independently learning new technologies as they emerge and assessing their impact, which is exactly what she is not doing.
- **C.** She is an information have-not, because she cannot use the replacement system without being trained — Information have-nots are people with limited or no computer access or skills, contrasted with information haves who have almost unlimited access to information. Waiting for training on one new system is not the same as having no access or skills at all.
- **D.** Neither term applies, because literacy and fluency describe an organization's technology rather than one employee's skills — Both terms describe a person, not a system: literacy is knowing how to use a computer and certain applications, and fluency is that person independently learning new technologies as they emerge and judging their impact on her work and life.

SECTION 1–2A

What an information system actually is

Ask most people what an information system is and they point at the computers. The chapter means something wider and far more exact, and the rest of the course rests on it: an **information system (IS)** is the combination of people and information technology that creates, collects, processes, stores, and distributes useful data. Every clause in that sentence is load-bearing, so take it apart before you use it.

Take the definition apart, clause by clause

People and information technology. The people half is not politeness. The chapter is blunt about it: many of today's technologies operate autonomously, but they do not build themselves and they do not exist for their own sake — they are created to serve a useful purpose for people.

A server humming in a closet, with nobody deciding what it should do and nobody acting on what it produces, has the technology half and nothing else, so it does not meet the definition.

Creates, collects, processes, stores, and distributes. Those five verbs are the job description, and a working system does all five. Watch them run in order.

1. **Creates** — tap your card at a store and the register creates a record of the sale.
2. **Collects** — the network collects that record into a central system.
3. **Processes** — software processes those records into the day's totals.
4. **Stores** — a database stores the result, so it outlives the moment it was made.
5. **Distributes** — a report distributes it to a manager who decides what to reorder.

Drop any one verb and the chain breaks: data that are created but never distributed help nobody, and data that are distributed but never processed are just noise arriving faster.

Useful data. The word **useful** is the tell. Any information system involves data that are useful for someone, somewhere — and often for more than one someone at the same time. The chapter offers three examples.

- Transactional data are useful for businesses, which is why a store keeps every single sale.
- Status updates in your news feed on Facebook are useful for your friends and for Facebook itself — one post, two audiences.
- Scores in a computer game are useful for the player and for the game developers alike.

So whenever you evaluate a system in this course, the first question is: useful to whom, for what decision?

One naming note so the vocabulary never trips you up. The same phrase, **information systems**, also names the field — the people who develop, use, manage, and study these systems in organizations. Different schools and companies call that field management information systems, business information systems, computer information systems, or simply systems. Same subject.

COMPLETE IT **Rebuild the two definitions word for word**

Choose the option that makes each sentence match the chapter's definition exactly.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- An information system is the combination of people and **information technology** that creates, collects, processes, stores, and distributes useful data. The definition pairs people with information technology as a whole. Hardware is only one of IT's three parts, and business processes are what the system supports, not a component of it.
- Information technology includes hardware, software, and **telecommunications networks**. An operating system is itself software, and a database is something software manages and hardware stores, so neither is a separate component. The third named component of IT is the telecommunications networks that link systems together.
- A program or set of programs that tell the computer to perform certain tasks is **software**. Hardware is the physical equipment; the instructions telling that equipment which processing functions to perform are software.
- A group of two or more computer systems linked together with communications equipment is **a telecommunications network**. An information system would also need people and useful data, and systems integration is the work of connecting separate systems, not the linked equipment itself.
- Physical computer equipment such as a computer, tablet, or printer, plus components like a monitor or keyboard, is **hardware**. Hardware is the physical equipment, and the chapter extends it to sensors, cameras, and actuators as well as traditional computer components.
- An information system creates, collects, processes, stores, and distributes data that are **useful** for someone, somewhere. Accuracy and timeliness are two of the five parts of data quality, which is a separate idea; the word in the definition itself is useful, as in transactional data that are useful for businesses and game scores that are useful for the player and the developers alike.

The technology half: hardware, software, telecommunications networks

Information technology (IT) includes hardware, software, and telecommunications networks. Three parts, no more and no fewer.

Hardware is physical computer equipment — a computer, a tablet, a printer — along with components like a computer monitor or a keyboard. It has grown well past the desktop to include a variety of other input and output devices such as sensors, cameras, and actuators, which is why a thermostat and a warehouse robot arm both count as hardware.

Software is a program or set of programs that tell the computer to perform certain tasks. Software is how an organization gets its business processes and its competitive strategy out of hardware, because it supplies the hardware with instructions on what processing functions to perform. Two identical laptops running different software run two different companies.

Telecommunications networks are a group of two or more computer systems linked together with communications equipment. Networks allow computers to share data and services, and that sharing is what enables the global collaboration, communication, and commerce we see today — without it every system is an island holding data no one else can reach.

Hardware is easier to place once you know what it replaced. Before the first computers — which worked on a mechanical basis using punch cards — the work ran on physical artifacts.

- Almost all business and government information systems consisted of file folders, filing cabinets, and document repositories.
- A calculating device such as an abacus or a slide rule did the arithmetic.

Those filing cabinets were real information systems. Computer hardware replaced those physical artifacts and gave us technologies to input and process data and output useful information. The purpose never changed; only the machinery did.

IS or IT? Traditionally the term information technology referred to the hardware, software, and networking components of an information system. That difference is shrinking, and many people now use IS and IT synonymously. When a question forces the contrast, use the strict reading: IT is the technology, while an IS is that technology together with people, aimed at producing useful data.

CHECK YOURSELF

The definition and its technology components

Pick the best answer, then read every explanation, including the ones for options you did not choose.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Which statement matches the chapter's definition of an **information system**?

- **A.** A computer program that stores and retrieves an organization's records. — That describes a piece of software, which is one part of the information technology half of an information system. It leaves out the people, the hardware the program runs on, the network it uses, and the useful data that is the point of the whole thing. It would be right if the question had asked what software is.
- **B.** The hardware, software, and telecommunications networks an organization owns. — That is the chapter's definition of information technology, not of an information system. Add the people and the useful data the system produces and you would have the IS definition.
- **C.** The combination of people and information technology that creates, collects, processes, stores, and distributes useful data. **— correct.** and it is the chapter's wording exactly: two halves (people and information technology), five verbs (creates, collects, processes, stores, distributes), and an ending that names the output as useful data.
- **D.** The department that maintains an organization's computers and user accounts. — The chapter does note that information systems is also used to name the field of people who develop, use, manage, and study these systems, so this is a real part of the IS world. But a support department is a group of people doing a job, not the system itself.

2. A hospital installs a small sensor that reads a patient's heart rate and passes each reading over the hospital's wireless link to a monitoring program running on a server in another building. Which part of that description is **hardware**?

- **A.** The monitoring program that reads each number and raises an alarm. — A program is software: a set of programs that tell the computer what processing functions to perform. It runs on hardware but is not hardware itself.
- **B.** The sensor on the patient and the server that receives the readings. **— correct.** Hardware is physical computer equipment, and the chapter explicitly extends it beyond traditional computer components to input and output devices such as sensors, cameras, and actuators. A server is physical equipment too.
- **C.** The wireless link connecting the patient's floor to the other building. — That is a telecommunications network: two or more computer systems linked together with communications equipment. The equipment carrying the link is physical, but what the option names is the connection between systems, which the chapter counts as the third IT component rather than as hardware.
- **D.** The heart-rate readings the sensor produces. — Readings are data, the raw symbols the system creates and collects. They are what the hardware produces, not the hardware.

3. The chapter says the difference between IS and IT is shrinking and that many people use the terms synonymously. What distinction does it still draw between them?

- **A.** Information technology is what an organization buys from vendors; an information system is what it develops itself. — Where a technology came from has nothing to do with the definitions. A purchased payroll package and an in-house one are both software, and both sit inside an information system that also has people and data.
- **B.** Information technology is technology used by consumers; an information system is technology used inside businesses. — The chapter draws no consumer-versus-business line here. Its own examples of useful data include a personal news feed and scores in a computer game right alongside business transactional data.
- **C.** Information technology is the newer term, and it has replaced information system in modern organizations. — This has the history backwards. Information technology is the older, narrower term for the hardware, software, and networking components; the two terms are now often used interchangeably, but neither replaced the other.
- **D.** Information technology is the hardware, software, and networks; an information system adds people and produces useful data. **— correct.** Traditionally IT named the hardware, software, and networking components of an information system, while the IS definition wraps those components together with people and points them at producing useful data.

Data: the root and the purpose of every information system

The definition begins and ends with data, so start there. Unformatted data, or simply **data**, are raw symbols, such as characters and numbers. They have no meaning in and of themselves and are of little value until processed.

The chapter's test case is the string 465889727. If someone asked you what it meant or stood for, you could not tell them — not because you lack a skill, but because there is nothing there to know.

Because everything downstream is built out of these symbols, the old adage garbage in, garbage out applies to data as well. That is why **data quality** is a key consideration in assessing whether data are reliable for making decisions. It consists of five named parts, and the plain reading of each is worth having in front of you.

> **Completeness** — are any values missing? A customer file with blank zip codes cannot answer a question about shipping regions, no matter how many rows it has.

> **Accuracy** — do the values match reality? A mistyped price is worse than a missing one, because a wrong number still looks trustworthy.

> **Timeliness** — are the values current enough for this decision? Last quarter's inventory count cannot tell a store manager what is on the shelf tonight.

> **Validity** — do the values take the form they are supposed to take? A birth date recorded as 30/30/2001 is not a date at all, so nothing can safely compute an age from it.

> **Consistency** — do the values agree across the places they are stored? If two systems hold different addresses for the same customer, at least one is wrong and nothing in the data tells you which.

A system is only ever as trustworthy as the symbols it starts with.

Information: the same data, plus context

Data can be formatted, organized, or processed to make them useful, and when that happens they are transformed into **information**, which is a representation of reality that can help answer questions about who, what, where, and when. Watch it happen to those same nine symbols.

1. Present them as 465-88-9727, and the grouping alone looks familiar.
2. Add that they sit in a certain database, in John Doe's record.
3. Add a field labeled SSN, and now you might rightly surmise that the number is the Social Security number of someone named John Doe.

Nothing about the digits changed. Contextual cues, such as a label, are what turn data into information that is familiar and useful to the reader.

The chapter's second example is closer to your phone. A raw list of all the transactions you made over the course of a month in a peer-to-peer mobile payment system such as Venmo would be fairly useless data — a wall of names and amounts in whatever order they happened.

A table that divided those same payments into two categories, sent and received, would be incredibly useful information: you could manage a monthly budget with it and make better decisions about your finances over time. Without information systems, it would be difficult to transform raw data into useful information at all.

Knowledge: the part that lives in a person

Information still sits on a screen, and a screen decides nothing. To actually use information, something else is required. **Knowledge** is the ability to understand information, form opinions, and make decisions or predictions based on the information. The chapter gives the word two faces.

> **An ability a person brings in** — you must have knowledge to be aware that only one Social Security number can uniquely identify each individual, because the database never told you that; you brought it with you.

> **A body of governing procedures** — guidelines or rules used to organize or manipulate data to make them suitable for a given task, such as the rule that two records sharing one SSN are either the same person or an error.

Both faces live in a person, not in a file.

	DATA	INFORMATION	KNOWLEDGE
Example	465889727	465-88-9727	465-88-9727 → John Doe
What it is	Raw symbols	Formatted data	Data relationships
Meaning	???	SSN	SSN → unique person

> **The one distinction to get right.** Data are raw symbols. Information is data given form and context so that it answers who, what, where, and when. Knowledge is the ability to understand that information and act on it. The trap is usually a middle option that sounds like the top one: a well-formatted, clearly labeled, freshly generated report is still information, however impressive it looks. It becomes knowledge only inside someone who can read it, judge what it means, and decide.

INTERACTIVE DIAGRAM

From raw symbols to a working system

Select each stage to see what it contains and what has to be added to reach the next one.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Data — Raw symbols, carrying no meaning yet

465889727 (Nine characters) → No label (Nothing states what it is) → No answer (You cannot say what it stands for)

- **Data** are raw symbols, such as characters and numbers, and they have no meaning in and of themselves.
- Data are of little value until processed, which is why the chapter calls them the root rather than the result.
- **Data quality** decides whether they are reliable enough to make decisions on: completeness, accuracy, timeliness, validity, and consistency.
- Garbage in, garbage out — every later stage inherits whatever defects are in the symbols.

Information — Formatted data that answers who, what, where, and when

465-88-9727 (Formatted) → Field: SSN (A label supplies context) → Record: John Doe (It now points at someone)

- **Information** is a representation of reality that helps answer questions about who, what, where, and when.
- Data become information when they are formatted, organized, or processed — the digits themselves never changed.
- Contextual cues, such as a label, are what make data familiar and useful to a reader.
- The month of payments split into **sent** and **received** is the same move: useless list in, usable budget table out.

Knowledge — The ability to understand information and act on it

SSN to person (A relationship, not a value) → One number, one person (A governing rule) → Decide or predict (Approve, flag, or reject)

- **Knowledge** is the ability to understand information, form opinions, and make decisions or predictions from it.
- You need knowledge to be aware that only one Social Security number can uniquely identify each individual.
- Knowledge is also a body of governing procedures — guidelines or rules — used to organize or manipulate data for a given task.
- It is the only stage that lives in a person rather than in a file, which is why the people half of the IS definition is not optional.

The whole system — People plus information technology, producing useful data

People (Build, manage, and use it) → Hardware (Equipment, sensors, actuators) → Software (Instructions for the hardware) → Networks (Two or more systems linked)

- These four together make an **information system** that creates, collects, processes, stores, and distributes useful data.
- The three technology boxes are exactly what the chapter means by **information technology**: hardware, software, telecommunications networks.
- **Systems integration** connects separate, often modular systems and their data using technologies such as APIs, so the boxes of different systems can work as one.
- Useful means useful to somebody: transactional data serve the business, a news feed serves your friends and the platform, game scores serve the player and the developers.

SORT

Data, information, or knowledge?

Place each item in the bucket that matches what it is at that moment, not what it could eventually become.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Data — raw, unformatted symbols with no meaning attached

- The string 465889727, with no label and nothing around it — These are raw symbols, and the chapter's whole point is that you cannot say what they mean or stand for.
- A list of every payment you sent and received last month, in the order they happened — The chapter calls exactly this a fairly useless list until it is organized, because nothing about the order answers a question you care about.
- The value 98.6 arriving from a thermometer sensor with no label attached — A number produced by real equipment measuring a real body is still a raw symbol until something says what it is a measurement of.
- The characters a barcode scanner sends to the register when an item is scanned — The scanner creates raw symbols; software has to process them against a product file before anyone learns a price or a product name.

Information — formatted or organized so it answers who, what, where, when

- 465-88-9727 stored in a field named SSN inside John Doe's record — Formatting plus a label plus a record is exactly the context that turns the same digits into a representation of reality.
- That same month of payments shown in a table split into sent and received — Dividing the transactions into categories is the formatting step, and the chapter calls the result incredibly useful for managing a budget.
- A chart line reading Temperature 98.6 F, taken 3:14 p.m. — The label and the timestamp answer what and when, which is precisely the work information does.

Knowledge — understanding, rules, and the ability to decide

- Being aware that only one Social Security number can uniquely identify each individual — No database supplied that; it is the understanding a person brings, and it is what lets someone spot a duplicate as an error.
- Your conclusion, after reading that table, that you should stop paying for weeknight delivery — Forming an opinion and making a decision based on information is the chapter's definition of knowledge.
- The standing rule that any temperature above 100.4 F must be reported to a physician — The chapter describes knowledge as a body of governing procedures, such as guidelines or rules, used to make data suitable for a task.

Systems integration: making the pieces speak to each other

Hardware, software, and networking components evolve rapidly, which makes the ability to tie everything together ever more important. Two chapter terms name that work.

> **Internetworking** is connecting host computers and their networks together to form even larger networks, like the internet.

> **Systems integration** is connecting separate and often modular information systems and data, using technologies such as APIs, to improve business processes and decision making.

Integration is also why the tidy question "what kind of system is this?" often has no tidy answer any more. Ten to 15 years ago it would have been typical to see systems that fell cleanly into one category, and two developments since then blurred the lines.

- Today many organizations have replaced stand-alone systems with enterprise systems that span the entire organization.
- With internetworking and systems integration in play, it is difficult to say that any given information system fits into only one category.

Systems integration also sits on the chapter's list of skills IS professionals need, described there as connectivity, compatibility, and integrating subsystems and systems.

CHECK YOURSELF **The distinction that gets tested most**

Answer, then read all four explanations, because the wrong options here are the ones students actually pick.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A fitness band records the value **72** every ten seconds and stores each one with no label attached. At that moment, what is the value 72?

- **A.** Information, because the band captured the value and wrote it to storage, and being held in a system is what makes a number information. — Storing something changes where it lives, not what it is. Data become information when they are formatted, organized, or processed, and none of those has happened here; a stored value with no label is still data.
- **B.** Data, because it is a raw symbol that carries no meaning until something formats, organizes, or processes it. **— correct.** Unformatted data are raw symbols such as characters and numbers, with no meaning in and of themselves and little value until processed. That is precisely the state of an unlabeled 72.

- **C.** Knowledge, because the band acts on the value to decide when to buzz your wrist, and acting on a number is what knowledge means. — An automated threshold is a rule applied to data, not knowledge in the chapter's sense. Knowledge is the ability to understand information, form opinions, and make decisions or predictions from it, so it needs a person who can judge what the reading means.
- **D.** Information, because 72 sits in the normal heart-rate range, so a reader can work out its meaning without a label. — This is the most tempting wrong answer, because you personally can guess what 72 means. But the chapter is explicit that contextual cues, such as a label, are needed to turn data into information. Attach the label heart rate, in beats per minute, at 2:15 p.m., and the same value becomes information.

2. The chapter contrasts a raw month of peer-to-peer payment transactions with a table that divides the same payments into sent and received. Which reading matches the chapter?

- **A.** The raw list is data and the table is information, because sorting them into categories gives them a form a person can use. **— correct.** The chapter calls the raw list fairly useless data and the categorized table incredibly useful information, and the only thing that changed between them is organization.
- **B.** Both are information, because both are records of payments that really happened. — Being true does not make something information. Truth is a data-quality question (accuracy); the data-versus-information question is about whether the values have been given form and context.
- **C.** The raw list is information and the table is knowledge, because the table can be used to make budgeting decisions. — Half right and half wrong, which is what makes it tempting. The raw list has no organization at all, so it is data, not information. And a table cannot be knowledge: knowledge is the ability to understand the table and decide, which lives in the person reading it.
- **D.** The raw list is data and the table is knowledge, because the app processed the transactions to build it. — Processing is how data become information, not how they become knowledge. The app can produce the table all by itself; it cannot form the opinion that your delivery spending is too high.

3. Which of these is **knowledge** in the chapter's sense?

- **A.** The nine digits 465889727, with nothing attached to them. — That is the chapter's example of data: raw symbols you cannot interpret. It would become information the moment a label and a record put it in context.
- **B.** The value 465-88-9727 sitting in a field labeled SSN in John Doe's record. — That is information: formatted data with contextual cues that answer who and what. It is one step short of knowledge, which is the step most students skip.

- **C.** Understanding that a Social Security number identifies exactly one person, so two records sharing one must be the same person or an error. **— correct.** Knowledge is the ability to understand information, form opinions, and make decisions or predictions from it, and it also covers governing rules such as one number, one person, used to organize or manipulate data for a task.
- **D.** A printed report listing every customer's Social Security number beside their account balance. — That is a lot of information at once, cleanly formatted and labeled, but volume and polish are not knowledge. The report becomes knowledge only in a person who can read it, judge what it means, and act — flagging a duplicate, catching an error, approving a loan.

If you keep one thing from this section, keep the shape of the definition: people and information technology (hardware, software, telecommunications networks) doing five things — create, collect, process, store, distribute — to data, so that what comes out is useful to someone. Every system named in the rest of the chapter is a variation on that single sentence.

SECTION 1–2B

The people who build, manage, and use information systems

Hardware, software, and networks do nothing on their own. The IS field is a vast collection of people who develop, maintain, manage, and study information systems. But a system does not exist in a vacuum, and the chapter is blunt about the other half: it is of little use if it were not for you — the user.

Why IS jobs keep landing at the top of the rankings

Because data has become so valuable for competitive advantage, the chapter argues that every company can now be considered a technology company, needing people who can optimize its business processes and find new ways of using information systems.

The forecasts agree. The **Occupational Outlook Handbook** from the U.S. Bureau of Labor Statistics (2025) predicts employment for computer and IS managers will grow **17 percent through 2033**, much faster than the average for all other occupations, and in nearly every industry, not just computer hardware and software companies.

Independent rankings point the same way, from three different directions.

> **Glassdoor** — among the 50 best jobs in America ranked by the job site, almost all of the top 10, and nearly half of the top jobs overall, were technology related.

> **U.S. News & World Report** — ranked **IT manager** the second best job in America, with software developer, information security analyst, and data scientist also among its top 10.

> **The degree itself** — an information systems degree can also provide the foundation for becoming a **data scientist**, currently one of the jobs with the highest demand.

RANK	CAREER	JOB SCORE (OUT OF 5.0)	MEDIAN PAY (US$)
1	Enterprise Architect	4.1	144,997
2	Full Stack Engineer	4.3	101,794
3	Data scientist	4.1	120,000
4	Dev Ops Engineer	4.2	120,095
5	Strategy Manager	4.2	140,000

RANK	CAREER	JOB SCORE (OUT OF 5.0)	MEDIAN PAY (US$)
6	Machine Learning engineer	4.3	130,489
7	Data engineer	4.0	113,960
8	Software Engineer	3.9	116,638
9	Java Developer	4.1	107,099
10	Product Manager	4.0	125,317

Table 1.1 · Best Jobs in America (2022), as ranked by the job site Glassdoor.

Read that table honestly. It is evidence, not proof.

- Rank is not pay: Java Developer sits at rank 9 earning US$107,099, ahead of Product Manager, who earns more at US$125,317.
- Rank is not the job score either: Enterprise Architect leads at 4.1 while Full Stack Engineer and Machine Learning engineer both score 4.3. Rank blends factors the table never prints.
- It is one job site's 2022 snapshot, and the chapter says rankings differ by source. What survives is the demand.

Pay follows the demand, though the figure depends on who is counting.

Bureau of Labor Statistics (2024) — put median annual earnings for these managers in May 2024 at **US$187,990**, with the top 10 percent earning more than US$239,200.

Salary.com — reported a 2025 median of US$149,316 for IT managers, well below the federal figure for the same kind of work.

National Association of Colleges and Employers (2025) — expects a mean starting salary of **US$71,556** for management information systems graduates.

Association for Information Systems, with Temple University's Institute for Business and Information Technology — put the average starting salary for IS graduates holding a master's degree at US$96,164, higher than business majors such as accounting, finance, or marketing.

Careers in information systems

The field includes those who design and build systems, those who use them, and those who manage them: **systems analysts**, **systems programmers**, **systems operators**, **network administrators**, **database administrators**, **systems designers**, **systems managers**, and **chief information officers**.

The table below samples these positions rather than listing them all, and titles are slippery: firms use one title differently, or different titles for one function.

IS ACTIVITY	JOB TITLE	JOB DESCRIPTION	MEDIAN SALARY (US$)
Develop	IS analyst	Analyze business requirements and select information systems that meet those needs	73,581
Develop	Software developer I	Code, test, debug, and install programs	81,690
Develop	Software architect IV	Create customized software for large corporations	145,613
Develop	Systems analyst consultant	Provide IT knowledge to external clients	149,100
Develop	Senior database engineer	Develop, modernize, and streamline databases	125,800
Maintain	IT auditor I	Audit information systems and operating procedures for compliance with internal and external standards	70,790
Maintain	Database administrator II	Manage database and database management software use	104,390
Maintain	Webmaster	Manage a firm’s website	84,300
Manage	IT department manager	Manage existing information systems	135,760

IS ACTIVITY	JOB TITLE	JOB DESCRIPTION	MEDIAN SALARY (US$)
Manage	IS and cyber security manager	Manage security measures and disaster recovery	146,790
Manage	IT quality assurance manager	Ensure availability and security of information stored on networks and in the cloud	121,510
Manage	E-commerce technical manager	Manage development, maintenance, and strategy related to e-commerce systems	142,020
Manage	Chief information officer (CIO)	Highest-ranking IS manager; oversee strategic planning and IS use throughout the firm	344,404
Manage	Chief digital officer (CDO)	Executive focused on converting traditional analog businesses to digital; oversee operations in rapidly changing digital sectors like mobile apps and social media	298,890

Table 1.2 · IS management titles, grouped by develop, maintain, manage.

MATCH **Who actually does what**

Pair each IS job title from Table 1.2 with the work that person really does; read the description first, then find the title it belongs to.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

IS analyst
: Analyze business requirements and select information systems that meet those needs This is a develop role, but the work starts with the business problem, not with code — which is why business competency matters even in an entry-level analyst job.

Software architect IV
: Create customized software for large corporations Median salary US$145,613, well above Software developer I at US$81,690, because architecture decisions shape everything built afterward.

IT auditor I

Audit information systems and operating procedures for compliance with internal and external standards Auditing is filed under maintain, not manage: the auditor checks that systems and procedures match the rules the firm is held to.

Database administrator II

Manage database and database management software use Do not confuse this with senior database engineer, who develops, modernizes, and streamlines databases; the administrator keeps the running database healthy and used properly.

IS and cyber security manager

Manage security measures and disaster recovery Note that disaster recovery sits in the same job as security — the same pairing the chapter recommends against ransomware.

Chief information officer (CIO)

Highest-ranking IS manager; oversee strategic planning and IS use throughout the firm At US$344,404 median, the highest-paid title in Table 1.2, and the job is strategy across the whole firm rather than any one system.

Chief digital officer (CDO)

Executive focused on converting traditional analog businesses to digital The CDO exists because of transformation work — taking a business built on paper and processes into mobile apps, social media, and other fast-changing digital sectors.

What makes IS personnel so valuable?

The chapter retires an old stereotype: IS departments are no longer filled only with nerdy men wearing pocket protectors, many more women hold IS positions now, and the IS professional is commonly a polished businessperson fluent in both business and technology.

Studies asking what makes them valuable agree on the answer. Good IS personnel possess valuable, **integrated** knowledge and skills in three areas — **technical**, **business**, and **systems** — and integrated is the operative word, because the chapter says twice that it is the business and systems areas that set the IS professional apart from people who have only technical knowledge and skills.

DOMAIN	DESCRIPTION
Technical knowledge and skills	
Hardware	Hardware platforms, infrastructure, cloud computing, virtualization, peripherals, mobile devices
Software	Operating systems, application software, non-relational databases, AI and machine learning, prompt engineering, mobile apps, APIs
Networking	Network administration, wireless networks, 5G, cybersecurity

DOMAIN	DESCRIPTION
Business knowledge and skills	
Business integration, industry	Business processes, functional areas of businesses and their integration, industry characteristics
Managing people and projects	Planning, organizing, leading, controlling, managing people and projects
Social	Interpersonal, group dynamics, political
Communication	Verbal, written, and technological communication and presentation
Systems knowledge and skills	
Systems integration	Connectivity, compatibility, integrating subsystems and systems
Development methodologies	Steps in systems analysis and design, systems development life cycle, alternative development methodologies
Critical thinking	Challenging one's and others' assumptions and ideas
Problem solving	Information gathering and synthesis, problem identification, solution formulation, comparison, choice

Table 1.3 · Core competencies — note that social and communication skills sit under **business**.

Technical competency

The **technical competency** area covers hardware, software, networking, and security — the "nuts and bolts." Students often get the next part backwards.

> **Just enough, not expert** — the IS professional is not required to be a technical expert in these areas, but must know just enough to understand how they work, what they can do for an organization, and how they can and should be applied.

> **Direct the specialists** — typically the IS professional manages or directs those who have deeper, more detailed technical knowledge, rather than out-building them.

> **Hardest area to keep current** — it is perhaps the most difficult area to maintain, because the pace of technological innovation is so rapid.

> **The outsourcing scare did not play out** — programming and support jobs once looked destined for outsourcing to providers abroad; instead demand rose for application development skills, especially combined with sound business analysis and project management.

The hot skills the chapter names for the next decade point the same way: many of them sit in the business domain rather than the technical one, with the rest spread across technology infrastructure and services, security, applications, Internet, and business analytics/data science.

Business competency

Business competency separates the IS professional from someone with only technical knowledge, and in an era of increased outsourcing the chapter says it may well save a person's job. Even as low-level technology jobs move offshore, the Bureau of Labor Statistics (2025) reports increased need for IS managers as organizations embrace cloud computing, cybersecurity, and artificial intelligence.

Three demands sit inside that competency.

> **Know the business, not only the systems** — it is vital to understand both the technical areas and the nature of the business those areas exist to serve.

> **Manage people and projects, not technology alone** — these are the business skills that propel IS professionals into project management and, ultimately, high-paying middle- and upper-level management positions.

> **Count people skills as business skills** — note what is filed here in Table 1.3: interpersonal skill, group dynamics, politics, and presentation ability are business skills, not extras.

Systems competency

Systems competency is the other area that sets the IS professional apart, and the table above breaks it into four parts.

> **Systems integration** — joining subsystems into systems so that they connect and stay compatible with one another.

Development methodologies — the steps in systems analysis and design, the systems development life cycle, and the alternative development methodologies that compete with it.

Critical thinking — challenging your own and others' assumptions and ideas rather than accepting them.

Problem solving — gathering and synthesizing information, identifying the problem, formulating and comparing solutions, and making the choice.

Those who understand how to build and integrate systems and how to solve problems ultimately manage large, complex systems projects, and manage those in the firm who have only technical knowledge and skills.

Add the social skills to work well with and motivate others and you have the chapter's whole answer to why IS professionals are valuable.

CHECK YOURSELF

Rankings, titles, and the three competencies

Pick the best answer, then read every explanation — the wrong options are the misreadings this section is designed to prevent.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. In the Best Jobs in America table, Enterprise Architect ranks 1st with a job score of 4.1 while Full Stack Engineer ranks 2nd with a **higher** score of 4.3; Java Developer ranks 9th on a median of US$107,099, ahead of Product Manager, who earns US$125,317. What does that pattern tell you?

- **A.** Rank blends several factors at once, so a career can rank higher without having the top score or the top pay **— correct.** The two columns disagree with the ordering in both directions, so the rank must combine factors the table does not print. That is why the chapter says the rankings differ between sources while the overall demand story stays clear.
- **B.** Rank is a straight ordering by median pay, so the pay column should fall steadily from row 1 to row 10 — Ordering by pay is the most natural reading of a jobs list, but the pay column does not fall steadily: rank 2 pays US$101,794 while rank 9 pays US$107,099. This would be right only if rank and pay moved together, and here they cross.
- **C.** Job score measures how demanding the job is, so a lower score earns a higher rank — Job score is a rating of the career, not a difficulty index — the chapter reports it as a score out of 5.0 alongside pay. If it measured difficulty, the highest-scoring careers would sit at the bottom, and they do not.

- **D.** Rank counts openings in each career, so the score and pay columns describe the job without affecting its position — Hiring volume genuinely is one ingredient such lists use, so this is a fair guess. But the table publishes only rank, job score, and median pay; it reports no opening counts, so this evidence cannot support the claim.

2. The chapter says the IS professional must know "just enough" about hardware, software, and networking. What does that phrase actually mean?

- **A.** Technical knowledge is optional once you reach management, since business skills are what get people promoted — Business skill does drive promotion, which is what makes this tempting. But the chapter's whole argument is that the three areas are *integrated*; a manager who cannot evaluate the technology is the person vendors sell to. This would be right only if the chapter described two competencies instead of three.
- **B.** The technical area is the easiest of the three to keep current, because the fundamentals rarely change — The chapter says the opposite: technical is perhaps the most difficult area to maintain, because of the rapid pace of technological innovation. This would be right if the hot-skills list stayed the same each decade, and it does not.
- **C.** Enough to understand how these technologies work, what they can do for an organization, and how they should be applied **— correct.** Enough understanding to judge fit and application, not enough to out-build the specialists — the IS professional typically manages or directs those with deeper, more detailed technical knowledge.
- **D.** Expert-level depth in each technical area, because you cannot direct specialists unless you know more than they do — The instinct that authority requires out-knowing your specialists is common, and it is the one the chapter denies: the IS professional typically manages or directs those who have deeper, more detailed technical knowledge. This would be right for a role defined by a single technology, but it leaves no room for the business and systems competencies.

3. Table 1.3 lists **Social** (interpersonal, group dynamics, political) and **Communication** (verbal, written, technological communication and presentation). Which of the three competency domains are those two filed under?

- **A.** Technical knowledge and skills — Technical covers hardware, software, networking, and security — cloud computing, virtualization, prompt engineering, APIs, 5G. This answer would be right if the question had asked where non-relational databases or wireless networks sit.
- **B.** Business knowledge and skills **— correct.** Business knowledge and skills has four entries: business integration and industry, managing people and projects, social, and communication. Filing people skills here is deliberate — they are treated as business capability that earns money, not as decoration.
- **C.** Systems knowledge and skills — Systems covers systems integration, development methodologies, critical thinking, and problem solving — how the pieces fit and how problems get solved. This would be right for "connectivity, compatibility, integrating subsystems and systems."

- **D.** A fourth domain of soft skills, kept separate from the three core competencies — Calling them soft skills is exactly the habit the table pushes back on. The chapter names three domains and no fourth; putting social and communication inside business is how it insists they are core, not supplementary.

You — the user

Technology is changing how products and services are produced, distributed, marketed, and sold, so in almost any business field — finance, accounting, operations, human resources, business law, marketing — you will use information systems constantly and be pulled into decisions about them.

Three habits are what the chapter says are likely to set you apart from your competition.

> **Understanding what systems can do** — and, just as importantly, what they cannot do, so a vendor's promise can be measured against what the technology delivers.

> **Being able to communicate with the technical staff** — describing a business need in terms the people who build the system can act on.

> **Making educated IS-related decisions** — taking part in the choice itself rather than waiting to be handed someone else's pick.

The risk is sharpest in smaller organizations with no dedicated IS department. There you are likely involved in IS investment decisions, and lacking a basic grasp of IS infrastructure, systems analysis and design, or security leaves you at the mercy of consultants or, worse, vendors acting out of their own interests and selling their "technology of the week."

That is the responsibility a user carries: not building the system, but being hard to fool about it.

EXPLORE **Three competencies, plus you**

Open each card and read all four facets; the fourth card is the one that applies to you no matter what you major in.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Technical competency · The nuts and bolts

- **What it covers:** Knowledge and skills in hardware, software, networking, and security. Hardware means platforms, infrastructure, cloud computing, virtualization, peripherals, and mobile devices; software means operating systems, application software, non-relational databases, AI and machine learning, prompt engineering, mobile apps, and APIs; networking means network administration, wireless, 5G, and cybersecurity.
- **Where it shows up in real work:** Knowing enough to tell whether a vendor's cloud proposal actually fits the firm, or why a wireless network keeps dropping in half the building.
- **What goes wrong without it:** You cannot judge what is being promised to you, and you cannot tell a hard technical problem from an excuse. But being *only* technical is exactly what the chapter says leaves a job exposed to outsourcing.
- **Why the chapter says it pays:** It is the hardest area to maintain, because innovation is so rapid. The IS professional usually manages or directs the deeper specialists rather than out-coding them.

Business competency · What the work is for

- **What it covers:** Business processes, the functional areas of a business and how they integrate, and industry characteristics. It also covers planning, organizing, leading, and controlling, managing people and projects, social skills (interpersonal, group dynamics, political), and communication (verbal, written, technological, and presentation).
- **Where it shows up in real work:** Running a project to a deadline, writing a proposal a non-technical manager will approve, reading the politics of a group, presenting a recommendation to the people who control the budget.
- **What goes wrong without it:** Someone with only technical skill builds exactly what was requested instead of what the business needed, and then cannot argue for the funding to fix it.
- **Why the chapter says it pays:** The chapter says this competency sets IS professionals apart and, in an era of increased outsourcing, may well save a person's job; it is what propels them into project management and upper-level management.

Systems competency · Making the pieces fit

- **What it covers:** Systems integration (connectivity, compatibility, joining subsystems into systems) and development methodologies (steps in systems analysis and design, the systems development life cycle, alternative methodologies). It also covers critical thinking, meaning challenging your own and others' assumptions and ideas, and problem solving: gathering and synthesizing information, identifying the problem, formulating and comparing solutions, choosing.
- **Where it shows up in real work:** Getting a new checkout system to exchange data with an inventory system written years earlier by a different vendor, and deciding which of three candidate fixes is genuinely worth doing.
- **What goes wrong without it:** You end up with components that are each correct and still will not work together, or a system that gets built before anyone identified the real problem.

- **Why the chapter says it pays:** Those who understand how to build and integrate systems and how to solve problems ultimately manage large, complex systems projects — and manage the people who have only technical knowledge.

You — the user · Every major, every job

- **What it covers:** Not a job title but a role you already hold: the person who uses the system and takes part in decisions about it. The chapter says an information system is of little use if it were not for you, the user.
- **Where it shows up in real work:** Choosing your next phone or operating system, deciding how you feel about sharing personal information online, securing the wireless network at home, keeping files in sync across devices — and at work, sitting in on a decision about what software to buy.
- **What goes wrong without it:** In a small firm with no IS department you still make the investment call. Without a basic grasp of IS infrastructure, systems analysis and design, and security, you are at the mercy of consultants or vendors selling their technology of the week.
- **Why the chapter says it pays:** Understanding what systems can and cannot do, and being able to communicate with the technical staff, is what the chapter says will set you apart from your competition in any business field.

Security matters: ransomware. Ransomware is a virus that, once it infects a system or network, encrypts the data it finds in place, in a format that renders it impossible for the victim to access. The attacker then demands a ransom for the decryption keys — though many victims have painfully learned that paying does not guarantee regaining the data.

- Victims range from individuals, who may lose years of family photos or personal records, to businesses large and small, which may lose customer records, financial data, or intellectual property.
- It usually arrives as an attachment to a spam email, or is downloaded in the guise of a video or other content from a website — which is why the user is part of the security system.
- Early demands were sized to be paid: attackers often targeted individuals and small businesses, and the average demand in 2018 was just US$530, with an upper limit of about US$1,000. Small ransoms still exist, but the shift is toward larger, higher-value companies, government agencies, and organizations — Change Healthcare was forced to pay a **US$22 million** ransom in early 2024.
- Attackers increasingly threaten to publish the data; for organizations holding personally identifiable information, lawsuits and reporting duties can cost more than the lost access.
- The chapter's answer has two parts: raise security awareness and vigilance to avoid infection, and improve backup and disaster recovery so infection is survivable.

CHECK YOURSELF **You are part of the system**

Answer each question from the chapter's account of the user's role and the Security Matters box on ransomware.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. You take a marketing job at a 30-person company with no IS department, and the owner asks you to help choose a new customer database. Why does the chapter say your IS knowledge matters here?

- **A.** You will be part of the investment decision, and without basic understanding you are at the mercy of consultants and vendors **— correct.** The chapter singles out smaller organizations without dedicated IS departments: you are likely involved in the investment decision, and lacking a grasp of IS infrastructure, systems analysis and design, or security leaves you exposed to people acting in their own interests.

- **B.** Small organizations face few IS decisions, so the stakes are low enough that any reasonable choice works out — The chapter argues the reverse — the absence of an IS department raises your exposure rather than lowering it, because there is no specialist to catch a bad decision. This would be right only if small firms did not depend on their systems, and they do.
- **C.** Hand the decision to an outside vendor, since their technical knowledge is deeper than anyone at the firm has — Vendors do have deeper technical knowledge; that is precisely the problem. The chapter warns that vendors are likely to act out of their own interests and sell their "technology of the week." Deferring to them would be reasonable only if their incentives matched yours.
- **D.** IS knowledge here is really product training, so what matters is learning whichever database the owner buys — Training on the tool you end up with does matter, but it comes after the decision this scenario is actually about. The chapter asks users to understand what systems can and cannot do so they can take part in the investment decision itself; this would be right only if someone qualified were reliably making that call for you.

2. What does ransomware actually do to a victim's data?

- **A.** It deletes the files permanently, which is why payment cannot restore them — Permanent deletion describes a destructive wiper, and it would make a ransom pointless — there would be nothing to sell back. The chapter's point is subtler: the data is still there, just unreadable, and paying still may not recover it.
- **B.** It encrypts the data in place in a format the victim cannot access, and the attacker demands payment for the decryption keys **— correct.** The chapter defines ransomware as a virus that, once it infects a system or network, encrypts the data it finds in place so the victim cannot access it, after which the attacker demands a ransom for the decryption keys.
- **C.** Its defining move is copying the data out and publishing it publicly — Data theft and public release is real and growing — the chapter says attackers increasingly threaten to release data if not paid, and for firms holding personally identifiable information the lawsuits and reporting duties can cost more than the lost access. But that is an added pressure tactic, not the definition.
- **D.** It locks the screen behind a warning message while the files stay readable underneath — Screen-locking scareware exists and this is a common mental picture, which is why it is a trap. Ransomware attacks the files themselves through encryption, so removing the malware does not give the data back.

3. Given the chapter's account, what should individuals and organizations actually do about ransomware?

- **A.** Pay promptly, since paying is what gets the decryption keys released — Paying feels like the direct fix, and the ransom is designed to look cheaper than the outage. But the chapter states that many victims have painfully experienced that paying the ransom does not guarantee regaining access to the data.

- **B.** Budget for it as a routine cost, since the 2018 average demand was about US$530 — That US$530 figure is real, but it is the 2018 average demand from a period when attackers targeted individuals and small businesses. The chapter documents the shift to larger, higher-value targets — Change Healthcare paid US$22 million in early 2024 — so budgeting from the old number understates the exposure badly.
- **C.** Stop storing personally identifiable information, since that is the data ransomware goes after — Personally identifiable information does raise the stakes, because public release brings lawsuits and reporting requirements. But ransomware encrypts whatever it finds: family photos, personal records, customer records, financial data, intellectual property. Limiting what you store is not a defense against encryption.
- **D.** Raise security awareness and vigilance to avoid infection, and strengthen backup and disaster recovery so an infection is survivable **— correct.** The chapter closes the box with exactly this pair: increase security awareness and vigilance to better avoid infections, and improve backup and disaster recovery preparations to become more resilient to such attacks.

The same reasoning covers private life, where IS-related decisions already abound.

- Which mobile phone to purchase next, and which operating system you prefer on your personal computer.
- How to best secure the wireless network you run at home.
- Whether to share personal information online, and how much of it.
- How to keep your various files in sync across different computers and mobile devices.

They feel like preferences, but underneath sit real differences in privacy, security, and apps. And a startup idea is only the first step — information systems appear in every phase of bringing it to market.

SELF-CHECK **The competencies you are building**

Rate each statement honestly; anything you cannot do yet has a pointer to the exact place in this section to reread.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- I can use the chapter’s own numbers to explain why IS careers rank so highly — and also say what the Best Jobs table does not prove. The 17 percent BLS growth forecast through 2033, and the honest-reading callout under Table 1.1.
- I can name at least six IS management job titles and say what each of those people actually does. Table 1.2, which groups titles by whether the work is to develop, maintain, or manage.
- I can name the three core competencies and give two sub-skills for each without looking. Table 1.3, then the three short sections that follow it.

- I can explain why the chapter files social and communication skills under business rather than treating them as extras. The Business knowledge and skills block in Table 1.3, and the closing lines of the systems competency section.
- I can explain what “just enough” technical knowledge means, and why the chapter calls the technical area the hardest to maintain. The Technical competency section, down to the hot skills for the next decade that sit in the business domain.
- I can describe what ransomware does to data and name the two defenses the chapter recommends. The Security Matters box: encryption in place, ransom for keys, then awareness plus backup and disaster recovery.
- I can explain what a non-IS major at a small company risks by not understanding information systems. The You — the user section, on consultants and vendors selling their technology of the week.

SECTION 1–2C

Information systems inside an organization

Information systems do not exist in a vacuum; they are built and used inside a context, and that context is almost always an organization. It is the last piece of the definition.

Organizations put information systems in place for four blunt reasons, and not one of them is "because the technology is new."

To become more productive and profitable — the system must pay for itself in work or money.

To gain competitive advantage — it does something rivals cannot, or does it faster.

To reach more customers — a website reaches anyone with a browser.

To improve customer service — answering a question faster is its own reason.

None of those motives is unique to business: the same four hold for professional, social, religious, educational, and governmental organizations, and for industries from medical to legal to manufacturing.

A government example, not a Silicon Valley one. The U.S. Internal Revenue Service launched its own website for exactly those reasons. Roughly **220,000 users** visited in the first 24 hours and **more than one million** in the first week — before the address had even been officially announced. Facebook.com and WSJ.com now take millions of visitors a day.

One company, three questions

People at different heights in a company need different kinds of information.

A cashier — she needs the price of the item in her hand right now, because a customer is waiting.

Her store manager — he needs to know which items ran out last week across the store, so next week's order is right.

A senior leader — she needs to know whether a store in a new state will pay for itself.

Same data, three questions — and the catalogue below sorts largely by which question a system answers.

INTERACTIVE DIAGRAM **Which class of system serves which kind of work**

Pick a layer of work to see what happens there, what the people there need to know, and which class of system serves them.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Operational level — The people who run the day, one event at a time

An event happens (A customer checks out; a student registers) → A TPS records it (Scanner, register, registration screen) → The data are stored (Every layer above lives on this)

- A **transaction processing system** processes day-to-day business event data at the operational level of the organization — the grocery checkout register connected to a network is the classic case.
- The person using it is not making a strategic decision; they are recording something that just happened, accurately and fast.
- The by-product is the real prize: a TPS generates a tremendous amount of data the firm can use to learn about customers and changing product trends.
- Skip this layer and nothing above it can work, because there is no record of what actually occurred.

Managerial level — Managers running a department or a store week to week

TPS data pile up (Thousands of recorded events) → An MIS organizes them (Detailed reports on the unit) → A manager acts (Reorder, restaff, reschedule)

- A **management information system** produces detailed information to help manage a firm or part of a firm — an inventory management and planning system, or student enrollment management.
- TPS data are sorted and organized to support managerial decision making, and the MIS is the most common system doing that work.
- The question changes from "what happened just now?" to "what has been happening across my whole area, and what should I change?"
- Alongside it sits the **functional area information system**, which supports the activities within one specific functional area of the firm — a planning system for personnel training and work assignments, for instance.

Analysis level — The people modeling a choice before the company commits to it

Decision support system (Models a specific choice) → Business intelligence system (Analyzes Big Data) → Intelligent system (Emulates human capability)

- A **decision support system** provides analysis tools and access to databases to support quantitative decision making — product demand forecasting, or loan and investment analysis.
- A **business intelligence system** analyzes Big Data to better understand various aspects of a business, using online analytical processing (OLAP) and data visualization.
- An **intelligent system** emulates or enhances human capabilities: analyzing bank loan applications, self-driving cars, Siri, Alexa, ChatGPT, Gemini.
- Modern systems are built to present integrated data to busy decision makers along with the tools to manipulate and analyze those data.

Systems that span every level — Enterprise systems that refuse to sit on one floor

ERP (Integrates all facets of the business) → CRM (Firm to customer) → SCM (Suppliers to production to delivery) → E-commerce (Selling from the firm's website)

- **Enterprise resource planning** supports and integrates all facets of the business, including planning, manufacturing, sales, and marketing, so it touches the clerk and the chief executive on the same day.
- **Customer relationship management** supports interaction between the firm and its customers, through things like sales force automation and lead generation.
- **Supply chain management** supports the coordination of suppliers, product or service production, and distribution — procurement planning is the everyday example.
- These three cannot be filed under one category, which is exactly the point: many organizations replaced stand-alone systems with enterprise systems that span the whole organization.

The catalogue: major categories of information systems

Read the middle column first: the category names are only labels for the job described there.

CATEGORY OF SYSTEM	PURPOSE	SAMPLE APPLICATION
Transaction processing system (TPS)	Processes day-to-day business event data at the operational level of the organization	Grocery checkout register on a network; student registration
Management information system (MIS)	Produces detailed information to help manage a firm or part of a firm	Inventory management and planning; student enrollment management

CATEGORY OF SYSTEM	PURPOSE	SAMPLE APPLICATION
Decision support system (DSS)	Provides analysis tools and access to databases to support quantitative decision making	Product demand forecasting; loan and investment analysis
Intelligent system	Emulates or enhances human capabilities	Analyzing bank loan applications; self-driving cars; Siri, Alexa, ChatGPT, Gemini
Business intelligence system	Analyzes Big Data to better understand various aspects of a business	Online analytical processing (OLAP); data visualization
Office automation system (personal productivity software)	Supports a wide range of predefined day-to-day work activities of individuals and small groups	Word processor, spreadsheet, presentation software, email client
Collaboration system	Enables people to communicate, collaborate, and coordinate with each other	Email system with an automated, shared calendar
Knowledge management system	Enables the generation, storage, sharing, and management of knowledge assets	Knowledge portal for common questions
Social software	Facilitates collaboration and knowledge sharing	A social network connecting colleagues and friends
Geographic information system (GIS)	Creates, stores, analyzes, and manages geographically referenced data	Route planning system
Functional area information system	Supports the activities within a specific functional area of the firm	Planning system for personnel training and work assignments

CATEGORY OF SYSTEM	PURPOSE	SAMPLE APPLICATION
Customer relationship management (CRM)	Supports interaction between the firm and its customers	Sales force automation; lead generation
Enterprise resource planning (ERP)	Supports and integrates all facets of the business, including planning, manufacturing, sales, and marketing	Financial, operations, and human resource management
Supply chain management (SCM)	Supports the coordination of suppliers, product or service production, and distribution	Procurement planning
Electronic commerce system	Enables customers to buy goods and services from a firm's website	Amazon, eBay, Nordstrom.com
Mobile app	Performs a well-defined function, typically on a mobile device	Instagram, Snapchat, WhatsApp, Office Mobile, Google Pay, Lyft

The transaction processing system is the foundation. Besides processing customer transactions efficiently, it generates a tremendous amount of data the firm can learn from, and two everyday examples show it.

- Your grocery store scans bar codes at the register, then prints discount coupons on the back of the receipt for products related to what you just bought.
- Amazon processes thousands of transactions an hour from around the world, feeding large data warehouses that are analyzed to produce purchase recommendations for future customers.

TPS data are then sorted and organized to support managerial decision making — most often through a management information system — and they also feed decision support, intelligent, business intelligence, and knowledge management systems, social software, geographic information systems, and functional area information systems.

MATCH **System type to the job it does**

Pair each category of information system with the purpose it serves inside an organization.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Transaction processing system

Processes day-to-day business event data at the operational level of the organization The checkout register that scans bar codes is the standard example, and the data it captures feed nearly every other system.

Decision support system

Provides analysis tools and access to databases to support quantitative decision making Product demand forecasting and loan or investment analysis are the sample applications: numbers in, modeled answer out.

Knowledge management system

Enables the generation, storage, sharing, and management of knowledge assets The everyday form is a knowledge portal where people find answers to common questions instead of asking a colleague again.

Geographic information system

Creates, stores, analyzes, and manages geographically referenced data Data with a location attached behaves differently from ordinary data, which is why route planning gets its own category.

Customer relationship management

Supports interaction between the firm and its customers Sales force automation and lead generation live here; the direction of travel is outward, from the firm to the buyer.

Supply chain management

Supports the coordination of suppliers, product or service production, and distribution This one points the other way, upstream toward the people who supply the firm, with procurement planning as the sample application.

Office automation system

Supports a wide range of predefined day-to-day work activities of individuals and small groups Word processing, spreadsheets, presentation software, and an email client are personal productivity tools, not enterprise systems.

Electronic commerce system

Enables customers to buy goods and services from a firm's website Amazon, eBay, and Nordstrom.com are the named examples; the system exists to complete a purchase, not to analyze one.

CHECK YOURSELF

Naming the right class of system

Four options, one best answer; read every explanation, including the ones for answers you did not choose.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. A grocery store's checkout lane scans bar codes as items are rung up, and the register is connected to the store network. Which category of information system is that register?

- **A.** A management information system, because the store uses the register data to manage the store — A management information system produces detailed information to help manage a firm or part of a firm, such as an inventory management and planning system. It consumes the register's data; it is not the register. Change the question to 'which system tells the manager what to reorder?' and this becomes correct.
- **B.** A transaction processing system, because it processes day-to-day business event data at the operational level **— correct.** The register captures a business event as it happens at the operational level, which is the definition of a transaction processing system. The coupons printed on the back of the receipt are a bonus built on that same captured data.
- **C.** A business intelligence system, because the scanned data are later analyzed for product trends — A business intelligence system analyzes Big Data to better understand aspects of the business, using tools like OLAP and data visualization. That analysis happens downstream, on data the TPS generated. This would be right if the question described the analytics team studying years of scanner data.
- **D.** An office automation system, because it is a predefined day-to-day work activity performed by one employee — An office automation system is personal productivity software — word processor, spreadsheet, presentation software, email client — supporting an individual's or small group's own work activities. A cashier ringing up a customer is processing a company transaction, not doing personal desk work.

2. A classmate says: "Categories like TPS and DSS are obsolete. Nobody uses those labels anymore." What is the accurate correction?

- **A.** They are right: organizations replaced categorized systems with enterprise systems, so the labels no longer describe anything real — Enterprise systems that span the organization did replace many stand-alone systems, which is why it is hard to say a system fits only one category. But the categories did not stop describing the work; they stopped being mutually exclusive. This would be right only if the categories had been abandoned, and they have not.
- **B.** The labels are exact if you go by a product's main feature, so a modern system still belongs in a single row of the table — This is the opposite error. Ten to 15 years ago it was typical to see systems that fell cleanly into one category, but internetworking and systems integration ended that: customer relationship management, supply chain management, and enterprise resource planning carry so many features and data types that no single row holds them. Naming a main feature hides the other jobs the system is doing.

- **C.** The labels still name the goals, features, and functions a system delivers, even though one modern system usually spans several of them **— correct.** Understanding the categories lets you see the myriad approaches, goals, features, and functions of modern systems, even though modern systems tend to span several categories at once.
- **D.** The labels described systems a company housed itself; cloud systems are described instead by the vendor and the subscription plan — Where a system runs and what job it does are separate questions. Many systems are no longer housed within organizations but sit in the cloud and are reached through a browser when needed, and they still process transactions, support decisions, or manage customer relationships. Hosting changes who maintains the servers, not the category of work.

3. An analytics team wants to study several years of purchase history to understand emerging product trends. Where do those data come from, and which class of system does the analysis?

- **A.** Managers type the monthly figures in by hand, and a management information system does the analysis — Managers do read MIS output, but they are not the data source. Transaction processing systems generate the raw event data automatically as business happens, which is why the volume is large enough to be worth analyzing at all.
- **B.** The data are generated by transaction processing systems and analyzed by a business intelligence system **— correct.** A TPS generates a tremendous amount of data as it processes transactions — Amazon feeds thousands of transactions per hour into large data warehouses — and a business intelligence system analyzes that Big Data to better understand the business.
- **C.** The data come from the company's office automation software, and a collaboration system does the analysis — Office automation software is word processing, spreadsheets, presentations, and email for individuals and small groups, and a collaboration system provides email, shared calendars, and threaded discussions. Neither one is where years of company purchase data live.
- **D.** The data come from the company's e-commerce website, and a customer relationship management system does the analysis — An e-commerce system is a real source of transaction data, but a firm with physical stores captures far more at the register, and CRM supports interaction with customers rather than large-scale trend analysis. This would be closer for a purely online retailer studying its own customer relationships.

Why the boxes leak

Ten to 15 years ago it was typical to see a system that fell cleanly into one of these categories. Three developments ended that.

> **Enterprise systems** — many organizations replaced stand-alone systems with ones that span the entire organization.

Internetworking — connecting host computers and their networks into larger networks, the internet being the giant example, let systems reach each other.

Systems integration — connecting separate, often modular systems and their data using technologies such as APIs, to improve business processes and decision making, stitched the pieces into one flow.

Many systems no longer sit in the building at all; they live in the cloud, reached through a browser when needed.

So a modern system usually spans several categories at once: collecting data from across the firm and from customers, integrating it from diverse sources, and presenting it to busy decision makers with tools to analyze it.

Customer relationship management, supply chain management, and enterprise resource planning are the clearest cases — each carries so many features and data types that it refuses to sit in one row. The categories still matter, because they name the goals, features, and functions a system delivers.

How to use the catalogue. Do not ask "what category is this product?" Ask "what job needs doing?" Recording an event, summarizing events for a manager, and modeling a decision are three different jobs. Name the job; the category follows.

DECIDE

Choosing the right system for the problem in front of you

You are the new operations manager at a 16-store regional grocery chain; pick the class of system that actually solves each problem.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Leadership has money to spend and a list of complaints. Your job is not to name a vendor — it is to name the kind of system each complaint calls for.

Decision 1. Checkout is slow and every store rings sales on a standalone register that never talks to the network, so nobody outside the store knows what sold today.

- Install networked checkout registers that scan bar codes and record each sale — a transaction processing system **— the stronger call.** Right. This processes day-to-day business event data at the operational level, and it fixes the deeper problem: the chain finally has a record of what actually happened, which every other system will need.

- Buy a business intelligence system so leadership can analyze store performance — A business intelligence system analyzes Big Data to understand the business — but there is no Big Data yet. Nothing is capturing the transactions. You would be buying a telescope before there is anything in the sky.
- Give each store manager spreadsheet and email software to track daily sales — That is office automation: personal productivity software supporting one person's day-to-day work. It would produce 22 hand-typed, inconsistent files instead of one automatic, reliable stream of transaction data.

Decision 2. Transactions are now captured, but the regional manager still cannot tell which stores are about to run out of which products until a shelf is already empty.

- Add an inventory management and planning system that turns the transaction data into detailed stock reports — a management information system **— the stronger call.** Right. A management information system produces detailed information to help manage a firm or part of a firm, and inventory management and planning is its textbook application. The raw events become something a manager can act on.
- Launch an e-commerce site so customers can order out-of-stock items online — An electronic commerce system enables customers to buy goods and services from the firm's website. Useful someday, but it does not tell your manager what is running low — it just moves the empty shelf online.
- Roll out a collaboration system so store managers can email each other about shortages — A collaboration system enables people to communicate, collaborate, and coordinate — email with a shared calendar. It helps them talk about the problem; it does not produce the detailed stock information they would be talking about.

Decision 3. Finance must decide whether to borrow money to build a 23rd store, and wants to model demand and repayment under three different economic scenarios before committing.

- Give finance analysis tools and database access so they can model the forecast — a decision support system **— the stronger call.** Right. A decision support system provides analysis tools and access to databases to support quantitative decision making; product demand forecasting and loan and investment analysis are exactly its sample applications.
- Pull the answer out of the transaction processing system, since it holds all the sales data — The TPS holds the raw record of what already happened, and it is the input you need — but it processes events, it does not model futures. Data alone is not a forecast.
- Search the company knowledge portal for what the firm did last time it expanded — A knowledge management system enables the generation, storage, sharing, and management of knowledge assets, so a portal of past answers is genuinely useful context. It just cannot run three quantitative scenarios against current demand data.

Decision 4. Suppliers keep shipping late, two trucks arrive at the same dock at once, and produce spoils while a store two towns over has none.

- Put in a system that coordinates suppliers, production, and distribution, starting with procurement planning — supply chain management **— the stronger call.** Right. Supply chain management supports the coordination of suppliers, product or service production, and distribution. The problem is upstream of the customer, so the system has to be upstream too.
- Deploy customer relationship management so shoppers get told when produce is unavailable — CRM supports interaction between the firm and its customers — sales force automation, lead generation. It points downstream, toward the buyer. It would communicate the failure rather than prevent it.
- Deploy a geographic information system to plan better delivery routes — A geographic information system creates, stores, analyzes, and manages geographically referenced data, and route planning is its sample application, so this genuinely helps a truck get somewhere efficiently. It still does not coordinate suppliers, procurement, and distribution as a whole.

Organizing the IS function

The people who run technology inside a company are the **IS function**, and their old reputation was earned. Old-school IS personnel held three beliefs about their own job.

- They owned and controlled the computing resources, so any change was theirs to grant or refuse.
- They knew better than users what those users needed.
- Their job was to tell users what they could and could not do.

Early IS departments carried huge backlogs and delivered systems that were over budget, late, hard to use, and unreliable.

Technology became too pervasive for that to survive. Fast-paced competition forced firms to treat IS as an enabler that streamlines business processes, improves customer service, and connects stakeholders inside and outside the company. Many organizations also realized that some of the best ideas for solving business problems come from the employees using the system.

So IS units moved into a **consulting relationship** with their users, built on four habits.

> **Reach out first** — they seek user input instead of waiting for complaints.

> **Change fast** — they modify systems at a moment’s notice to meet a need.

> **Welcome ideas** — they celebrate new ideas rather than explaining why they will not work.

> **Hand over ownership** — they treat the technology and the information as belonging to the customer, and they build **help desks, hotlines, information centers, and training centers** to support them.

A firm is unproductive when IS staff and everyone else are at odds and remarkably productive when they work hand in hand: technology is potentially the great lever, but it works best when people use it together rather than against each other.

The pervasiveness of technology

Technology is now entrenched within the business units themselves — accounting, sales, marketing — so the seam between the technology and the business is hard to find.

In many organizations, especially those using agile approaches such as **scrum**, the builders and managers of a system spend most of their time out in the business unit with its users, often permanently placed there with an office, desk, phone, and PC. Systems staff commonly have education, training, and experience in information systems **and** in the functional area the system supports, such as finance.

Because systems are used so broadly, IS personnel often have **dual-reporting relationships**, reporting both to the central IS group and to the business function they serve. Firms want the benefits of decentralizing the IS function, but they are not willing — and not able — to forgo the benefits of centralizing it, so they try to hold both at once and keep a coordinating centre alongside the local staff.

> **What decentralization buys** — flexibility, adaptability, and systems responsiveness, because builders sit beside users.

> **What centralization buys** — coordination, economies of scale, compatibility, and connectivity firm-wide.

> **Why central planning survives** — some centralized planning still has to exist, to achieve economies of scale in acquiring and developing systems and to keep systems integration and enterprise networking coherent.

The clock is speeding up too: IS departments once thought in five-year time frames, but new devices now arrive every 6–18 months, so firms need people who understand the technology side and the business side at once. That is why how information systems are managed matters no matter which career you choose.

When things go wrong: technology addiction

Beginning in **2018**, online gaming was designated by health experts as a real, diagnosable addiction. The numbers are blunt.

- Average adults consume more than **11 hours** of media a day.
- Fifty percent of 18- to 24-year-olds check their phone within five minutes of waking up.
- A third of 25- to 34-year-olds open social media sites or apps more than ten times a day.
- A majority of adolescents already claim to be addicted to technology, and too much screen time can affect memory and lead to a decline in academic performance.
- The average human attention span has fallen from **12 seconds** at the turn of this century to **8 seconds** — less than a goldfish.
- University of Pennsylvania researchers have shown that social media use decreases overall health and well-being.

The mechanism is chemical: **dopamine**, the brain chemical associated with pleasure, is released when we are stimulated, and the hit from a like or a status update can have the same addicting effect as drugs like cocaine or heroin. Being plugged in constantly also leaves the brain hyper-aroused, so we walk around in a constant state of distraction.

There is a bright side: younger generations, immersed in this technology their whole lives, may be better able to adjust and adapt. The suggested fixes are unglamorous.

- Start by turning the gadgets off for a while each day.
- Improve your overall health, which helps as well: eat right, stay hydrated, and work out regularly.
- Get enough sleep, and do not fall asleep to Netflix or Instagram.

CHECK YOURSELF

The IS function, and what it costs us

Three questions on how technology staff are organized inside a firm and on the human side of being always connected.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Which description matches the service-oriented mindset that replaced the old-school IS department?

- **A.** IS personnel own and control the computing resources and tell users what they may and may not do with them — This is the old-school mindset precisely: IS believed it owned the computing resources and knew better than users. It came bundled with huge backlogs and systems that were over budget, late, and hard to use. It is what the service orientation replaced.
- **B.** IS personnel act as consultants, seek user input before complaints arrive, and run help desks, hotlines, and training centers **— correct.** Personnel in many IS units took on a consulting relationship with users, reaching out proactively, modifying systems quickly, and building help desks, hotlines, information centers, and training centers to support them.
- **C.** IS personnel step back entirely, so each business unit buys and runs whatever technology it wants with no central coordination — Serving users is not the same as abandoning coordination. Even organizations that decentralize technology decisions still need centralized or centrally coordinated staff for economies of scale, systems integration, and enterprise networking.
- **D.** IS personnel wait for users to bring in systems complaints, then work through that queue in the order the complaints arrived — This is the reactive posture the service orientation replaced. Service-oriented units reach out to their internal customers and proactively seek their input and needs rather than waiting for customers to come in with complaints, and they modify systems at a moment's notice to meet a need. Working a complaint queue in arrival order is how the old project backlogs grew.

2. Firms increasingly place systems people inside business units with a desk, a phone, and a dual-reporting relationship. Why do they still keep centralized IS staff as well?

- **A.** Because business units are not permitted to make technology decisions of their own — Business units make plenty of technology decisions — that is the whole point of decentralizing and of embedding systems staff alongside users. The tension is about coordination, not permission.
- **B.** Because a central group costs less than embedded staff, so firms are gradually ending the embedded arrangement — Cost is not the driver described, and the embedded arrangement is expanding rather than ending: systems personnel are often permanently placed in the business unit, frequently with training in both IS and the functional area they support.
- **C.** Because centralization delivers the flexibility, adaptability, and responsiveness that decentralization cannot — This swaps the two lists. Flexibility, adaptability, and systems responsiveness are the benefits of decentralization. Centralization buys something different, which is why firms want both at once.
- **D.** Because centralization preserves coordination, economies of scale, compatibility, and connectivity across the whole firm **— correct.** Organizations want decentralization's flexibility, adaptability, and responsiveness, but they will not forgo centralization's coordination, economies of scale, compatibility, and connectivity — so they keep a centrally coordinated IS staff.

3. Which statement matches the evidence in the technology-addiction discussion above?

- **A.** Average adults consume more than 11 hours of media a day, and the average human attention span has fallen from 12 seconds to 8 **— correct.** Both figures are reported: more than 11 hours of media consumption a day for the average adult, and an attention span that dropped from 12 seconds at the turn of the century to 8 — less than a goldfish.
- **B.** Half of 18- to 24-year-olds open social media more than ten times a day, and a third of 25- to 34-year-olds check their phone on waking — The two age groups and the two behaviors are swapped. Fifty percent of 18- to 24-year-olds checked their phone within five minutes of waking up, and a third of 25- to 34-year-olds visited social media sites or used mobile apps more than ten times a day.
- **C.** The dopamine released by a like or a status update works through a different mechanism than addictive drugs do — The argument is the opposite: dopamine is released when we are stimulated, and the hit from a like, a status update, or a piece of gossip can have the same addicting effect on the body as drugs like cocaine or heroin.
- **D.** Younger generations are the group least able to adjust, precisely because they have been immersed in technology their whole lives — This inverts the one hopeful note. Because younger generations have been immersed in the technology their entire lives, they may be better able to adjust and adapt, developing a feel for their limits — even though a majority of adolescents already claim to be addicted.

SECTION 1–3

The dual nature of information systems

Information systems have become so important, and so expensive, that the chapter reaches for a blunt image: information technology is **like a sword**. You can use it effectively as a competitive weapon — but, as the old saying goes, those who live by the sword sometimes die by the sword. That is the **dual nature of information systems**, taught here through two cases: one system that went wrong, one that works.

A **competitive weapon** is anything that makes a customer pick you over the business across the street: a live delivery map, say, when the pizza place down the road still makes you phone and ask. It cuts both ways, because on the night the map fails, the customers who chose you for it feel most let down.

Case in Point: an information system gone awry — Zoom outages disrupt (almost) everyone

The COVID-19 pandemic of early 2020 left people facing stay-at-home orders, travel bans, and other mobility restrictions, so they turned to information systems for everything from working from home to the most mundane tasks. Activities that had never been online moved online at once:

- **Delivery apps surged**, because errands that once meant going somewhere now had to be ordered.
- **Companies scrambled** to move documents and workflows to the cloud, since the office was out of reach.
- **Entire social lives moved onto Zoom**, from exercise classes to church service.

Videoconferencing was not new, because many companies had long used it to replace face-to-face meetings. What was new is how far it reached: Zoom's daily meeting participants surged to over **300 million**.

Such a tremendous unforeseen increase in users can quickly cause problems, and on **Sunday morning, May 17, 2020** (U.S. time), it did: users suddenly started experiencing issues accessing the service. Comparatively few businesses and professionals were impacted, because it was a Sunday.

The largest impact was for **churches**, which had moved their services online; with many worshippers unable to connect, churches across the United States were forced to end their services early. Zoom restored the systems by the afternoon, and the lesson stands: relying on a **single provider** can quickly disrupt many activities in unforeseen ways.

> **Notice what the chapter does not blame.** No hacker, no sabotage, no blunder — the trouble came from success. The growth that made Zoom indispensable made its bad Sunday everybody's bad Sunday.

SEQUENCE **How the Zoom outage actually unfolded**

Drag the six moments into the order the chapter describes, from the pressure that built up to the lesson left behind.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

The chapter's failure case does not begin with a broken server. It begins with the world changing outside the company.

1. **Early 2020: stay-at-home orders, travel bans, and other mobility restrictions push people to use information systems for everything from working from home to the most mundane tasks.** — This is the trigger outside the company. Nothing about Zoom's software changed first - demand did.
2. **Activities that had never been online move online: delivery apps surge, companies move documents and workflows to the cloud, and entire social lives, from exercise classes to church service, shift onto Zoom.** — Videoconferencing was old news for business meetings. What was new was ordinary life depending on it, which is what set up the damage.
3. **Zoom's daily meeting participants surge to over 300 million, a tremendous and unforeseen increase.** — The chapter names the surge itself as the hazard: such a tremendous unforeseen increase in users can quickly cause problems.
4. **On Sunday morning, May 17, 2020, U.S. time, users suddenly start experiencing issues accessing the service.** — The failure lands after the growth, not before it, and the day of the week decides who it hurts.
5. **Comparatively few businesses and professionals are impacted; the largest impact falls on churches, whose worshippers cannot connect, so services across the United States are ended early.** — It was a Sunday morning, so the users with something scheduled at that exact hour and no fallback absorbed the loss.
6. **By the afternoon Zoom restores the systems to working order, leaving the lesson that relying on a single provider can quickly disrupt many activities in unforeseen ways.** — The recovery was fast, and the lesson still stands: what makes this a failure case is the dependence it exposed, not the length of the outage.

Case in Point: an information system that works — FedEx

FedEx, an US$87.7 billion family of companies (2024 data), is the world’s largest express transportation company, delivering millions of packages and millions of pounds of freight to over 220 countries and territories each business day, and it continuously updates and fine-tunes its extensive, interconnected information systems to improve its services and sustain a competitive advantage.

WHAT THE SYSTEMS COORDINATE	THE CHAPTER'S FIGURE
People and fleet	More than 500,000 employees, almost 700 aircraft, more than 200,000 ground vehicles
FedEx.com traffic	More than 80 million unique visitors a month; more than 500 million tracking requests per day
Time to divert a package after it passes an overhead scanner	Between 1 and 2 seconds; decisions made in a few hundred milliseconds
Performance reengineered and improved	On average, twice a year
Packages delivered within one business day	A quarter of all daily packages handled

Accuracy, not the web page, is the product: FedEx strives to provide the most accurate tracking information to each visitor.

The divert window is the binding constraint, because each package typically travels through at least one **sorting facility** where conveyor belts scan and reroute it, and no human decides a route in a few hundred milliseconds. That is why the chapter calls automation in the ground hubs another **enabler of competitive advantage** that helped position FedEx as the global leader in express transportation.

What the two cases share, and what separates them

Both are typical of systems that are pervasive in today's life or used in large, complex organizations, so they share three traits:

> **So large in scale and scope that they are difficult to build** — which is why the chapter insists on getting such development right the first time around.

> **Critical to the success of the organizations that built them** — they were and continue to be, so trouble inside the system is trouble in the business.

> **Strategic in their intent** — both were developed, and are continuously updated, to gain or sustain competitive advantage over rivals.

What separates them is the direction the blade turned. At Zoom, the dependence that made the service valuable meant everything resting on it stopped at once. At FedEx, the systems are reengineered twice a year, so the edge is renewed rather than assumed.

EXPLORE

One system that failed, two that worked

Open each card and read the failed system against the two successful ones facet by facet, rather than case by case.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Zoom videoconferencing · The system gone awry

- **What the system is:** The videoconferencing software that, during the 2020 pandemic, absorbed not just business meetings but entire social lives, from exercise classes to church service.
- **What actually happened:** Daily meeting participants surged to over 300 million, and on Sunday morning, May 17, 2020, users suddenly could not access the service. Churches across the United States ended their services early. Zoom restored the systems by that afternoon.
- **Why it went that way:** The trouble came from a tremendous, unforeseen increase in users rather than from an obvious blunder. Everything that had moved onto one platform went down with that platform.
- **The lesson the chapter draws:** Relying on a single provider can quickly disrupt many activities in unforeseen ways - and the activities with no fallback at that exact hour take the hardest hit.

FedEx tracking · The system that works, part one

- **What the system is:** The customer-facing side of FedEx's extensive, interconnected systems: FedEx.com, which draws more than 80 million unique visitors per month and answers more than 500 million tracking requests per day.
- **What actually happened:** FedEx strives to provide the most accurate tracking information to each visitor, and continuously updates and fine-tunes the systems behind it to improve its services and sustain a competitive advantage.
- **Why it went that way:** Accuracy at that volume is not a by-product of shipping packages; it is something the company deliberately keeps investing in, which is what makes the choice strategic in intent.
- **The lesson the chapter draws:** A system can be the reason customers choose you, so it is developed to gain or sustain competitive advantage rather than merely to cut cost.

FedEx ground-hub automation · The system that works, part two

- **What the system is:** The sorting facilities each package typically travels through, where an extensive network of conveyor belts scans a package multiple times and reroutes it toward its intermediate and final destinations.
- **What actually happened:** Once a package passes an overhead scanner there is between 1 and 2 seconds to divert it, so decisions must be made in a few hundred milliseconds. On average FedEx reengineers and improves the performance twice a year and now manages to deliver a quarter of all daily packages handled within one business day.

- **Why it went that way:** No human can decide a route in a few hundred milliseconds, so the automation is not a convenience laid over manual sorting - it is the only way the service exists at all.
- **The lesson the chapter draws:** Automation in the ground hubs is another enabler of competitive advantage, and an advantage reengineered twice a year is one being renewed faster than rivals can copy it.

What “strategic” means in this chapter

In everyday speech, calling something strategic just means calling it important. This chapter is narrower: a choice is **strategic** when it is made with the intent of gaining or sustaining an advantage over rivals, not merely to make an existing task cheaper. Switching to a cheaper email provider saves money but changes nothing about who picks you; tracking accurate enough that a hospital hands you its deliveries does.

The advantage itself has a name the chapter borrows from Michael Porter: **competitive advantage** over rivals (Porter, 1985; Porter & Millar, 1985). Do not let this notion slip by, because it asks three things of technology:

> **Efficiency** — the use of technology can enable the same work with less time and expense.

> **Return on investment** — information systems must provide one, so a system that never pays back what it cost fails the test.

> **Competitive advantage** — technology use can **also** be strategic and a powerful enabler of it.

Nor is that a big-company privilege: whether a small mom-and-pop boutique or a large government agency, every organization can find a way to use information technology to beat its rivals.

CHECK YOURSELF **The sword, the two cases, and the word strategic**

Pick the best answer, then read every explanation, because the wrong options are all positions students actually take.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. The chapter says information technology is **like a sword**. What is that image meant to convey?

- **A.** That the money sunk into information systems is the main danger, so the risk to manage is overspending on technology that underdelivers — Cost is real here - the chapter opens by noting how important and expensive information systems have become, and it insists that information systems must provide a return on investment. But that is one edge of the blade. The image is about the two-sided outcome: the same system that wins customers can be the one that takes an organization down.
- **B.** That information systems cut through the layers of an organization, letting any employee reach any other directly — This describes a genuine effect of information systems on communication, and it is a real thing that happens in organizations - but it is not the sword metaphor. To be right, the image would have to be about flattening hierarchy, and instead it is about a weapon that can wound the one holding it.
- **C.** That the same technology can be used effectively as a competitive weapon and can also be the thing that brings an organization down **— correct.** The chapter's framing is that you can use information technology effectively as a competitive weapon, but those who live by the sword sometimes die by the sword. Zoom and FedEx are the two edges of that same blade.
- **D.** That a system has to be kept sharp with constant updates, or it gradually stops doing its job — This is half true and it rests on a real chapter fact: FedEx reengineers and improves performance on average twice a year, and advantage from IS can be fleeting because competitors eventually do the same thing. But that is the maintenance point, not the metaphor. The sword is about which direction the blade turns, not about sharpening.

2. According to the chapter, who was hit hardest by the May 17, 2020, Zoom outage, and why?

- **A.** Churches, because their services had moved online and the outage came on a Sunday morning, leaving many worshippers unable to connect **— correct.** The chapter reports that the largest impact was for churches, who had moved their services online; with many worshippers unable to connect, churches across the United States were forced to pull the plug and end their Sunday services early.
- **B.** Businesses and professionals, because the outage struck in the middle of the workweek when most meetings were scheduled — This is the intuitive guess, and it is exactly backwards: the chapter says comparatively few businesses and professionals were impacted. The outage was on a Sunday morning, U.S. time, so the workweek was not running. This would be the right answer to an outage at 10 a.m. on a Tuesday.
- **C.** Schools, because students in remote classes could not attend their lessons for most of the day — Remote schooling is a real part of the pandemic shift the chapter describes, so this is a reasonable thing to expect - but the case in point does not name schools among those affected. It would be right only if the chapter had reported it, and reporting what the chapter actually says is the point of a case in point.
- **D.** Zoom itself, because the outage reversed the surge that had carried it past 300 million daily meeting participants — The 300 million figure is real, but it is the surge in daily meeting participants that preceded the outage, not a loss. Zoom had restored the systems by the afternoon, and the harm the chapter describes falls on the people depending on the service. This would be right if the chapter had reported a customer exodus, and it does not.

3. What do the Zoom and FedEx cases have in common in the chapter's telling?

- **A.** Both systems eventually failed under load that their builders had not anticipated — Failure is what separates the two cases, not what unites them: the FedEx case reports no outage at all. This would be right if the section were about two failures, but the chapter deliberately pairs one system gone wrong with one gone right to show the dual nature.
- **B.** Both companies keep their systems in-house rather than depending on outside providers, which is what made the systems dependable — The chapter never says who built which piece, and its competitive-advantage argument runs the other way: firms from Amazon to Zoom win by combining commoditized technologies with proprietary systems and business processes. Ownership of the parts is simply not the link the chapter draws between the two cases.
- **C.** Both were built to reduce operating costs, and any effect on competition was a welcome side benefit — This is the cost-only view of IT that the chapter reports as an argument and then rejects. It would be right if the chapter had said the systems were built for cost reduction and risk mitigation, but it says the opposite: the choices were strategic in their intent, aimed at competitive advantage.
- **D.** Both are large, hard-to-build systems, critical to their organizations, and both were developed with strategic intent **— correct.** The chapter says these systems are so large in scale and scope that they are difficult to build, that they were and continue to be critical to the success of the organizations that built them, and that the choices made in developing them were strategic in their intent.

"But hasn't IS become a commodity?"

In everyday use a **commodity** is something so standardized that one supplier's version is interchangeable with another's, which is why nobody wins customers by selling better electricity. At the start of the millennium, as information systems became standardized and ubiquitous, some argued three things:

- Information systems are now **more of a commodity**, absolutely necessary for every company.
- Firms should focus IT strictly on **cost reduction and risk mitigation**, not on winning customers with it.
- Investing in it **for differentiation or competitive advantage is futile**, because whatever you buy is on sale to your rivals too.

The chapter answers rather than dismisses that argument. As evidenced by advances in smartphones, the emergence of social networks, and changes in creative industries, IT is changing rapidly, and many companies have gained competitive advantage by innovatively using increasing digital density:

> Companies from Amazon to Zoom — they created advantages by combining certain **commoditized technologies with proprietary systems and business processes**, so the standard parts are shared and the combination is not.

Google and Facebook — customer-generated data create value, and how those data are gathered, processed, and used can be a source of **sustained** competitive advantage.

Amazon again — it sells cloud computing services to other businesses, generating revenue directly from its IT investments.

There is a hard limit on all of this: **companies with bad business models tend to fail regardless of whether they use information technology**, while companies with good business models that use IT successfully to carry them out tend to be very successful. A **business model** is a summary of a business's strategic directions that outlines how the objectives will be achieved.

Technology carries out a business model; it does not supply one.

The advantage does not stay bought. Competitive advantage from the use of information systems **can be fleeting, as competitors can eventually do the same thing**. That is why FedEx reengineers performance twice a year: an edge never renewed is an edge being copied.

CHECK YOURSELF **Commodity, advantage, and business models**

Choose the answer that matches the chapter's position, then read why each of the other three is a position the chapter does not take.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Some argue that because information systems are standardized and ubiquitous, they are now a **commodity**, and that firms should therefore spend on IT strictly for cost reduction and risk mitigation. What is the chapter's answer?

- **A.** It agrees, which is why it warns that advantage from information systems can be fleeting — This runs two separate statements together. The chapter does say advantage from information systems can be fleeting because competitors eventually do the same thing, but fleeting means the advantage must be renewed, not that pursuing it is pointless. Its stated conclusion is that IS can be an enabler of competitive advantage.
- **B.** It holds for small firms, but companies the size of FedEx and Amazon can buy an advantage outright — Size is exactly what the chapter denies matters here: whether a small mom-and-pop boutique or a large government agency, every organization can find a way to use information technology to beat its rivals. The chapter also describes advantage as built through combination, not purchased ready-made.

- **C.** It is right that bought technology cannot differentiate a firm, so a firm should rebuild the standardized parts itself — This gets the recipe backwards. The winning combination the chapter describes uses commoditized technologies for the standard parts and keeps the proprietary systems and business processes as the differentiator, so rebuilding standard parts spends scarce effort where it cannot set you apart.
- **D.** IT keeps changing rapidly, and firms from Amazon to Zoom win by pairing commoditized technologies with proprietary systems and processes **— correct.** The chapter grants that information systems are absolutely necessary for every company, then rebuts the rest: IT is changing rapidly, digital density is increasing, and companies from Amazon to Zoom created competitive advantages by combining certain commoditized technologies with proprietary systems and business processes.

2. The chapter warns that competitive advantage from the use of information systems can be **fleeting**. What reason does it give?

- **A.** Competitors can eventually do the same thing, so an edge that is not renewed gets matched **— correct.** The chapter's sentence is precise: the competitive advantage from the use of information systems can be fleeting, as competitors can eventually do the same thing. That is why FedEx reengineers and improves performance on average twice a year rather than treating its systems as finished.
- **B.** New devices come out every 6 to 18 months, so a system is outdated before it earns back what it cost — The 6-to-18-month device cycle is a real fact from this chapter, used where it explains that IS departments once planned in five-year time frames and must now adjust to a faster pace. But it explains why organizations have to keep up, not why an advantage fades. Advantage fades because of what rivals do.
- **C.** A technology that becomes standardized leaves the firms using it looking much alike to their customers — This is the commodity argument again, and the chapter rejects it: firms from Amazon to Zoom create advantage by combining commoditized technologies with proprietary systems and business processes, and Google and Facebook turn customer-generated data into sustained advantage. Standard parts do not make firms identical.
- **D.** Large information systems tend to cost more to run over time than the savings they produce — The chapter does say information systems must provide a return on investment, but it never claims large systems usually fail to deliver one - its position is that IS are necessary, can create efficiencies, and can enable advantage. This would be right only if the chapter had made a claim about typical project economics, and it does not.

3. A regional retailer buys the same analytics and logistics technology its most successful rival uses, but its stores stock merchandise its customers do not want at prices they will not pay. What does the chapter predict?

- **A.** It will close the gap, because in the chapter's view the technology is what creates the advantage — This reverses the chapter's causal order. Technology carries out a business model; it does not supply one. The chapter's success condition is a good business model plus information technology used successfully to carry that model out - both, in that order.

- **B.** It will likely fail anyway, because companies with bad business models tend to fail regardless of whether they use information technology **— correct.** The chapter states that companies with bad business models tend to fail regardless of whether they use information technology, while companies with good business models that use IT successfully to carry them out tend to be very successful. Merchandise and pricing are business-model problems, and buying software does not touch them.
- **C.** It will succeed once it layers proprietary systems on top of the technology it bought, the way Amazon and Zoom did — Layering proprietary systems on commoditized technology is genuinely the chapter's route to competitive advantage, and it is what companies from Amazon to Zoom did. But it is the route for a firm whose business model already works. Proprietary software on top of merchandise nobody wants still sells merchandise nobody wants.
- **D.** The chapter takes no position, because it treats the business model and the technology as separate subjects — The chapter links them explicitly: information systems conceived, designed, used, and managed effectively and strategically, together with a sound business model, are what let an organization gain or sustain competitive advantage - and a bad business model is its stated way to fail with good technology.

In sum, information systems are a necessary part of doing business, they can be used to create efficiencies, and they can also be used as an enabler of competitive advantage. Managed effectively and strategically and combined with a sound business model, they let an organization be more effective and more productive, expand its reach, and gain or sustain competitive advantage over rivals.

DECIDE

Rolling out a system that cuts both ways

Work through five decisions in order; after each one, read what the chapter says actually follows from your choice.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

You are on the team at a growing regional courier company that has just won a contract to carry deliveries for a large hospital network. Every choice below is the same sword the chapter describes: the system you are about to build can win the account or lose it.

Decision 1. Leadership asks what the new package-tracking system is actually for, so they know how to judge it later. What do you tell them?

- Its purpose is to do something rivals cannot easily match - hospital-grade, package-level tracking - so the choice is strategic in intent, and it still has to earn a return. **— the stronger call.** This matches the chapter. The choices made in developing the systems at Zoom and FedEx were strategic in their intent: developed and continuously updated to gain or sustain competitive advantage over rivals (Porter, 1985; Porter and Millar, 1985). The chapter also insists information systems must provide a return on investment - technology use can be efficient and strategic at the same time.

- Its purpose is strictly cost reduction and risk mitigation, because information systems are a commodity now and chasing differentiation with IT is futile. — You have adopted the commodity argument the chapter reports and then rejects. IT keeps changing rapidly, and companies from Amazon to Zoom created advantage by combining commoditized technologies with proprietary systems and business processes. Judged on cost alone, the tracking accuracy the hospital actually cares about will be the first thing cut.
- Its purpose is strategic, so return on investment is not the right question to ask about it. — The chapter does not let you off that hook. It says plainly that information systems must provide a return on investment and can also be a powerful enabler of competitive advantage. Dropping the ROI test is how a strategic project becomes an unaccountable one.

Decision 2. The fastest route to launch is to run everything - scanning, routing, and customer notifications - on one outside provider's platform.

- Use the provider, but plan for the hours it is unavailable, because when one provider carries every function, all of them stop together. **— the stronger call.** This is the exact lesson the chapter draws from Zoom: relying on a single provider can quickly disrupt many activities in unforeseen ways. Note what you did not do - you did not refuse the provider. You refused to let one outage take down scanning, routing, and notifications at the same moment.
- Go all in on the single provider, since it is the fastest path and the provider is far larger and better staffed than you are. — Size did not protect anyone on May 17, 2020. Zoom was serving over 300 million daily meeting participants when the service became unreachable on a Sunday morning, and the churches with nothing to fall back on had to end their services early. Your hospital deliveries would have no fallback either.
- Build every piece in-house from scratch so that the company never depends on an outside provider. — This overcorrects past the chapter's point. Its advice is to combine commoditized technologies with proprietary systems and business processes, the way companies from Amazon to Zoom did. Rebuilding standard, interchangeable parts spends your scarce effort where it cannot differentiate you, and delays the part that can.

Decision 3. The contract assumes a steady daily volume. A flu surge across the hospital network could triple it with about a week of warning.

- Design and test for a spike you have never actually seen, because the failure case in the chapter was caused by unforeseen growth rather than by a bug someone shipped. **— the stronger call.** Right. Zoom's outage followed a tremendous unforeseen increase in users, and the chapter presents that surge itself as the hazard. Systems this large are difficult to build, which is why the chapter insists on handling their development the right way the first time around.
- Plan for the contracted volume; if demand triples, that is a good problem, and capacity can be added when it happens. — This is the assumption the Zoom case punctures. Demand arrived faster than anyone planned for, and the moment the system faltered was the moment its users needed it most. A week of warning is not enough time to rebuild something you did not design to stretch.

- Cap how many customers can use live tracking at once, so that the system cannot be overloaded during a surge. — You have protected the servers by damaging the product. Accurate tracking is the reason the hospital chose you, and rationing it during the busiest week converts your competitive weapon into a complaint. FedEx answers more than 500 million tracking requests per day rather than limiting who may ask.

Decision 4. Year one goes well. Sorting is automated, the hospital renews, and the board asks whether IT spending can now drop to simple maintenance.

- No - keep reengineering the system on a schedule, because an advantage from information systems can be fleeting once competitors work out how to do the same thing. **— the stronger call.** This is the chapter's warning and FedEx's practice: FedEx reengineers and improves the performance on average twice a year and now manages to deliver a quarter of all daily packages handled within one business day. The edge is renewed, not banked.
- Yes - the system works and the contract is signed, so the advantage is locked in and further spending is waste. — The chapter says the competitive advantage from the use of information systems can be fleeting, as competitors can eventually do the same thing. A frozen system does not hold its lead; it simply waits to be matched.
- Yes, and redirect the savings into advertising the tracking feature, so customers remember who has it. — Advertising an advantage does not renew it. The claim in the ad stays the same while a rival's capability improves, and the chapter's mechanism for fading advantage is rivals doing the same thing, which no amount of promotion prevents.

Decision 5. Fourteen months later a rival launches tracking that matches yours. In the same month, a well-funded startup with slicker software than either of you shuts down.

- Read both events the chapter's way: advantage from IS is fleeting because rivals eventually do the same thing, and companies with bad business models tend to fail regardless of whether they use information technology. **— the stronger call.** Exactly. The two facts are the two halves of the section. Good technology on a sound business model tends to succeed; good technology on a broken one tends to fail anyway; and any advantage you do build has to be renewed because it can be copied.
- Conclude that the tracking investment was wasted, since a rival now offers the same thing. — Fleeting is not futile. The chapter's position is that information systems are a necessary part of doing business, can be used to create efficiencies, and can also enable competitive advantage - the fourteen months of exclusivity, the renewed hospital contract, and everything you learned building it were all real.
- Conclude that the startup failed because its technology was not good enough, and study its software for what to add next. — The chapter points somewhere else entirely: companies with bad business models tend to fail regardless of whether they use information technology. Copying features from a company that could not make its model work is how you inherit its problem.

Ethical dilemma: the social and environmental costs of the newest gadgets

An **ethical dilemma**, sometimes called a moral dilemma, occurs when you must choose between two options, each of which involves breaking a moral imperative — not right against wrong, but two harms. There is usually no definite solution, so the chapter gives two steps, in order:

1. Weigh the **consequences** of each option in benefits and harms, considering degree and time horizon.
2. Then judge the **actions** themselves, irrespective of consequences, for honesty, fairness, and respect.

Apply it to the device in your pocket. Tiny silver letters on the back of an iPhone say: "Designed by Apple in California—Assembled in China." Globalization lets Apple design the electronics consumers crave while contract manufacturers worldwide handle components and assembly, so Apple controls the designs but not always how its suppliers work.

Foxconn, one of Apple's primary Chinese assembly partners, was scrutinized following complaints of poor working conditions, because huge production volumes and tight deadlines pushed workers to their limit. The toll took several forms:

- **Twitching hands**, and the uncontrollable mimicking of the motion after work, from repeating it for hours.
- **A rapid burnout rate**, visible in the resignation of 50,000 workers each month.
- **Up to 14 suicides**, the harm that drew the scrutiny.
- **An independent audit** that confirmed excessive overtime and health and safety issues.

The cost is not only human. In search of design aesthetic, Apple also designs its products as non-repairable, so degraded lithium-ion batteries send spent AirPods to the bin.

The other side is what makes this a dilemma rather than a scandal:

> **Profit maximization** — Apple pursues it for its shareholders, the obligation every other choice answers to.

> **Supplier scarcity** — few suppliers worldwide can produce its volumes on short notice, so shifting suppliers is not easy.

> **Margin** — reducing hours, raising salaries, or adding fringe benefits cuts the profit margin.

> **The alternative for workers** — for many young Chinese, a few months at Foxconn beats tilling a family farm or not working at all, as evidenced by the thousands lining up for recruiting sessions every week.

Each option trades one harm for another.

The chapter leaves you two questions. If you were in Tim Cook's shoes, what would you do? As a consumer, what are your own ethical dilemmas associated with the ever-increasing desire for new gadgets?

SECTION 1–4

Computer ethics, privacy, and intellectual property

Every system in this chapter runs on data, and much of it is about people, including you. This section separates three questions that overlap without being identical: what is ethical, what an organization promised, and what applicable law permits.

Ethics is not the same thing as law

Something can be permitted by law and still be the wrong thing to do. **Computer ethics** describes moral issues and standards of conduct as they pertain to the use of information systems.

The chapter uses the sale of customer data to expose that ethical question. Do not turn its ownership framing into a universal legal rule: whether an organization may share or sell personal data depends on the jurisdiction, the kind of data, the notice and consent provided, and the promises the organization made.

Mason's four questions: PAPA

In 1986 **Richard O. Mason** reduced the debate to four issues — information privacy, accuracy, property, and accessibility — known ever since as **PAPA**. Almost every argument about technology and people is one of these four.

Information privacy asks what an individual should have to reveal to others in the workplace or through other transactions, such as online shopping — for example, a site you bought from once greets you by name and recommends the product you were about to look for, because you told it more than you meant to.

Accuracy asks about the authenticity and fidelity of information: the software to rearrange and otherwise change a photograph is ordinary now, so an image can look authentic and still misrepresent what happened, and somebody has to answer for a record about a person that turns out to be false.

Property asks who owns information about individuals and how it may be sold and exchanged — one purchase puts your name and address on a list that is sold onward, which is why unwanted solicitations arrive from credit card companies, department stores, magazines, and charities you have never dealt with.

Accessibility asks what a person or organization has the right to obtain about others, and how that information may be accessed and used — for example, whether every clerk in a company should be able to open any customer file, or only the ones hired, trained, and supervised for it.

EXPLORE **The four PAPA issues, one card at a time**

Open each of Mason's four issues and read all four facets before moving to the next.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Information privacy · What must I reveal?

- **What the question is:** What information an individual should have to reveal to others in the workplace or through other transactions, such as online shopping.
- **Where you already meet it:** An online store greets you by name and recommends the product you were about to search for, and the ads that follow you look aimed at you personally.
- **What goes wrong when nobody asks it:** Social Security numbers, credit card numbers, medical histories, and family histories end up online, where a coworker, a spouse, or a future employer can find them with a search.
- **The hard part:** You control only your half. A friend can post a photo of you, an abandoned profile keeps the address you typed years ago, and search engine caches keep pages reachable long after they come down.

Accuracy · Is the record true?

- **What the question is:** Ensuring the authenticity and fidelity of information, and settling who answers for a record about a person that turns out to be wrong.
- **Where you already meet it:** The shopping habits and credit history a company compiles about you, and photographs, which anyone with ordinary software can now rearrange and otherwise change before publishing them.
- **What goes wrong when nobody asks it:** Decisions get made about you from a record you have never seen, and an altered picture circulates as though it were evidence, because nothing in the system distinguishes a faithful record from a convincing one.
- **The hard part:** The remedy exists on paper as Access/Participation, the right to see your data and request correction, but you have to learn the record exists before you can challenge it.

Property · Who owns the data?

- **What the question is:** Who owns information about individuals, and how that information can be sold and exchanged.
- **Where you already meet it:** The unwanted solicitations from credit card companies, department stores, and charities that follow one purchase, and the answer to the question of how you got on another mailing list.
- **What goes wrong when nobody asks it:** Separate pieces of you get joined. Demographic data, psychographic data, and card transactions combine into a profile precise enough to support targeted manipulation, discrimination, or a damaging breach.
- **The hard part:** The chapter frames database control as a property question, but control is not unlimited: promises, consent, the kind of data, and applicable law can all constrain sale or exchange.

Accessibility · Who may reach it?

- **What the question is:** What information a person or organization has the right to obtain about others, and how that information can be accessed and used.
- **Where you already meet it:** Which employees inside a company can open the customer database, and what a search engine hands a stranger who types your name, since almost anything posted by or about you can be found and sits in a long-term cache.
- **What goes wrong when nobody asks it:** Unsupervised staff browse records they have no business opening, and personal information such as Social Security numbers, credit card numbers, and medical histories becomes reachable by anyone who thinks to look for it.
- **The hard part:** It has to be answered twice: at a strategic and ethical level (should we be doing this?) and at a tactical level (hiring, training, supervision, and software and hardware safeguards).

SORT

Which PAPA issue is this?

Drop each situation into the one Mason issue it raises first and most directly.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Privacy — what an individual should have to reveal to others

- A restaurant makes you type an email address to use its free Wi-Fi, and never says what happens to the address. — Giving an email address for Wi-Fi is one of the collection moments the chapter names, and the question is what you should have to reveal in an ordinary transaction.

- You give a birth date and home address to a social network, abandon the profile a year later, and the data stay out there. — You revealed personal information for one purpose and lost any say over how long it lives, which is the privacy question rather than an ownership dispute.

Accuracy — authenticity and fidelity of the information

- An insurer's file lists a surgery you never had, so your claim is denied. — Nothing was leaked or sold here; the record is simply not true, which is the authenticity and fidelity problem.
- Someone edits a photograph of a public figure to show an event that never happened, then posts it as news. — The chapter raises photo manipulation as an ethical problem precisely because the image no longer faithfully represents what happened.

Property — who owns the data and how it may be sold or exchanged

- A magazine shares its subscriber list with a department store, which starts mailing you catalogs. — Who controls a subscriber list and under what conditions it may be exchanged is exactly the property question; legality depends on the surrounding notice, consent, promises, and law.
- A company promised survey answers would be used strictly internally, then sold them to another firm years later. — The exchange raises the property question, and breaking the stated purpose is also an ethical failure that may trigger legal enforcement against deceptive promises.
- A retailer combines what you searched for, what you bought, and what you liked into one profile that predicts your next purchase. — Each piece was given separately, so the ethical question is ownership of the combined file, the aggregation the chapter calls the Database of Intentions.

Accessibility — who has the right to obtain it, and who can reach it at all

- A claims clerk with no reason to look opens a neighbor's file, because the system lets any employee open any record. — The data are accurate and legitimately held; the issue is who has the right to obtain them and how access is controlled.
- A hiring manager searches your name and reads a question you posted years ago on a page that has since been taken off the web. — Nothing here is inaccurate and nothing was sold; the question is what a person has the right to obtain about you and how they may reach it, and search engine caches keep pages available long after the original comes down.
- A bank ties access to its customer database to hiring, training, and supervision, backed by software safeguards. — This is the tactical half of the accessibility question, the controls deciding who may reach the data and how.

One more ethical problem sits alongside those four without being one of them. Where those with access to information systems hold great advantages over those without, there is a **digital divide**, and it is one of the major ethical challenges facing society because computer literacy now decides who can compete.

Mason's four ask about information that already exists about you; this one asks who reaches the technology at all.

Information privacy up close

Most personal data is handed over voluntarily and quietly, in three ordinary moments.

> **Credit card purchases** record what you bought, and you indirectly allow that collection simply by using the card.

> **Surveys** gather more than the answer asked for, whether you fill one out to apply for a card or to rate a restaurant.

> **The email address you type to reach a Wi-Fi network** is the price of the connection, so the network is free only in name.

Providing data enables the immediate transaction; it does not grant unlimited permission for every later use. Notice, choice, an organization's promises, and applicable law still matter.

One survey alone is harmless; the combination is not, because two kinds of data work together.

> **Demographic data** answers who am I, and where do I live, the part of you that appears on any form.

> **Psychographic data** answers what do I like, what are my tastes, the part you give away by choosing rather than declaring.

Put the two beside each other and a company can piece the bits into a highly accurate profile. The aggregation has a name, the **Database of Intentions**: what you want, what you buy, what you like, what you are interested in, what you are doing, where you are, who you are, and whom you know.

Such profiles feed predictive models that allow manipulating people through targeted advertising, and invite discrimination and damaging breaches.

> Providing data at separate moments never meant agreeing they be combined into one picture.

So who controls it? The chapter frames a company-maintained database as a property issue, then emphasizes limits created by what the company said. Current law may add rights to know, correct, delete, limit, or opt out, depending on the person, data, organization, and jurisdiction. A firm that promised to use marketing data strictly inside its own business and then sold them would at minimum break its ethical commitment, and regulators may also enforce deceptive privacy promises.

Each company answers the question twice.

At a strategic and ethical level, should we be doing this, which is where a promise about the data is kept or broken.

At a tactical level, how do we ensure the security and integrity of the data, which is answered through hiring, training, and supervision of employees with access, plus software and hardware security safeguards.

Things you cannot control. A friend's photo of you stays up whether you like it or not, and search engines keep pages in a long-term cache after they come off the web.

In 2014 the European Court of Justice ruled that individuals have a **right to be forgotten**, so search engines may have to remove links to personal information that is inaccurate, inadequate, irrelevant, or excessive, though that raises a censorship problem of its own.

Two newer rules go further.

The General Data Protection Regulation (GDPR), applied across the European Union from 2018, establishes principles including transparency, data minimization, integrity, confidentiality, and protection of personal data.

The California Consumer Privacy Act (CCPA) gives covered California consumers defined rights over personal information held by covered businesses.

CHECK YOURSELF

Ethics, PAPA, and who owns your data

Pick the best answer, then read every explanation, including the ones for options you did not choose.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. What does the term **computer ethics** refer to?

- **A.** The laws a government passes to punish computer crime and protect data — Those are statutes, such as the ECPA. Law and ethics overlap but differ: governments play catch-up, so much of what is legal is still an open ethical question. This would be right if the question asked what computer crime legislation does.
- **B.** The technical controls that keep unauthorized people out of a company database — That is Integrity/Security, one of the five fair information practices and the tactical half of the accessibility question. It is a way of acting ethically, not the field that decides what acting ethically means.

- **C.** The moral issues and standards of conduct that pertain to the use of information systems **— correct.** This is the chapter's definition, and it is deliberately broad: it covers conduct that may be legally permitted in a particular context, such as combining purchase data into a customer profile, and still worth arguing about.
- **D.** A company's written policy listing which software employees are allowed to install — That is an acceptable-use policy, one product of ethical thinking inside one organization. Ethics is the set of standards such a policy gets written from, which is why schools and businesses each write their own.

2. A hospital system lists the wrong blood type for a patient because a clerk mistyped it, and treatment decisions follow that record. Which of Mason's four issues does this raise first?

- **A.** Accuracy, because the authenticity and fidelity of the information have failed **— correct.** Accuracy is Mason's question about authenticity and fidelity, and it is the issue that can fail even when data remain private and access is correctly restricted.
- **B.** Information privacy, because medical data about an individual is exposed — Privacy concerns what an individual should have to reveal to others. Nothing was revealed to anyone who should not see it; the record is simply untrue. This would be the answer if a correct record had been leaked or sold.
- **C.** Property, because the hospital controls the record and decides how it is maintained — Property asks who controls information about individuals and how it may be exchanged. Control of the record is not what failed here, and nothing was transferred.
- **D.** Accessibility, because clinicians can open the record and act on what it says — Accessibility asks who has the right to obtain information and how it is accessed. The clinicians reading the file are exactly who should have access; the problem is what the file says.

3. You buy a lamp from an online retailer. Which statement best connects the chapter's property question to current privacy rules?

- **A.** Because the record is about you, the retailer needs affirmative written permission before repurposing it — Some laws require consent for particular uses, but written permission before every internal use is not a universal rule. This would be correct only where the applicable law or a prior promise specifically required that consent.
- **B.** Because the retailer maintains the record, its ownership interest controls unless it made a specific promise — The chapter uses database control to introduce the property question, not to erase privacy obligations. This would be correct only if no law, notice, consent choice, contract, or promise limited the proposed use.
- **C.** Because the card issuer processed payment, its rules control how the retailer may use the transaction record — A card issuer maintains its own payment record, while a retailer maintains its customer and order records. Processing one part of a transaction does not give the issuer exclusive control over every resulting record.

- **D.** The retailer controls its customer record, but notice, consent, promises, data type, and applicable law can limit later use or sale **— correct.** This preserves the chapter's property question while recognizing that current legal rights and duties vary by jurisdiction, data type, notice, consent, and the organization's promises.

What a privacy policy is supposed to do

Because consumers rank privacy a top concern, governments pressured businesses to post policies. The accepted list of **fair information practices** comes from the U.S. Federal Trade Commission.

PRACTICE	WHAT IT MEANS IN PLAIN TERMS
Notice/Awareness	What data are gathered, what for, who gets access, whether giving them is required or voluntary, and how confidentiality is ensured — the job of a site's **data privacy statements**.
Choice/Consent	Options about what is done with the data: you **opt in**, signaling agreement to collection or further use, for example by checking a box, or **opt out**, signaling that data cannot be used otherwise.
Access/Participation	Means to see the data collected about you, check it for accuracy, and request correction.
Integrity/Security	Using reputable sources of data, and controls against unauthorized access, disclosure, or destruction.
Enforcement/Redress	Means to enforce these practices and give customers a remedy, through self-regulation or law.

Here is the catch: a privacy statement is notice, not a technical shield. Organizations must honor their promises and applicable law, yet a policy can still be vague or hard to use while a vendor records the shape of a visit.

- Which pages you look at.
- Which products you buy.
- Where it is delivered.
- Which products you examine in detail.
- How you pay.

A vendor may seek to share or sell those records where its notice, consent, promises, and applicable law allow it, and an unscrupulous one may ignore those limits. Review the privacy policy of every company you do business with, use the choices it offers, and prefer businesses that state their practices clearly.

DECIDE

One evening of ordinary privacy decisions

Work through the evening one decision at a time and read what each choice actually costs you.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

You are at a coffee shop on your laptop, planning to buy a pair of headphones. Nothing here is dramatic, and that is the point: this is where most personal data gets given away.

Decision 1. The shop's free Wi-Fi will not connect until you enter an email address on its login page.

- Enter the personal address you use for family, school, and banking. — The address can now be recorded in connection with a place and time, subject to the notice, choices, promises, and law that apply. It is also the same address your important password resets use, so unrelated marketing and security messages share one crowded channel.
- Enter a separate email account you keep only for sign-ups and online purchases. **— the stronger call.** This is the chapter's advice applied early. The address may still be collected and used as disclosed and permitted, but the marketing that follows lands where it cannot bury anything important and is not tied to your banking or work identity.
- Make up an address that does not exist, so nothing can be sent to you. — It hides you tonight, but you cannot receive the confirmation link many networks send, and the first real order needs a real address anyway. A separate real account solves it permanently instead of once.

Decision 2. The store you found is one you have never used, and you cannot find a privacy policy anywhere on the site.

- Order anyway; a missing policy just means a small company that has not written one yet. — A missing policy leaves you without clear notice of what is gathered, why, who receives it, or how it is protected. Applicable law may still impose duties, but you cannot evaluate the vendor's stated practices or choices before handing over the data.
- Buy from a competitor that publishes a clear policy, ideally one audited by an independent organization such as TRUSTe. **— the stronger call.** This is the chapter's rule: review the privacy policy of every company you do business with, and refuse those that have no clear policy. Outside monitoring adds a check the site cannot perform on itself.
- Order, but pay with a card that has a low limit so the damage is capped. — That addresses payment exposure, which is a different risk. It changes nothing about the vendor's collection of the purchase, the browsing that led to it, or the shipping address, nor does it tell you how those records may be used.

Decision 3. At checkout a box is already checked: 'Yes, share my information with our trusted partners.' A link beside it says Privacy.

- Open the privacy statement, see what partners means and whether the data are required, then uncheck the box. **— the stronger call.** You used two practices in one minute: Notice/Awareness, which is what a data privacy statement is for, and Choice/Consent, by exercising the opt out. Reading alone would not have protected you; unchecking is what changed the outcome.
- Leave the box as it is, since it was checked by default and is probably required to complete the order. — It is not required; it is an opt-out arrangement, where sharing happens unless you act. Leaving it checked is how your details reach partners you never chose, and how the catalogs start.
- Uncheck the box and skip the policy, since the box is the only thing that matters. — Better than leaving it checked, but the statement is where you learn what else is collected, who has access, and how confidentiality is handled. The checkbox governs one use of your data, not all of them.

Decision 4. The order is placed. You keep browsing other stores on the same laptop, and those headphones start appearing in ads on unrelated sites.

- Log out of the store account, which should stop it from following you. — Logging out ends your session but leaves the cookies on your machine, and those are what let a site owner monitor where you go and what you do. The tracking continues while you are signed in to nothing.
- Rely on the store's privacy policy, since it promised not to misuse your information. — A policy states intentions, not capabilities, and the chapter is blunt that data privacy statements often do not protect consumers. The tracking here is being done by a file sitting on your own machine.
- Manage the browser's cookie settings, use a private window, switch to a search engine such as DuckDuckGo, and turn off ad personalization. **— the stronger call.** These are the chapter's suggested privacy steps. Each limits part of the trail between what you shop for and what follows you, but none should be mistaken for complete anonymity.

Decision 5. The headphones arrive late and damaged. You draft a furious post naming the clerk who answered the phone, with a photo of the receipt showing your address.

- Post it; it is all true, and you can delete it tomorrow if you regret it. — Truth is not the constraint. The internet never forgets: posts are stored, cached by search engines, and reshared, and content stays somewhere on the web after the original page is gone. The receipt also publishes your home address to strangers.
- Post it without the photo, since the text alone cannot identify anyone. — The photo was the worst part, so this is an improvement, but a named employee and an angry tone are exactly the regrettable material that can be devastating for a career later. The permanence problem is unchanged.

- Send a factual complaint through the company's support channel and keep the receipt and the employee's name out of public view. **— the stronger call.** It solves the actual problem and leaves nothing a future employer or stranger can dig up. The chapter's warning is about permanence, and this is the version of the complaint that does not follow you.

Email at work is not your mail

Most companies provide internet and email access, and many monitor some activity on the accounts or systems they supply. Do not assume a work message is private: monitoring rules vary with jurisdiction, the communication, the employer’s policy, and the notice given.

> **The Electronic Communications Privacy Act (ECPA)** of 1986 regulates interception and access to electronic communications, but its rules and exceptions do not create a blanket guarantee that messages on an employer’s system are private.

> **State laws and workplace policies** may require notice or restrict particular monitoring practices, so the applicable rule cannot be inferred from one chapter example or one state alone.

A clear monitoring policy helps establish expectations, but applicable law still controls. The practical lesson is narrower: treat work accounts and systems as organizational resources, read the policy, and send sensitive personal messages through a personal account on a personal device.

Steps that actually maintain your privacy online

None of these makes you invisible, but each closes one leak.

> **Choose websites monitored by independent organizations** such as TRUSTe, because an outside group audits privacy practices a site cannot credibly audit itself.

> **Avoid having cookies left on your machine**, since commercial sites leave **cookies** so the owner can monitor where you go and what you do there; manage your browser’s cookie settings or use cookie management software.

> **Visit sites anonymously** with private browser windows, search engines such as DuckDuckGo, and ad personalization off, which keeps marketers, identity thieves, and coworkers at bay.

> **Use caution when requesting confirmation email** by keeping a separate account for online purchases, apart from your work address, so order mail and the marketing behind it lands somewhere harmless.

> **Beware what you post or say online**, because the internet never forgets: content stays somewhere on the web after the page is gone, and one regrettable item can wreck a career.

SELF-CHECK

Your own privacy habits

Rate each statement honestly; anything you cannot claim points straight at what to re-read.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

- I can define information privacy in one sentence and name a moment this week when I traded some of it for convenience. Re-read the definition: what an individual should have to reveal to others in the workplace or through transactions such as online shopping.
- I know whether the last account I created asked me to opt in, or quietly relied on my not opting out. See Choice/Consent in the fair information practices table: opting in signals agreement, opting out signals refusal.
- I could find the data privacy statement on a site I use weekly, and I know the five things notice is supposed to tell me. Notice/Awareness covers what is gathered, what for, who gets access, whether it is required or voluntary, and how confidentiality is ensured.
- I have deliberately managed the cookie settings in my browser, and I can say what a cookie lets a site owner observe. See the steps for maintaining privacy online: cookies let the site owner monitor where you go and what you do on the site.
- I can name something I have posted that I would not want a future employer to read, and explain why deleting it may not remove it. Search engines keep long-term caches, and most content stays somewhere on the web after the original page is gone.
- I would be comfortable if every message I sent from a work account this month were read aloud by my employer. Re-read the email section: the ECPA protects voice mail far more than email, and the working rule is to send only what would not embarrass you if made public.
- I can list four behaviors the Computer Ethics Institute prohibits and the two habits it recommends, without looking them up. See the responsible computer use list, including the two positive recommendations that follow the prohibitions.

Intellectual property in a world of perfect copies

The other half of the property question is about what people create. **Intellectual property (IP)** means creations of the mind that have commercial value: a song, a photograph, an article, a program.

Analog copies lost quality, which protected creators by accident; digital files removed that friction, so you can duplicate a friend’s music library losslessly, pass it to strangers over peer-to-peer networks, or repost a photograph without asking.

The cases are ordinary ones. Your school may license software to you as a student, and you forget to uninstall it after graduating or lend it to family; or you download a pirated program you cannot afford.

Lossless copying has reached physical objects: **3D printing** builds objects from digital models by adding thin layers instead of milling a part from a block and binning up to 90 percent of the material, and cheap printers make counterfeits easy, causing tremendous losses of intellectual property.

People defend all of this in three ways.

> **No real loss occurred**, the argument runs, since the person would have used a free alternative or bought nothing at all.

> **Students cannot afford expensive software**, which is offered as a reason the rule should not apply while somebody is studying.

> **Copying counts as praise** in many non-Western societies, where using someone else's work honors the creator, so putting a famous song under a video or dropping another person's writing into a blog post reads as a compliment rather than a theft.

Either way the act is the same: using intellectual property without permission, attribution, or compensation.

Why organizations write a code of ethical conduct

The internet age left governments playing catch-up on computer crime, privacy, and security, and left everyone else with questions no statute answers.

- Is it ethical to rearrange and otherwise change a photograph, when it still looks like a genuine record?
- Is it ethical to take computer time at work for personal business, when the equipment and the hours belong to somebody else?
- Is it ethical to compile a customer's shopping habits and credit history for sale, even where the specific use is legally permitted?

Because nobody settles those in the moment, many businesses and most universities publish guidelines telling users to act responsibly, ethically, and legally and to follow online etiquette and the law.

Responsible computer use

The best-known set comes from the **Computer Ethics Institute**, a research, education, and policy study organization that studies how advances in information technology have affected ethics and corporate and public policy. Its widely quoted guidelines prohibit eight things:

- Using a computer to harm others, since one post or one program reaches further than any pair of hands could.
- Interfering with other people's computer work, because deleting a shared file or downing a shared service stops work that was never yours to stop.
- Snooping in other people's files, since being able to open a file has never been the same as having a reason to.
- Using a computer to steal, because theft is no milder when the thing taken is a number in a database instead of cash in a drawer.
- Using a computer to bear false witness, since a rearranged photograph carries the look of a genuine record and outlasts a spoken lie.
- Copying or using proprietary software without paying for it, because digital copying is lossless and free, so only your own decision stops you.
- Using other people's computer resources without authorization or compensation, since the storage, bandwidth, and processing time were paid for by somebody else.
- Appropriating other people's intellectual output, because a creation of the mind with commercial value carried a price you did not pay.

Beyond the prohibitions it recommends two habits: think about the social consequences of the programs you write and the systems you design, and use a computer in ways that show consideration and respect for others.

When in doubt, review the ethical guidelines published by your school, your place of employment, or your professional organization, because those are the rules you will actually be held to.

The anonymity is imaginary. People drawn to illegal or unethical behavior count on being anonymous, but we leave electronic tracks as we wander the web, and many have been traced and prosecuted while sure their trail was hidden. Post objectionable material, draw complaints, and your internet service provider can ask you to remove it — or remove you.

CHECK YOURSELF **Email, consent, IP, and the code of conduct**

Answer all four, then read every explanation before you move on.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. Your employer has never mentioned monitoring. What is the soundest approach to email you send from your work account?

- **A.** Treat it as private until the company publishes a policy, because notice normally defines when monitoring begins — Notice can be required by a policy or particular law, but it is not safe to infer privacy from silence. This would be correct only where the applicable rule made advance notice a condition of the specific monitoring at issue.

- **B.** Treat the ECPA as the controlling protection, comparable to the rule for a private phone conversation — The ECPA regulates electronic communications, but different communications, access methods, exceptions, and jurisdictions can produce different rules. This would be correct only if one protection applied identically to every channel, which it does not.
- **C.** Treat personal messages as exempt from monitoring, because their subject matter is not company business — Monitoring commonly turns on the account, device, network, policy, and applicable law rather than whether a message feels personal. This would be correct only if the governing policy or law created that subject-matter exemption.
- **D.** Read the policy and applicable rules, avoid sensitive personal use, and assume an employer-provided account may be monitored **— correct.** This does not claim that every form of monitoring is lawful; it recognizes that rules vary and applies the chapter's practical lesson to keep sensitive personal communication off employer-provided accounts and systems.

2. A checkout page shows a box that is **already checked**: 'Yes, send me offers from our partners.' Which consent model is the page using?

- **A.** Opt in, because a checked box is a signal of agreement to further use of the data — Opting in is when you take the action that signals agreement, such as checking an empty box or answering a request for permission. Had the box started empty and you checked it, this would be right; pre-checking reverses who has to act.
- **B.** Opt out, because the data will be used unless you take an action to refuse **— correct.** Opting out means signaling that data cannot be collected or used in other ways, and here the sharing proceeds by default unless you uncheck the box.
- **C.** Notice/Awareness, because the wording on the box tells you what will happen to your data — Notice/Awareness is the disclosure of what data are gathered, what for, who gets access, and whether providing them is voluntary, normally in the data privacy statement. The box is the choice mechanism the notice describes.
- **D.** Access/Participation, because you can see the setting and change it yourself — Access/Participation means seeing the data a company has collected about you, checking it for accuracy, and requesting correction. Toggling a marketing preference is a consent setting, not access to your record.

3. You keep using software your school licensed to you after you graduate, and you pass the installer to a cousin. Why does the chapter treat this as an intellectual property problem?

- **A.** There is no real problem, since the publisher lost no sale: you would have used a free alternative if you had to pay — This is exactly the argument the chapter says people make, alongside the claim that students cannot afford expensive software. The chapter reports it seriously and still concludes the work is being used without permission and without compensating the creator.
- **B.** Because each copy degrades the original, so the publisher's product is worth less afterward — This is analog thinking. Copying a song to tape once cost quality, but digital duplication is lossless, which is precisely why the chapter says digital media made this problem so much larger.

- **C.** Because IP means creations of the mind with commercial value, and this uses one without permission or payment **— correct.** The license let you install and use the software while enrolled; using it afterward and handing it on goes past what was granted, and intellectual property is a creation of the mind with commercial value.
- **D.** Because the school owns the software, so passing it on takes school property rather than the publisher's — A license is permission to use, not ownership. The school holds an agreement with the vendor allowing enrolled students to install the software, and the vendor keeps the intellectual property throughout.

4. Besides its list of prohibitions, what does the Computer Ethics Institute's guidance actually recommend that people do?

- **A.** Publish an organizational code of conduct so that every user knows the rules in advance — Writing guidelines is what businesses, universities, and public school systems do, and those guidelines tell users to act responsibly, ethically, and legally and to follow online etiquette and the law. That is an organization's job, not one of the two habits the institute recommends to an individual user.
- **B.** Secure your accounts and files so that nobody can reach them without authorization — Good practice, and close to the Integrity/Security fair information practice, but the institute states it from the other side, as a prohibition on using other people's computer resources without authorization. It is not one of the two recommendations.
- **C.** Think about the social consequences of programs you write and systems you design, and show consideration for others **— correct.** These are the two positive recommendations that follow the prohibitions, and they are the only items asking you to consider people you will never meet rather than simply to refrain from an act.
- **D.** Avoid using a computer to harm others or to snoop in other people's files — This is genuinely on the institute's list, which is what makes it tempting, but it sits among the prohibited behaviors. The question asked what the guidance recommends beyond the prohibitions, which is the two positive habits.

APPLICATION SUPPLEMENT · 1–5

Turning information systems into competitive strategy

Imagine two grocery chains buying the same checkout software from the same vendor. One uses it to lower labor cost; the other uses it to learn what households buy and make offers the first cannot match. Same software, different **intent**. This supplemental section applies two strategy tools to that second kind of use.

> **How this supplement relates to the chapter.** The chapter defines competitive advantage and cites Porter's work, but it does not present Five Forces or the value chain as a fifth chapter learning objective. The frameworks below come from the Porter works named in the chapter's references and extend Objectives 1.1–1.4 into a manager's practice exercise.

What "strategic" means here

When something your company has or does makes a customer choose you instead of a rival, and the rival cannot easily copy it, you have a **competitive advantage**.

That is not the same as being good: two coffee shops can both make good coffee, but a shop that learns when to alert you about your usual bean has an advantage because a rival would first need to build comparable customer history.

Of the systems built at **Zoom** and **FedEx**, the chapter says the choices made in developing them were **strategic in their intent**: both were developed, and are continuously updated, to help the companies **gain or sustain** a competitive advantage over their rivals.

Continuously matters: on average FedEx reengineers and improves their performance twice a year, in ground hubs where a package passing an overhead scanner leaves between one and two seconds to be diverted, so the decision must be made in a few hundred milliseconds.

There are three reasons to spend on a system, and only the third is strategic.

> **Efficiency** — the system does the same work with fewer people or less waste, because the process itself is unchanged and only its cost moves.

> **Return on investment** — the system pays back what it cost, which the chapter presents as something a system must do, not as the most it can do.

> **Strategic use** — the system changes the customer's reason to choose you, because it is no longer supporting the offer; it is part of it.

Size does not decide which one you get: whether it is a small mom-and-pop boutique or a large government agency, every organization can find a way to use information technology to beat its rivals.

The commodity argument, and the chapter's answer. At the start of the millennium, as information systems became standardized and ubiquitous, some argued they were now a commodity every company must have, that IT should target cost reduction and risk mitigation strictly, and that investing for differentiation was futile. The chapter states that argument and rejects it on four counts:

> **IT is still moving** — smartphones, social networks and the creative industries show information technology changing rapidly, and many companies gained advantage by innovatively using increasing **digital density**.

> **The combination is the advantage** — companies from Amazon to Zoom created advantage by **combining commoditized technologies with proprietary systems and business processes**.

> **The data can be the value** — for Google and Facebook, customer-generated data create the value, and how those data are gathered, processed, and used can sustain advantage.

> **The IT itself can be the product** — Amazon sells cloud computing to others, earning revenue directly from its IT investment.

The purchased software is never the advantage; what you build on it is.

A clearly hypothetical practice situation

Suppose you advise a regional grocer. The following assumptions exist only for this practice situation, and four of them drive everything that follows.

- Its shoppers are loyal out of habit, not because of anything the grocer built, so nothing they would lose keeps them from shopping somewhere else.
- Managers phone produce orders to a small supplier pool, so nobody can say chain-wide what is bought or what is paid for it.
- Inventory sits in a separate system in each store, so headquarters cannot say what is on hand chain-wide.
- A members-only warehouse club has entered some local markets, and a meal-kit service advertises across the region.

Porter's Five Forces: where the pressure comes from

Before you can say what technology a company needs, you have to say what is squeezing it. An **industry** is the set of companies selling the same kind of product to the same kind of customer, and Porter's argument is that profit in any industry is pressed by five separate forces. Name the wrong one and you buy the wrong system.

> **Competitive rivalry** is how hard the firms already in your industry fight for the same customers, because they sell nearly the same thing to a limited pool of shoppers — two gas stations across the street changing their signs the same morning.

> **Threat of new entrants** is how easily a company outside your industry today could compete tomorrow, because whatever keeps outsiders out — cost, scale, know-how, loyalty, together called **barriers to entry** — is low: opening a lemonade stand is easy, starting an airline is not.

> **Threat of substitutes** is a different product meeting the same underlying need, so the customer never enters your industry; for a grocer that is not another grocer but the meal kit, the delivery app, the restaurant.

> **Bargaining power of buyers** is your customers' ability to push your price down or your service up, because leaving costs them little — one shopper with a price app and a nearby rival has more power than you think.

> **Bargaining power of suppliers** is the reverse: the firms you buy from can raise prices or set terms, because you have few alternatives. The chapter reports this pressure without naming it a force — few suppliers worldwide can, on relatively short notice, produce the numbers needed to meet demand for Apple's products, so shifting suppliers is not easy for Apple.

Each force also has a **signal** that it is the strong one, and a kind of IT initiative that relieves it.

INTERACTIVE DIAGRAM **The five forces, one at a time**

Pick a force to redraw it: what the pressure looks like, the signal that it is the strong one, and the kind of IT initiative that relieves it.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Competitive rivalry — Pressure from the companies already selling what you sell

The practice grocer (Several locations, familiar brands) → Two rival chains (Same aisles, same suppliers) → One shopper (Fills a single cart each week) → The usual weapon (Price cuts that thin both margins)

- **What it is** - how hard the firms already inside your industry fight one another for the same finite set of customers. Two gas stations across the street changing their signs on the same morning is rivalry you can watch happen.
- **Signal it is strong** - discounting is the only tool anyone reaches for, a promotion is matched within days, and unit volume stays flat while margin falls. Nobody is winning; everyone is paying.
- **What IT does about it** - moves the fight off price. One chain-wide inventory record so the advertised item is actually on the shelf, faster checkout so the line is shorter, and personalized offers built from purchase history a rival does not have.
- **The catch** - anything a rival can also buy becomes table stakes within a season. The chapter's point holds here: advantage sits in the proprietary systems and business processes you build on top of commoditized technology, not in the software licence.

Threat of new entrants — Pressure from companies not in your industry yet

Outsiders (Have capital, want your customers) → Barriers to entry (Cost, scale, know-how, loyalty) → A warehouse club (Enters some local markets) → The practice grocer (Suddenly shares its shoppers)

- **What it is** - how easily a company that does not compete with you today could start tomorrow. What holds outsiders back is called barriers to entry: the money, scale, expertise, or customer loyalty a newcomer would have to match first.
- **Signal it is strong** - newcomers keep appearing, starting up takes little capital, and nothing you own would take a competitor years to reproduce. If your only barrier is that customers are used to you, you have no barrier.
- **What IT does about it** - builds the barrier that money alone cannot buy: years of purchase history, saved shopping lists, subscription reorder, and a loyalty record that makes leaving cost the customer something real.
- **The catch** - a well-funded entrant can buy scale and buy price. The barrier that actually holds is the one that takes time to accumulate, which is why the data asset matters more than the store count.

Threat of substitutes — Pressure from a different answer to the same need

The real need (Feed a family on Tuesday) → Your answer (Groceries from an aisle) → A different answer (Meal kits, delivery, restaurants) → The result (The category shrinks, not just your share)

- **What it is** - a product or service from outside your industry that meets the same underlying need, so the customer never enters your industry at all. For a grocer the substitute is the meal-kit box and the restaurant, not the grocer across town.

- **Signal it is strong** - your price cuts stop working, because the customer you lost is not comparing you against another grocer. Whole categories decline at once, and the shoppers who leave do not come back for a coupon.
- **What IT does about it** - extends your offer into the substitute's territory: online ordering with reliable delivery windows, subscription staples that reorder themselves, and prepared-meal ordering that competes for the same Tuesday evening.
- **The catch** - substitutes are the force companies notice last, because management watches competitors with the same sign on the roof. Nothing in your sales data names the substitute; you have to go look for it.

Bargaining power of buyers — Pressure from the people who hand you money

Shoppers (Phone in hand, price app open) → Switching cost (Near zero - a rival is nearby) → The practice grocer (Shelf price effectively set by the shopper)

- **What it is** - your customers' ability to force your price down or your service up, because walking away costs them almost nothing. Power grows when they can compare easily, when switching is trivial, or when a few large accounts are most of your revenue.
- **Signal it is strong** - shoppers compare your shelf price in seconds and leave over pennies, promotions buy visits but not repeat visits, and your busiest customers cannot be named because you have no record of who they are.
- **What IT does about it** - gives the customer something to lose by leaving: a loyalty record, a saved list, an order history, a personalized offer that only exists because the system knows this household.
- **The catch** - a loyalty program that is only a discount teaches shoppers to wait for the discount. The relief comes from the data and the switching cost, not from the coupon.

Bargaining power of suppliers — Pressure from the firms you buy from

A small supplier pool (Supplies nearly all the produce) → The practice grocer (Learns of a price rise on the invoice) → Alternatives (Unqualified, unpriced, unknown) → The result (Terms set on the other side of the table)

- **What it is** - the ability of the firms you buy from to raise prices, cut quality, or dictate terms, because you have few alternatives and switching is slow. The chapter reports this pressure without labelling it a force: there are few suppliers worldwide who can, on relatively short notice, produce the numbers needed to meet demand for Apple's products, so shifting suppliers is not easy for Apple.
- **Signal it is strong** - a handful of suppliers cover most of your purchasing, price changes arrive as surprises rather than negotiations, and nobody can produce a list of qualified alternates.
- **What IT does about it** - a purchasing system that puts every quote side by side, tracks contract expiry before renewal, keeps a qualified alternate list current, and shares demand forecasts so the supplier plans with you instead of guessing.

- **The catch** - a long exclusive contract feels like control and is usually the opposite; it lowers this quarter's price and raises next year's dependence.

One lost sale can be told as a story about any of three forces, which is what makes naming them hard.

CHECK YOURSELF **Name the force**

For each situation, decide which of the five forces is actually at work before you decide what to build.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. The practice grocer's housewares aisle is losing sales. Shoppers say they now rent a pressure washer from a tool-rental app instead of buying one. Which force is this?

- **A.** Competitive rivalry, because another company is taking sales that used to belong to the grocer — Rivalry is pressure from firms selling the same product inside the same industry - another housewares retailer down the road. The rental app does not sell pressure washers at all, which is precisely what makes it something other than a rival.
- **B.** Threat of substitutes, because renting meets the same need without buying the product at all **— correct.** A substitute satisfies the same underlying need - a clean driveway - by a different route, so demand leaves the whole product category rather than shifting between sellers inside it.
- **C.** Threat of new entrants, because the rental app is new to the market — A new entrant is a company that starts selling what you sell. This would be the right answer if the app opened a store selling washers. Being young or newly launched is not the test; entering your industry is.
- **D.** Bargaining power of buyers, because shoppers are choosing where to spend their money — Buyer power is the pressure customers apply while still buying from your industry, by forcing your price down or your terms up. Here they are not negotiating with anyone in the industry; they left the category.

2. The practice grocer buys nearly all its produce from a small supplier pool. Suppliers raise prices, and store managers discover the change only when invoices arrive. Which force is strong?

- **A.** Bargaining power of buyers, because the grocer is the buyer in this relationship — Buyer power names the pressure a company's own customers exert on it, not the fact that the company happens to be doing the purchasing. Every force is named for whoever holds the leverage, and here the growers hold it.
- **B.** Bargaining power of suppliers, because alternatives are few and suppliers set terms the grocer discovers after the fact **— correct.** Supplier power is strong when suppliers are few, switching is slow or costly, and terms change unilaterally. All three appear here, and the invoice surprise also shows the grocer has no visibility into its own purchasing.

- **C.** Competitive rivalry, because other grocers are competing for the same produce — Rivalry would be the right label if other grocers were undercutting the practice grocer on the shelf. Competing for scarce supply is real, but the pressure described - price set by the supplier, learned late - sits on the supply side of the business.
- **D.** Threat of new entrants, because new growers could enter and bring prices back down — This is a correct description of new entrants applied to the wrong side of the business, and it names a possible relief rather than the current pressure. Easy entry by new growers would weaken supplier power; nothing here says that is happening.

3. You have concluded that buyer power is the strongest pressure on the practice grocer. Which framework do you turn to next, and for what reason?

- **A.** The five forces again, run location by location, because exposure to the warehouse club differs across the region — Re-running the forces per location does sharpen a diagnosis, and it is worth doing when the evidence is thin. It still cannot produce a project, because the five forces describe conditions outside the company and never name an activity inside it.
- **B.** The value chain, because it maps the activities inside the company and shows where a system could actually be built **— correct.** The five forces say what is squeezing the firm from outside; the value chain breaks the firm into nine internal activities, so an external finding becomes a specific place inside the business to build something.
- **C.** The value chain, because it ranks the five forces from strongest to weakest — The value chain is the right framework named for the wrong reason. It contains no forces and performs no ranking - it divides the firm into five primary and four support activities.
- **D.** Neither - move straight to comparing vendors, because the diagnosis is finished and the frameworks have nothing left to contribute — Naming a force says what is wrong outside the firm, not what to change inside it, so a shortlist assembled now is a list of products in search of a problem. This would have been right only if the diagnosis had already identified the activity the system must strengthen.

Porter’s Value Chain: where inside the company to act

The five forces describe the weather outside the building; they never say which department to change. For that you need a map of the company itself: the **value chain**, which cuts a firm into nine activities, each of which either adds value a customer will pay for or destroys it. Those nine fall into two groups.

> **Primary activities** — the five stages a product passes through on its way to the customer, so a change to one of them eventually reaches the buyer.

> **Support activities** — the four that touch no single product but make the others possible, so their payoff shows up spread across all five primary ones.

The table takes all nine in turn, with a system that adds value in each.

ACTIVITY	WHAT IT COVERS	A SYSTEM THAT ADDS VALUE HERE
Inbound logistics primary	Receiving, storing and moving arrived goods, before they are sellable.	A dock scanner counting a pallet as it is unloaded, so nobody reorders what arrived.
Operations primary	Turning inputs into what the customer buys — a stocked, priced shelf.	Register scans becoming restock tasks, so the milk case refills before it empties.
Outbound logistics primary	Getting the finished product out to the customer.	Pickup and delivery batched by neighborhood, in slots a driver can meet.
Marketing and sales primary	Making customers aware and giving them a reason to choose you.	A loyalty program targeting the item a household stopped buying.
Service primary	Keeping the customer whole after the sale — returns, complaints, repairs.	A returns screen showing purchase history, so a refund can be resolved quickly.
Firm infrastructure support	General management, finance, accounting, legal and planning.	Margin by store by day, so a losing location appears early rather than after a reporting cycle.
Human resource management support	Recruiting, hiring, training, scheduling and keeping people.	Scheduling built on forecast traffic, so the busiest hour is not the thinnest.
Technology development support	Improving the product and the processes themselves — research, design, systems built in house.	Building a differentiated app instead of relying only on one every rival can rent.
Procurement support	Finding, qualifying and contracting suppliers — the buying decision for the whole firm.	A portal where approved growers quote on one order, with expiry dates visible.

The distinction most often missed. Procurement is deciding whom to buy from and on what terms; **inbound logistics** is what happens to the goods after that decision, from truck to shelf. A quote-comparison portal is procurement and a dock scanner is inbound logistics, and confusing them puts your recommendation in the wrong department.

SORT **Which value chain activity does this system improve?**

Place each proposed system for the hypothetical grocer in the value chain activity it actually strengthens - four of the nine activities appear here.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

Inbound logistics — primary: receiving, storing and moving goods that have arrived, before they are sellable

- A dock scanner that records every pallet the moment it is unloaded, so on-hand counts are right before goods reach the shelf — The goods were bought weeks ago; this system handles what happens to them between the truck and the shelf, which is inbound logistics.
- Temperature sensors in the refrigerated warehouse that raise an alert before a pallet of dairy spoils — The dairy has arrived but is not yet sellable, so protecting it in storage is part of receiving and handling incoming goods.
- A cross-dock routing system that decides which store should receive each incoming case — Directing arrived goods to their destination inside the company is internal handling of inputs, so it is inbound logistics rather than delivery to a customer.

Operations — primary: turning inputs into the thing the customer buys - for a grocer, the stocked and priced shelf

- A restock list that turns each register scan into a task telling a clerk exactly which shelf to refill — For a grocer, a stocked, priced, shoppable shelf is the product itself, so producing it is operations.
- A batch planner in the in-store bakery that says how many loaves of each kind to bake before opening — Producing the item the customer buys is operations, whether the input is flour or a pallet of packaged goods.

Marketing and sales — primary: making customers aware and giving them a reason to choose you

- A loyalty app that notices a household stopped buying diapers and sends that household an offer — It creates awareness and a reason to choose the grocer for a specific customer, which is marketing and sales.

- A campaign tool that measures which of two weekly circular headlines pulled more shoppers into stores — It measures and improves how customers are attracted, which is squarely marketing and sales.
- A basket report showing that shoppers who buy charcoal also buy paper plates, used to plan next month's promotions — The analysis exists to shape offers and promotions, so it strengthens marketing and sales even though the data comes from the register.

Procurement — support: finding, qualifying and contracting suppliers - the buying decision itself

- A bidding portal where approved growers quote a price on the same order of tomatoes — Choosing whom to buy from and on what terms is procurement - a support activity performed for the whole company, not the physical receiving that follows it.
- A contract database that flags supplier agreements approaching expiry — Renewal timing is a supplier and terms decision, so it lives in procurement even though no goods move.

From a finding to a recommendation you can defend

Turning a finding into a proposal an executive will fund has four parts, and they have to arrive in this order.

1. **Name the force or the activity** exactly, because a recommendation that opens with technology has skipped the diagnosis.
2. **Give the evidence**, the observable fact behind the finding — sales falling only at locations exposed to a new competitor — because evidence is what stops the meeting being two opinions.
3. **State the initiative** concretely: not “improve our digital capabilities” but “one chain-wide inventory record updated at the dock and the register.”
4. **Commit to a measurable outcome**, the number that will move and the date you check it, because a system with no defined result can only be defended, never judged.

Say it in one sentence. Because [force or activity] is [strong or weak], shown by [evidence], we should build [initiative], which we will judge by [measure] at [date]. If you cannot fill every slot, you do not have a recommendation yet — you have a preference.

DECIDE **Advising a regional grocer**

Work the four decisions in this hypothetical situation; after each one, compare every outcome before continuing.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

This is a hypothetical practice case. A regional grocer's sales are slipping, leadership wants to spend on technology, and nobody has yet said what problem the spending is meant to solve. You are the analyst in the room.

Decision 1. A members-only warehouse club enters some local markets. Sales fall only at the grocer's exposed locations. Leadership asks you to name the pressure before anyone names a product.

- Threat of new entrants - a company that was not competing here is now selling the same goods to the same shoppers **— the stronger call.** Right, and the hypothetical evidence is unusually clean: only locations exposed to the club declined, which ties the loss to the entrant rather than to a region-wide cause. Naming the force this precisely lets you reject any initiative that would not touch it.
- Threat of substitutes - shoppers have found a different way to meet the same need — A substitute meets the need without buying from your industry at all, such as a meal kit or a restaurant. The club sells the same groceries from the same suppliers, so it competes inside the industry, not outside it. Hold that thought - a real substitute is in this region too.
- Competitive rivalry - a competitor is taking share, which is what rivalry means — Understandable, and it is not far off, but rivalry is pressure from the firms already established in the industry. The distinguishing fact here is that this competitor was not in the market before, which is exactly what makes it an entrant. The difference matters: rivalry pushes you to defend, an entrant tells you your barriers to entry are low.

Decision 2. You have named the force. Now the response. The grocer knows little about who its shoppers are: no membership, no history, no way to distinguish a regular household from an occasional visitor.

- Match the club's advertised prices across every store — This starts a price war against an opponent with better buying power and lower overhead. Margin falls chain-wide, including at locations that were never threatened, and the club can outlast you. Price is the dimension where the entrant is strongest.
- Launch a loyalty program that records what each household buys, and use that history to make personalized offers **— the stronger call.** This raises the customer's switching cost and builds an asset the entrant cannot instantly buy: accumulated purchase history. It also fixes the blindness by helping the grocer recognize its regular customers.

- Expand the weekly circular's print run so more households see the grocer's specials — This creates more impressions of the same undifferentiated message. It costs money, reaches the club's members too, and produces no customer history, so the grocer still cannot say which households responded.

Decision 3. Separately, produce suppliers raise prices, and managers learn of it only from invoices. Purchasing happens through separate calls from individual stores.

- Sign a long-term exclusive contract with the cheapest supplier to lock in the price — This lowers near-term cost and raises supplier power later. Under exclusivity, the grocer has no alternate, no comparison price, and little leverage at renewal - the same trap the chapter describes when few suppliers can meet demand on short notice.
- Instruct store managers to cut produce orders until prices come down — Empty produce cases drive shoppers to the store that has tomatoes. The invoice falls only because the grocer reduced the reason to visit.
- Put purchasing on one system that shows every store's prices and volumes, and qualify additional growers **— the stronger call.** Now the grocer sees its own buying for the first time, can compare quotes side by side, and has credible alternates - which is what moves a negotiation. This is a procurement fix, not an inbound logistics fix, and saying so keeps the project in the right department.

Decision 4. The board will fund one initiative. It asks for a concise defense of the loyalty program.

- 'It will modernize us. Customers expect a digital experience from a grocer in this decade.' — No force, no evidence, no measure. Everything in that sentence would be equally true if the initiative were a new website, a mobile game, or nothing at all - which is why a board cannot tell whether it worked.
- 'Buyer power is high - switching costs are near zero and we cannot identify repeat customers. A loyalty program gives us purchase history for differentiated offers. We will judge it on repeat-visit rate and share of basket at affected stores at the scheduled review.' **— the stronger call.** That is the full shape: force, evidence, initiative, measurable outcome with a review point. It can be argued with, which is the point - a proposal that cannot be argued with also cannot be evaluated. Add one caution out loud: any advantage from this is fleeting, because rivals can eventually do the same thing, so the plan must include what gets improved after launch.
- 'Two of our competitors already run loyalty programs, so we are behind and need to catch up.' — This justifies parity, never advantage, and it hands your technology roadmap to competitors. It also tells the board nothing about why the program would work here or how anyone would know if it did.

Two cautions the chapter insists on. Both limit what any system, however well built, can do for you:

> **The business model comes first** — companies with bad business models tend to fail regardless of whether they use information technology, while companies with good business models that use IT successfully to carry them out tend to be very successful, because the system amplifies the idea rather than replacing it.

> **The advantage can be fleeting** — competitive advantage from the use of information systems fades because competitors can eventually do the same thing, which is why the goal is gain *or sustain*, and why FedEx keeps rebuilding systems it already leads with.

CHECK YOURSELF

Right framework, right place

Three situations that test whether you can place a system in the value chain and defend what it will do.

Lesson and answer summary — the interactive version of this activity loads when scripting is available.

1. The practice grocer wants a system that tells every store manager what is on the next truck and when it will dock. Which value chain activity does it improve?

- **A.** Procurement, because the system is about goods the grocer purchased — Procurement is choosing suppliers and agreeing on price and terms - the purchase decision. Those goods were procured weeks ago; what happens to them from the truck onward is a separate activity with separate systems.
- **B.** Inbound logistics, because it covers receiving and handling goods on their way to the shelf **— correct.** Inbound logistics covers everything that happens to incoming goods before they are sellable, and knowing what is arriving and when is exactly that problem.
- **C.** Operations, because store staff will be using it during the working day — Operations is transforming inputs into what the customer buys - the stocked, priced shelf. Who touches a system and when they touch it does not determine its activity; the activity it improves does.
- **D.** Outbound logistics, because it moves product closer to the customer — Outbound logistics is getting the finished product out to the customer, such as batching pickup orders or routing deliveries. This system handles the trip from supplier in, which is the opposite direction.

2. An applicant-tracking and shift-scheduling system substantially reduces cashier turnover at the practice grocer. Where does it sit in the value chain?

- **A.** Operations, because cashiers do their work on the sales floor — Operations is producing what is sold. Where employees physically stand is not the test - recruiting, scheduling and retaining people is its own activity that supports operations rather than being part of it.
- **B.** Service, because cashiers are the staff who deal directly with customers — Service is what keeps the customer whole after the sale - returns, complaints, repairs. Cashiers do face customers, but a system that recruits and schedules staff acts on the workforce, not on the post-sale experience.
- **C.** Human resource management, a support activity that makes the primary activities possible **— correct.** Human resource management covers recruiting, hiring, training, scheduling and retention, and it is classed as support because every primary activity depends on the right people being there.
- **D.** Firm infrastructure, because hiring and payroll are run centrally out of headquarters — Firm infrastructure is general management, finance, accounting, legal and planning. Being run from headquarters is not the test: the value chain gives people decisions their own support activity, and this would be firm infrastructure only if the system managed the company's finances or planning.

3. The practice grocer's board is told a new loyalty system will give the chain a permanent advantage because no rival has one. Given this chapter, what is the most accurate response?

- **A.** The claim holds, because a system built to gain or sustain advantage keeps producing that advantage for as long as it runs — Sustaining is work, not a property the system acquires at launch: FedEx sustains its advantage by reengineering and improving performance on average twice a year. This would have been right if the chapter said advantage persists once built, and instead it says advantage can be fleeting.
- **B.** Advantage from information systems can be fleeting, because competitors can eventually do the same thing **— correct.** This is the chapter's own caution, and it explains the phrase gain or sustain: the advantage lives in continuing to improve the system after launch, not in the launch itself.
- **C.** The claim fails because information systems are now a commodity, so IT spending belongs on cost reduction and risk mitigation — This is the commodity argument the chapter states and then rejects, pointing to rapidly changing IT and to companies from Amazon to Zoom that built advantage by combining commoditized technologies with proprietary systems and business processes. It is a real position, but it is not this chapter's conclusion.
- **D.** The claim fails because loyalty software is sold off the shelf, so a rival can license the same package next quarter — The software is buyable, which is true and is why the launch by itself is not the advantage. It still misses the chapter's point: advantage comes from combining commoditized technology with proprietary systems, processes and data, so accumulated purchase history is the part a rival cannot license.

REFERENCE

Chapter vocabulary

Every term the chapter defines, in one place, written so it makes sense on its own. Search by word, or filter by the objective the term belongs to. Use this while you read rather than after — most of the confusion in this material comes from two terms that sound alike meaning genuinely different things.

3D printing

1.4

A way of making a solid physical object by adding thin layers of material one on top of another from a digital model, instead of cutting the shape out of a block. Because it adds material rather than carving it away, it wastes far less and can reproduce almost any design cheaply.

For example: Printing a replacement plastic clip for a broken pair of headphones instead of buying the whole headset again.

APIs (application programming interfaces)

1.1

Standard connection points that let one company's software use another company's data or features over common web protocols, without needing intimate knowledge of how that software works inside. The chapter compares an API to a power socket: you plug in and get electricity without knowing how the utility produces or delivers it.

For example: Lyft shows you a live map inside its own app by plugging into Google Maps rather than building a mapping system from scratch.

apps

1.1

Software programs written to perform one particular, well-defined function, usually on a phone or tablet. Companies now build them because customers reach for a phone far more often than a laptop.

For example: Opening Venmo for fifteen seconds just to send a friend your half of the pizza, then closing it.

artificial intelligence

1.1

Using information technology to imitate the things human intelligence does, such as recognizing patterns, understanding language, and making judgments. It is what lets software make sense of enormous streams of data no person could read.

For example: Asking ChatGPT or Siri a question in plain English and getting an answer written the way a person would write it.

Big Data

1.1

Datasets so enormous and so tangled that ordinary tools struggle with them. Three things make them hard: volume, meaning sheer size; variety, meaning many different types of data mixed together; and velocity, meaning the data arrive and must be analyzed faster and faster.

For example: Every skip, replay, and playlist add from hundreds of millions of Spotify listeners piling up every second of the day.

BYOD

1.1

Short for bring your own device: the workplace trend of employees using their personal phones and laptops for work instead of equipment the company issues. Workers like it and it can raise productivity, but it makes security, compliance, and tech support much harder for the employer.

For example: Reading your work schedule and answering your manager's messages on the same phone you use for TikTok.

climate change

1.1

Large-scale, long-term shifts in temperatures and weather patterns, both regionally and worldwide. Its effects — rising seas, more severe storms, changed weather — create problems for individuals, businesses, and whole societies.

For example: A coastal town rebuilding after a severe hurricane far more often than it did a generation ago.

competitive advantage

1.3

Whatever lets a company beat the rival firms selling to the same customers — doing something better, faster, cheaper, or in a way nobody else can copy yet. Information systems can create it, but the chapter warns it is often fleeting because competitors eventually do the same thing.

For example: FedEx delivers a quarter of the packages it handles each day within one business day, helped by sorting hubs that decide in a few hundred milliseconds where to divert each scanned package.

computer ethics

1.4

The standards of right and wrong conduct that apply specifically to how people use computers and information systems. It covers questions such as who may read whose files, who owns data about you, and what counts as stealing when nothing physical is taken.

For example: Deciding not to log in with a classmate's credentials to reach a research database your account does not cover.

computer fluency

1.1

Being able to teach yourself new technologies as they appear and judge what they will do to your work and your life. It goes past knowing today's tools, because the tools keep changing.

For example: Picking up a new scheduling app your employer rolls out in one afternoon on your own, and noticing which parts of your shift it will change.

computer literacy

1.1

Knowing how to operate a computer and use common applications. The chapter says it can be the difference between being employed and being unemployed, but that it is no longer enough on its own.

For example: Being able to build a spreadsheet, attach it to an email, and find the file again next week.

consumerization of IT

1.1

The pattern where new technologies show up first in products sold to ordinary shoppers and only later get adopted inside companies. It forces businesses to keep watching the consumer market to see what will change how they operate.

For example: Tablets were built and sold as consumer gadgets first and only later became standard equipment in warehouses, showrooms, airplane cockpits, and hospitals.

data

1.2

Raw symbols such as characters and numbers, with no meaning attached to them. They stay nearly worthless until something processes them and gives them a context.

For example: The string 465889727 sitting by itself tells you nothing at all — you cannot say what it is.

data privacy statements

1.4

The posted notice on a website that says what data are gathered, what they are used for, who gets access, whether giving them is required or voluntary, and how confidentiality will be handled. It tells you what a company intends to do; it does not by itself protect you.

For example: The privacy policy link at the bottom of a checkout page that almost nobody opens before paying.

data quality

1.2

How trustworthy data are for making a decision, judged on five things: completeness, accuracy, timeliness, validity, and consistency. Bad inputs produce bad decisions no matter how good the system is.

For example: A mailing list holding an old address and a duplicate record sends your acceptance letter to an apartment you moved out of last year.

demographic changes

1.1

Shifts in the makeup of a population — how old people are, how many babies are born, and who moves where. Developed countries are aging fast while regions such as Africa are growing quickly, which changes where workers and customers will be.

For example: Your hometown high school closing a wing because far fewer children are being born there than in your parents' day.

digital density
1.1

The amount of connected data produced by each unit of activity. As it climbs, ordinary actions throw off more data that can be linked together, which is what makes new services and new business models possible.

For example: A run used to produce nothing; now a single run produces pace, route, heart rate, elevation, and a sleep score the moment your watch syncs.

digital divide
1.1

The gap between people who can get to information systems and people who cannot, where the first group holds a large advantage over the second. It falls hardest on rural communities, older people, people with disabilities, minorities, and developing countries that lack affordable internet access.

For example: A student who can only get online at the public library falls behind a classmate with home broadband and a laptop.

globalization
1.1

The knitting together of economies all over the world, made possible by innovation and technological progress. You see it in the way goods, money, information, and workers now move across borders.

For example: Your phone was designed in California, assembled in China, and supported by a call center on a third continent.

hardware
1.2

The physical equipment of an information system — the machines and parts you could actually drop. Today that includes not just computers and printers but sensors, cameras, and other input and output devices.

For example: The laptop in front of you, plus its screen and keyboard.

healthcare IS
1.1

Information systems built to support the work of medicine, from diagnosing and treating patients to analyzing patient and disease data to running the business side of a clinic or hospital.

For example: The patient portal where your doctor's office posts your lab results and lets you book the follow-up visit.

home automation
1.1

Monitoring and controlling a house's lighting, heating, and appliances from a distance instead of by hand at a switch or a dial. The chapter uses this as another name for smart home technology.

For example: Turning the porch light on from your phone while you are still on the bus home.

Industrial Internet of Things (IIoT)

1.1

The use of internet-connected devices inside manufacturing, which brings the information technology side and the factory-floor operations side together. It can improve efficiency, product quality, and flexibility, letting companies mass-produce customized products and watch their supply chains closely.

For example: Sensors on a plant's machines reporting wear so a part gets swapped out before the whole line stops.

information

1.2

Data that have been formatted, organized, or processed until they mean something. Information represents a piece of reality and answers questions about who, what, where, and when.

For example: The same digits shown as 465-88-9727 in a field labeled SSN inside John Doe's record — now the number tells you something.

information privacy

1.4

The question of how much personal detail you should have to hand over to others — an employer, a store, a website — through your job or through transactions such as online shopping, and what they may do with it afterward. It gets harder as organizations combine scattered pieces into an increasingly complete picture of you.

For example: A shopping site greeting you by name and showing you the exact jacket you looked at on a different site yesterday.

information system (IS)

1.2

The combination of people and information technology that creates, collects, processes, stores, and distributes useful data. The people are part of the definition, not an afterthought, because the technology exists to serve someone.

For example: Your school's registration system: the staff who run it, the software, the network, and the class-schedule data it hands back to you.

information technology (IT)

1.2

The technology portion of an information system: hardware, software, and telecommunications networks. It leaves out the people, although the chapter notes the gap between the terms IS and IT has been shrinking and many people now use them interchangeably.

For example: The servers, the app, and the campus Wi-Fi that let you check a grade — but not the registrar who maintains the records.

intellectual property (IP)

1.4

Creations of the mind that have commercial value, such as songs, software, writing, photographs, and product designs. Digital copying makes it easy to take and share these without permission, attribution, or payment to whoever made them.

For example: Copying a friend's entire music library with no loss in quality and passing it along to other friends.

Internet of Things (IoT)

1.1

A network of everyday physical objects that can share data over the internet on their own. Anything that generates or uses data can be connected, monitored, or controlled remotely.

For example: A car tire with a pressure sensor that reports itself as low, and a home meter that reports energy use back to the utility.

internetworking

1.2

Linking host computers and their separate networks together so they form one much larger network. The internet itself is the result of doing this on a global scale.

For example: Your dorm network joins the campus network, which joins your provider's network, which is how your message reaches a friend overseas.

knowledge

1.2

The ability to understand information, form an opinion about it, and make a decision or prediction from it. It also covers the rules and guidelines you apply to organize or manipulate data for a particular task.

For example: Knowing that one Social Security number identifies exactly one person, so two records sharing one must be an error.

knowledge society

1.1

A society in which what people know counts for as much as land, labor, or money once did, which makes education the thing that determines how far people can go. Peter Drucker predicted it would emerge as knowledge workers grew in number and importance.

For example: A graduate being considered for a job a classmate without a degree cannot even apply for, in an economy where median earnings for workers 25 and over run US$67,256 with a bachelor's degree against US$39,428 with only a high school diploma.

knowledge worker

1.1

A professional, usually well educated, whose core job is creating, changing, or pulling together knowledge rather than making or moving physical things. Peter Drucker coined the term in 1959 and predicted these workers would be paid better and hold more bargaining power than earlier agricultural and industrial workers.

For example: A financial analyst who spends the day reading reports and turning them into one investment recommendation.

management information systems

1.2

Systems that take the record of daily transactions and turn it into detailed reports that help someone manage a firm or one part of it. The same phrase is also used as a name for the academic field that studies information systems.

For example: The weekly report telling a store manager which sizes sold out so she knows what to reorder.

micro-moments

1.1

The short bursts when a person reaches almost instinctively for a mobile device to get one thing done — to buy something, know something, do something, or go somewhere. Customer contact now happens in these bursts rather than in long sessions at a desk.

For example: Pulling out your phone in the middle of an argument to settle who won a game in 2019.

network effect

1.1

The idea that a network, or a tool built on one, becomes more valuable as more people use it. It explains why the leading platform in a category is so hard to unseat.

For example: A messaging app is useless when none of your friends are on it and nearly impossible to quit once all of them are.

office automation systems

1.2

Software that supports the routine day-to-day work of individuals and small groups, sometimes called personal productivity software. Word processors, spreadsheets, presentation tools, and email clients all belong here.

For example: Writing a paper, building a budget in a spreadsheet, and emailing both to someone, all from the same suite of programs.

opt in

1.4

Signaling that you agree to have your data collected or used further, for example by checking a box. Nothing is supposed to happen until you actively say yes.

For example: Ticking the box that says yes, send me deal emails, while ordering a pair of sneakers.

opt out

1.4

Signaling that your data may not be collected or used in some further way. Here the collection is the default and the burden is on you to say no.

For example: Clicking unsubscribe at the bottom of a marketing email, or switching off ad personalization in your account settings.

outsourcing

1.1

Handing a business process or task — accounting, manufacturing, security, customer support — to another company or another country. Firms do it to cut costs, reach skills they lack, move faster, and stay focused on what they do best.

For example: A U.S. hospital sending X-rays overseas to be read by radiologists while doctors at home are asleep.

quantified self

1.1

Logging every part of your own daily life — activity, performance, what you eat, mood, heart rate — in order to improve your health and performance. Activity trackers such as the Fitbit, worn and passively used on a regular basis, are what make that logging practical.

For example: Wearing a Fitbit all day and checking your steps, resting heart rate, and sleep score every morning.

resource scarcity

1.1

The problem that fossil fuels and other natural resources are limited while population growth, global trade, and consumption keep pushing demand higher. Humans already consume beyond what the planet can supply.

For example: The lithium inside your phone battery gets harder and more expensive to source as every new device needs some.

robotics

1.1

Putting programmable machines to work on manual, physical tasks that a person would otherwise do by hand. Continuous input from sensors, paired with machine learning and artificial intelligence to make sense of that input, is what has driven the fastest advances here.

For example: Autonomous Caterpillar mining trucks hauling loads with nobody in the cab.

sensors

1.1

Devices that detect, record, and report changes in the physical world around them. Cheap sensors paired with wireless radios are the reason so many ordinary objects can now feed data to the internet.

For example: A strip in a road surface that reads the temperature and triggers a lower speed limit when ice becomes likely.

shifts in economic power

1.1

Long-term changes in which countries hold purchasing power and control over natural resources. Established economies are losing their dominant positions, which creates political struggles that have to be worked out.

For example: Companies that once wrote off developing countries now sell into them, because a fast-growing middle class there holds purchasing power it did not have a generation ago.

smart home technologies

1.1

Technology that lets someone monitor and control a home’s lighting, heating, or appliances from a distance. The market for it is expected to top US$250 billion by 2029.

For example: A Nest thermostat that learns your schedule and lets the house cool down while nobody is in it.

software 1.2

A program, or a set of programs, that tells the computer hardware what tasks to carry out. Without it the hardware sits there doing nothing useful.

For example: The operating system running your laptop and the browser you open on top of it.

strategic 1.3

Describes a choice made with the intent of helping a company gain or sustain an advantage over its rivals. The chapter contrasts this with using technology only to run existing work more efficiently or to earn a return on the investment.

For example: FedEx reengineering and improving its package-handling systems about twice a year so its delivery promise stays ahead of competitors.

sustainable development 1.1

Development that meets the needs of people now without wrecking the ability of future generations to meet their own needs. It is the standard the chapter applies to growth in a world of limited resources.

For example: A company signing long-term wind and solar contracts so its data centers can keep growing without burning more fossil fuel.

systems integration 1.2

Connecting separate, often modular information systems and their data together, frequently using APIs, so business processes run better and decisions get made on fuller information. It is why few real systems fit neatly into just one category anymore.

For example: A store's checkout feeding the inventory system automatically, so shelves get restocked without anyone retyping numbers.

telecommunications networks 1.2

Two or more computer systems linked together by communications equipment so they can share data and services. They are what make global collaboration, communication, and commerce possible.

For example: The campus Wi-Fi that lets your laptop reach the library servers and then the wider internet.

transaction processing systems (TPS) 1.2

Systems that handle the flood of routine day-to-day business events at the operational level of an organization, such as sales and registrations. They also generate huge amounts of data that other systems later analyze.

For example: The grocery checkout register that scans bar codes and prints coupons on the back of your receipt.

urbanization 1.1

The movement of people out of rural areas and into cities. Roughly 50 percent of the world's population now lives in cities, and keeping those cities livable as they grow is a serious challenge.

For example: Someone leaving a farming town for a city apartment because that is where the jobs are.

wearable technologies 1.1

Clothing or accessories with electronics built into them, such as a smartwatch. Their sensors record body data like movement and heart rate as well as surrounding conditions like light, orientation, and altitude.

For example: An Apple Watch tracking your heart rate during a workout and buzzing when a text lands on your phone.

PUT IT TOGETHER

Final challenge

Twenty-five situations rather than twenty-five definitions. Each one presents a documented chapter case or a clearly hypothetical organizational situation and asks what is actually going on, which is the form the material takes once you are the person being asked. Answer all of them, then score: the result breaks out by chapter objective and supplement, so if one area is weak you will see exactly which section to reread instead of a single number that hides it.

FINAL CHALLENGE

Final challenge

Answer key — the interactive, self-scoring version loads when scripting is available.

1. 1.1 A county buries temperature sensors in a stretch of highway. The readings travel over the network and automatically lower the posted speed limit when ice is likely. Road use that generated no connected record now generates a continuous stream that other systems act on. Which idea does this best illustrate?

- **A.** Big Data, because the sensors produce a high volume of readings at high velocity from many different points — Big Data names the datasets themselves: extremely large and complex, high in volume, variety, and velocity. These readings may well pile up into Big Data, and that would be the right answer if you were asked to classify the stored result rather than the change in the road.
- **B.** Increasing **digital density**, because the road now generates connected data per unit of activity **— correct.** Digital density is the amount of connected data per unit of activity. Traveling this road used to leave no connected reading; now it feeds a speed-limit decision, which is exactly the new value-added interaction that rising digital density enables.

- **C.** The network effect, because each sensor added to the road makes the system more valuable to every driver using it — The network effect is the notion that a network, or a tool or application based on one, increases in value with the number of other users. It would fit if drivers made the service better for each other; here the readings are collected whether one car passes or a thousand.
- **D.** The consumerization of IT, because sensor technology reached ordinary consumers before it reached government agencies — Consumerization of IT is the pattern of technologies appearing in the consumer marketplace first and being pulled into organizations later, as smartphones were. That would be the answer if the county had adopted a device its residents already had at home.

2. 1.1 A kitchen and bath retailer opens operations in three countries at once. In the first, supplier contracts are hard to enforce because commercial law works differently. In the second, warehouse broadband is unreliable and few local workers have logistics training. In the third, the product photography offends local norms. Which of these is a **geoeconomic** challenge as the chapter uses the term?

- **A.** The contracts that are hard to enforce, because a country's laws and regulations are part of how its economy functions — That is a governmental challenge: differences in political systems, regulatory environments, laws, standards, or individual freedoms. It would be the right answer to a question asking which obstacle comes from how a country governs commerce.
- **B.** The photography that offends local norms, because what customers are willing to accept is economic behavior — That is a cultural challenge: differences in languages, beliefs, attitudes, religions, or life focus, and different viewpoints about intellectual property. Choose it when the obstacle is what people find acceptable rather than what a country can support or permit.
- **C.** None of the three, because geoeconomic challenges are about currency exchange rates and tariffs on imported goods — Currency and tariffs are genuine trade obstacles, but the chapter defines geoeconomic challenges more broadly as differences in infrastructure, demographics, welfare, or workers' expertise. Reading the term as a synonym for exchange rates is the usual misstep.
- **D.** The unreliable broadband and the shortage of trained logistics workers, because that is infrastructure and expertise **— correct.** Geoeconomic challenges are differences in infrastructure, demographics, welfare, or workers' expertise. Unreliable broadband is infrastructure and a shortage of trained logistics staff is expertise, so both belong here rather than under law or culture.

3. 1.1 A rural clinic wins funding for a telehealth program, but enrollment stalls. Its own survey finds that the patients who would benefit most — older residents, people with disabilities, and households on unpaved roads with no broadband — are the least able to join. How does the chapter frame what the clinic has run into?

- **A.** The **digital divide**, because the groups named lag national averages for internet access and computer literacy **— correct.** The digital divide is the gap between those with access to information systems and those without, and the chapter names rural communities, the elderly, people with disabilities, and minorities as the groups still behind on internet access and computer literacy. It calls this an ethical challenge because computer literacy is tied to a person's ability to compete.
- **B. Resource scarcity**, because the clinic lacks the funding it would need to reach every household in its service area — Resource scarcity in this chapter means the limited availability of fossil fuels and other natural resources, a societal megatrend rather than one organization's budget. It would fit a question about pressure on the planet's finite supply.
- **C.** The **knowledge society**, because the program assumes patients have the education that knowledge work depends on — The knowledge society is Drucker's prediction that possessing knowledge would matter as much as land, labor, or capital, with education as its cornerstone. That explains why access matters so much; the thing physically blocking these patients is the access itself.
- **D. Demographic changes**, because the population this clinic serves is aging faster than the country as a whole — Demographic changes are shifts in the structure of populations related to age, birth rates, and migration. Age does appear in the scenario, but the barrier described is connectivity and skills, which the chapter names separately.

4. 1.1 A neighborhood tool-lending app is of little use while membership is small because few requested tools are listed nearby. As membership grows, requests are much easier to fill, and new members say they joined because so many others already had. What is happening?

- **A.** Increasing digital density, because every loan now generates connected data the app can act on — Digital density, the amount of connected data per unit of activity, is genuinely rising here as well. It explains what each loan leaves behind, not why the member who joined last is better off than the member who joined first.
- **B.** A micro-moment, because a member instinctively picks up a mobile device to accomplish a goal — A micro-moment is that instinctive reach for a mobile device to buy something, know something, do something, or go somewhere, and this app is used in exactly those moments. It describes one moment of use, not why the service as a whole got more useful as it grew.
- **C.** The **network effect**, because the value of the application increases with the number of other users **— correct.** The network effect is the notion that the value of a network, or of a tool or application based on a network, increases with the number of other users. That is precisely what the members describe, and it is the mechanism behind business models such as Uber and Airbnb.
- **D.** Consumerization of IT, because a tool built for consumers is now being used by contractors for paid work — Consumerization of IT means innovations are introduced in the consumer marketplace first and then adopted by organizations. It would be the answer if the story were about a company adopting the app because its staff already used it privately.

5. 1.1 An insurance firm never issued phones to its adjusters. Staff began checking work email on their own phones, and now they open customer relationship management records on them in the field. The IT director's memo lists security and compliance worries plus the cost of supporting many different devices, and in the same paragraph lists higher productivity, better retention of good adjusters, and happier customers. What is being described?

- **A. BYOD**, which the chapter calls a major concern of managers precisely because it carries real risks and real opportunities **— correct.** Bring your own device is the use of personal devices and applications for work-related purposes. The chapter frames it exactly as this memo does — worrying for security, compliance, and support, yet capable of raising productivity, retention, and customer satisfaction — and calls managing it a major concern of business and IT managers alike.
- **B.** Outsourcing, because work the firm used to equip and control is now being done on hardware the firm does not own — Outsourcing is the moving of business processes or tasks, such as accounting, manufacturing, or security, to another company or another country. These adjusters are still employees doing their own work; only the hardware changed hands.
- **C.** The **digital divide**, because adjusters with newer personal phones can do more of the job than adjusters with older ones — The digital divide is the gap between people with access to information systems and those without, which the chapter discusses at the level of communities and countries. Uneven phone models inside one staff is a support problem rather than that divide.
- **D. Systems integration**, because personal phones are being connected to the firm's customer relationship management system — Systems integration is connecting separate and often modular information systems and data, using technologies such as APIs, to improve business processes and decision making. Integration work may follow from this situation, but it does not name the trend the memo is about.

6. 1.2 A warehouse scanner emits an unlabeled identifier. A screen places it in a row labeled **SKU**, beside its aisle and a low on-hand quantity. The shift lead orders more because experience says that quantity will not meet weekend demand. Which part of that sequence is **knowledge**?

- **A.** The unlabeled identifier, because it is what the system actually captured and everything else is derived from it — That is data: raw symbols such as characters and numbers, with no meaning in and of themselves and little value until processed. Asked what the scanner produced, this would be the right answer.
- **B.** The labeled row showing SKU, aisle, and quantity, because a person can now read it and understand what it refers to — That is information: data formatted and organized so that it answers who, what, where, and when. The contextual cues, meaning the labels, are what turn the raw string into something readable, but reading it is not yet deciding anything.
- **C.** The scanner and the network that carried the reading, because without them none of the rest could happen — That is the technology component of the information system, the hardware and telecommunications networks. The chapter treats those as what carries and processes data, not as a form of understanding.

- **D.** The lead’s rule that the displayed quantity will not meet expected demand, because it lets her decide **— correct.** Knowledge is the ability to understand information, form opinions, and make decisions or predictions from it — a body of governing procedures, guidelines, or rules used to organize data for a task. The reorder rule lives in the lead, and without it the row on the screen would just sit there.

7. 1.2 A distributor spends heavily on servers, a database, a network, and licences. After implementation nobody has been trained, no report reaches a decision maker, and the sales team still keeps its own spreadsheets. A consultant tells the owner: you bought information technology, but you do not yet have an information system. On the chapter’s definition, what makes that a fair statement?

- **A.** The equipment is under-powered for the transaction volume the distributor runs, so it cannot process the data quickly enough — Capacity problems are real and would explain slow or failing reports. Nothing in the scenario says the machines cannot keep up; it says nobody uses what they produce.
- **B.** An information system is people plus technology producing **useful** data, and both the people and the usefulness are missing **— correct.** The definition has two halves. Information technology — hardware, software, and telecommunications networks — is one of them; the other is people, and the data have to be useful for someone, somewhere. Technology that produces no useful data for anyone has not become a system.
- **C.** The statement is unfair, because the chapter says many people use IS and IT synonymously, so buying one is buying the other — The chapter does note that the difference is shrinking and that many use the terms interchangeably. It also keeps the distinction that matters here: technologies do not exist for their own sake, they are created to serve a useful purpose for people.
- **D.** The distributor has hardware and software but no telecommunications network, the third component of information technology — A network is listed among the purchases, so this fails on the facts of the scenario. It would be the right objection if the pieces could not share data at all, since telecommunications networks are what let computers share data and services.

8. 1.2 A pharmacy chain wants every sale recorded at the register the moment an item is scanned, so the day’s takings reconcile at close and the shelf count drops as the goods leave the store. Which class of system is doing that work?

- **A.** A management information system, because it produces the detailed information a manager needs to run the stockroom — A management information system produces detailed information to help manage a firm or part of a firm, and inventory management and planning is the chapter’s own example of one. It normally runs on data the transaction system captured first, which is the step this question describes.
- **B.** A decision support system, because the chain will use the numbers to decide how much of each item to order next — A decision support system provides analysis tools and access to databases to support quantitative decision making, such as product demand forecasting. Deciding how much to reorder would use one; recording the scan that sold the item would not.

- **C.** A **transaction processing system**, because it processes day-to-day business event data at the operational level **— correct.** The chapter lists a grocery store checkout cash register connected to a network as a sample application of a transaction processing system. The defining trait is capturing day-to-day business events at the operational level as they occur.
- **D.** A business intelligence system, because it analyzes sales data to reveal patterns in what customers buy — A business intelligence system analyzes Big Data to better understand aspects of a business, using tools such as online analytical processing and data visualization. That is what gets built on top of years of sales records, not the record keeping itself.

9. 1.2 A regional manager wants to see what next quarter's demand would look like under three different pricing plans. The tool she uses pulls history out of several databases, lets her change the assumptions, and models each plan so she can compare outcomes side by side. Which category fits best?

- **A.** A **decision support system**, because it provides analysis tools and database access to support quantitative decision making **— correct.** This matches the chapter's definition line for line, and product demand forecasting is its own sample application. The giveaway is a person testing alternatives quantitatively rather than recording events or reading a standard report.
- **B.** A transaction processing system, because the historical sales it draws on were captured one transaction at a time — A transaction processing system processes day-to-day business event data at the operational level, and it is indeed the source of the history she is modeling. That is the chapter's point that transaction data feed a broad range of other systems, including this one.
- **C.** An office automation system, because she is working in a spreadsheet-style environment to produce her own analysis — Office automation systems — word processors, spreadsheets, presentation software, email clients — support a wide range of predefined day-to-day work activities of individuals and small groups. The interface may look similar, but the purpose here is modeling a decision, not producing a document.
- **D.** An enterprise resource planning system, because drawing on several databases means the system spans the whole organization — Enterprise resource planning supports and integrates all facets of the business, including planning, manufacturing, sales, and marketing. Reading from more than one database is not the same as integrating the business, which is a far larger claim.

10. 1.2 Two finalists for a systems analyst job. One has deeper hands-on expertise with the company's hardware and network than anyone currently on staff. The other knows enough about hardware, software, and networking to judge what each can do for the firm, understands how the order-to-cash process actually works, and has run projects to a deadline. Which candidate matches what the chapter says makes IS personnel valuable?

- **A.** The first, because technical competency is the hardest area to maintain given the pace of innovation, so depth there is scarcest — The chapter does say the technical area is perhaps the most difficult to maintain because of the rapid pace of technological innovation. It draws the opposite conclusion from that fact: the IS professional must know just enough about these areas to understand how they work and what they can do for an organization.

- **B.** Either one, because the chapter treats technical, business, and systems knowledge as interchangeable routes into the same career — The chapter treats the three domains as things one person integrates rather than as alternatives to choose among. It says directly that business and systems competency are what set the IS professional apart from someone with only technical knowledge and skills.
- **C.** The first, because business and project skills can be picked up on the job while technical depth has to be hired in — This inverts the chapter's argument. Business competency is the area it says may save a person's job in an era of increased outsourcing, and business skills are what propel IS professionals into project management and higher-paying management positions.
- **D.** The second, because good IS personnel integrate technical, business, and systems knowledge and direct the deeper specialists **— correct.** The core competency table covers technical, business, and systems domains, and the chapter states that the IS professional need not be a technical expert — typically that person manages or directs those who have deeper, more detailed technical knowledge.

11. 1.2 A manufacturer wants one system in which accepting a customer order immediately reserves the inventory, schedules the production run, posts the entries to the general ledger, and releases a shipment, so that sales, manufacturing, finance, and planning all work from the same record. Which category is that?

- **A.** A supply chain management system, because the whole point is coordinating what suppliers deliver and when they deliver it — Supply chain management supports the coordination of suppliers, product or service production, and distribution, with procurement planning as the chapter's example. It covers part of this picture and would be right if the question stopped at inbound supply, but it does not reach the general ledger or the sales side.
- **B.** An **enterprise resource planning system**, because it supports and integrates all facets of the business, planning through sales **— correct.** Enterprise resource planning is defined by exactly this breadth: financial, operations, and human resource management in one system spanning planning through sales. The chapter groups it with customer relationship management and supply chain management as systems that deliberately cannot be filed under one older category.
- **C.** A collaboration system, because four departments will work from shared information instead of emailing each other — Collaboration systems enable people to communicate, collaborate, and coordinate with each other, with email plus an automated shared calendar as the example. They coordinate people; they do not execute the business transactions described here.
- **D.** A customer relationship management system, because everything in the chain is triggered by a customer order — Customer relationship management supports interaction between the firm and its customers, with sales force automation and lead generation as sample applications. A customer starts this chain, but reserving stock and posting ledger entries are not customer-facing interactions.

12. 1.3 A hospital moves scheduling, charting, and pharmacy onto a single cloud vendor. For a sustained period it books more procedures than competitors and reduces records work. Then the vendor has a serious weekday outage and the hospital cancels scheduled elective procedures. Which reading of this matches the chapter?

- **A.** The hospital's business model was unsound, and bad business models tend to fail whether or not the firm uses technology — The chapter does say that companies with bad business models tend to fail regardless of whether they use information technology. Nothing here suggests that situation: the hospital's sustained operational gains are evidence that the model worked.
- **B.** This is a security failure, since consolidating three clinical functions gives an attacker a single point to strike — Concentrating systems does raise security questions, but an outage is an availability failure rather than a breach. This would be the right frame if data had been stolen, or encrypted by an attacker demanding a ransom.
- **C.** This is the **dual nature** of information systems: the system that gave the advantage also created the dependence **— correct.** The chapter compares information technology to a sword: effective as a competitive weapon, and dangerous to those who live by it. Its Zoom case makes the same point in almost the same words, noting that relying on a single provider can quickly disrupt many activities in unforeseen ways.
- **D.** The earlier gains should be discounted, because a system that can go down has not produced a genuine advantage — This treats any exposure to risk as proof there was never a benefit. The FedEx case shows large, complicated, failure-prone systems producing sustained advantage, which is the chapter's point that the advantage and the risk arrive together rather than cancelling out.

13. 1.3 A distributor's IT committee has two proposals and money for one. **A**: retire aging servers and move the same workloads to a hosted provider, producing substantial operating savings with nothing a customer would notice. **B**: build reordering that reads each customer's own consumption history and proposes the next order before they run out, which no competitor offers. Which is **strategic** in the chapter's sense?

- **A.** A, because the savings are certain while B's revenue is not, and a system should return its investment before it chases growth — A is an efficiency case, and the chapter agrees that information systems must provide a return on investment. It presents that as the floor rather than the ceiling, adding that technology use can also be strategic and a powerful enabler of competitive advantage.
- **B.** Both equally, because an investment decision of this size shapes the firm's direction whichever proposal wins — Spending decisions always matter, but the chapter reserves the word strategic for systems developed and continuously updated to gain or sustain competitive advantage, as at Zoom and FedEx, rather than for any purchase that keeps operations running.
- **C.** Neither, because the chapter reserves strategic use of information systems for very large organizations such as FedEx — The chapter says the reverse in as many words: whether it is a small mom-and-pop boutique or a large government agency, every organization can find a way to use information technology to beat its rivals.

- **D.** B, because it aims to gain or sustain competitive advantage over rivals rather than lower the cost of work already being done **— correct.** Strategic here means developed, and continuously updated, to gain or sustain competitive advantage over rivals. B changes the customer's reason to buy from this distributor, while A changes only what the distributor pays to do what it already did.

14. 1.3 A regional bank is first in its market with photo check deposit, and new account openings run well above trend. Later, every competing bank offers the same feature and the bank's growth returns to normal. What is the best explanation?

- **A.** Competitive advantage from the use of information systems can be **fleeting**, because competitors can eventually do the same thing **— correct.** This is the caution the chapter attaches to its own argument: information systems can enable advantage, but that advantage can be fleeting because competitors can eventually do the same thing. It is also why FedEx reengineers and improves its systems on average twice a year even while leading.
- **B.** The feature was not strategic to begin with, because it was built on standard technology rather than something invented at the bank — The chapter rejects that test. Companies from Amazon to Zoom created competitive advantage by combining commoditized technologies with proprietary systems and business processes, so standing on standard technology does not disqualify an initiative from being strategic.
- **C.** The bank stopped investing in the system after launch, so rivals caught up with a feature it had left unimproved — Continued investment does matter — FedEx reengineers and improves its systems on average twice a year — but nothing in the scenario says this system was left alone, and the growth ended exactly when rivals matched the feature rather than when the bank stopped work.
- **D.** The early growth came from the marketing around the launch, since a deposit feature by itself does not move account openings — Launch marketing might explain a brief spike, not sustained account openings above trend that return to normal precisely as competitors match the feature. The timing points at the rivals, not at the advertising.

15. 1.3 Two airlines licence the identical off-the-shelf revenue management package. The first runs it as delivered. The second loads years of its own booking, weather, and gate-turn history into it and rebuilds how crews are staged around what it learns, sustaining a margin lead. What does the chapter say accounts for the difference?

- **A.** The second airline must have licensed a more capable edition, since identical software would otherwise produce identical results — A more capable edition would be a difference in the purchased product, and both airlines licensed the identical package. The scenario holds the software constant on purpose, so the remaining difference has to be what each firm did with it.
- **B.** Nothing durable accounts for it, since standardized software is a commodity that every competitor can licence just as easily — That is the commodity argument the chapter states and then answers. It accepts that information systems have become standardized and ubiquitous, and it still points to Amazon, Zoom, Google, and Facebook building advantage on top of them.

- **C.** Advantage comes from combining commoditized technologies with **proprietary systems and business processes** on top **— correct.** The chapter’s phrasing is that companies created competitive advantages by combining certain commoditized technologies with proprietary systems and business processes. Accumulated history and a rebuilt staging process are things a competitor cannot licence.
- **D.** The first airline’s mistake was licensing at all, since a strategic system has to be developed in house to count as one — The chapter never requires in-house development, and its API discussion makes the opposite point: Uber built almost its entire app around functionality other companies provided. What matters is what gets built on top, not who wrote the base.

16. 1.4 A fitness app told users at sign-up that their workout data would be used to improve coaching inside the app. After a change of ownership, the company sells those data to an insurance broker. Which assessment best connects the chapter's ethics lesson to current privacy rules?

- **A.** Database control is the main issue, so agreement to the original collection is enough for the later sale — This is the ownership-only reading of the chapter's property question. It would fit only if the original notice, consent, promises, data type, and applicable law placed no limit on the new use; the stated inside-the-app purpose supplies such a limit.
- **B.** The changed purpose breaks the stated promise, creating an ethical problem and potentially a legal one under applicable rules **— correct.** The chapter's example treats a reversed purpose promise as an ethical failure. Current enforcement can also make deceptive privacy promises or particular data sales legally significant, though the exact rule depends on jurisdiction and facts.
- **C.** Jurisdiction is the main issue, so the original promise has no independent ethical weight — Jurisdiction determines legal rights and duties, but ethics and an organization's stated commitment remain relevant even when a specific statute does not decide the case. This would fit a question limited solely to which statute applies, not this ethical assessment.
- **D.** Removing names resolves the concern, because the broker receives records without direct identifiers — De-identification can reduce some privacy risk, but it does not erase the changed purpose or automatically remove a record from every privacy rule. This would fit only if the promise and applicable standard expressly permitted that form of sharing.

17. 1.4 A retailer combines three things it collected separately: loyalty-card purchase records, answers to a mall survey shoppers filled out for a free coffee, and transaction data from its store credit card. Merged, the file predicts household events accurately enough to time mailings to them. Which concern does the chapter raise about exactly this?

- **A.** This is a data breach, since data gathered for one purpose has ended up inside another system — A breach is unauthorized access, disclosure, or destruction of data, which is what the integrity and security practice exists to prevent. Everything here was done deliberately by the company that already holds the data.

- **B.** This is an intellectual property violation, because the survey answers are creations of the people who wrote them — Intellectual property is creations of the mind that have commercial value, such as music, software, articles, and product designs. Marketing survey answers are personal data about the respondent, which the chapter handles under privacy rather than property.
- **C.** There is no concern to raise, because each of the three datasets was provided by the shopper voluntarily — This is the argument companies rely on, and it is why the chapter says they are often walking a fine line. Voluntary provision at three separate moments is precisely the consent the chapter says was never given for the merged picture.
- **D.** Providing data at different points is not agreement that they be combined, which is what **aggregation** does **— correct.** The chapter says that just because people provide data at different points does not mean they agree for the data to be combined to create a holistic picture, and it warns that predictive models built this way allow manipulating people and reduce human beings to objects that can be quantified.

18. 1.4 An online retailer's privacy page states plainly what it collects, why, who will see it, and how confidentiality is maintained, and it offers a checkbox for marketing email. A customer who has moved twice writes in asking to see and correct the address on file, and is told there is no way to do that. Which fair information practice is the retailer missing?

- **A.** Notice/Awareness, because customers must be told what data are gathered and what the data are used for — Notice and awareness covers disclosing what is gathered, what it is used for, who will have access, whether provision is required or voluntary, and how confidentiality will be ensured. The privacy page in the scenario already does all of that.
- **B.** Choice/Consent, because customers must be given options about what will be done with their data — Choice and consent is about giving options on what will be done with the data, typically an opt-in or opt-out decision. The marketing checkbox is that practice, and it is present.
- **C.** Access/Participation, because customers must be able to see what is held about them and correct inaccuracies **— correct.** Access and participation is providing customers with means to access data collected about them, check for accuracy, and request correction of inaccuracies. A customer who cannot see or fix a stale address is being denied exactly that.
- **D.** Enforcement/Redress, because there must be a means to enforce these practices and to provide customers with remedies — Enforcement and redress is providing the means to enforce these practices, and remedies for customers, through self-regulation or appropriate laws and regulations. That is what someone would invoke after being refused; the practice being refused here is access.

19. 1.4 A hobbyist downloads a model file for a discontinued appliance part, prints copies on a home 3D printer with the manufacturer's logo raised on the housing, and sells them at a weekend market as genuine spares. Why does the chapter treat this as an **intellectual property** problem and not just a manufacturing story?

- **A.** Because affordable 3D printing lets anyone manufacture a copy, which causes tremendous losses in intellectual property **— correct.** The chapter puts 3D printing beside lossless digital duplication: as printers become better, faster, and more affordable, they open new avenues for quickly and inexpensively producing counterfeit goods, causing problems for consumers expecting the original and tremendous losses in intellectual property.
- **B.** Because copying a physical object requires a licence, while copying a digital file is a contract matter rather than an IP one — This has both halves backwards. Copying digital files is the chapter's central intellectual property example, and many physical objects can be reproduced freely; what matters is whose creation of the mind is being taken.
- **C.** Because the hobbyist never bought the original part, and intellectual property protects revenue on every unit sold — Lost revenue per unit is the argument the chapter shows people disputing, since buyers often claim they would have chosen a free alternative or bought nothing at all. The chapter still treats the behavior as a problem, on the ground that someone else's work is used without permission, attribution, or compensation.
- **D.** Because the printed part carries the maker's logo, and intellectual property is about the brand rather than the design — Trademark is one kind of intellectual property, and the fake logo does compound the harm. The chapter's point about 3D printing is narrower and harder: the design itself can now be copied by anyone, so an unbranded print sold as a genuine spare raises the same problem.

20. 1.4 Someone edits a photograph of a coworker so that it appears to show him somewhere he never was, then posts it to a group the whole department follows. Under the Computer Ethics Institute guidelines quoted in the chapter, which prohibition does this most directly break?

- **A.** Snooping in other people's files — Snooping is reading files that belong to someone else without authorization. It would be the answer if the photograph had been taken out of the coworker's own account; what is at issue here is what was published, not how it was obtained.
- **B.** Using a computer to bear false witness **— correct.** Bearing false witness is using a computer to present something untrue as fact about another person, which is what an altered photograph passed off as real does. The chapter raises this case directly when it asks whether the practice of rearranging and otherwise changing photographs is ethical.
- **C.** Appropriating other people's intellectual output — Appropriating other people's intellectual output means taking someone's creative or intellectual work and using it as your own. That would be the answer if the wrong were publishing a photographer's image without credit rather than falsifying what it shows.
- **D.** Interfering with other people's computer work — Interfering with other people's computer work means disrupting what someone else is doing with their machine or their systems. Nothing here stops the coworker from using a computer; the harm is to his reputation.

21. 1.5 A furniture maker's largest accounts supply most of its revenue. One demands longer payment terms and a lower price, and mentions that rival suppliers have quoted the same work. Which of the five forces is strongest here?

- **A.** Bargaining power of suppliers, because the builders are dictating the terms of the relationship to the maker — Supplier power is the pressure the firms you buy from apply to you, by raising prices or setting terms when you have few alternatives. The word supplier appears here because the furniture maker is the supplier, so choosing this means reading the situation from the wrong side of the table.
- **B.** Competitive rivalry, because two other suppliers have quoted the same work and are competing for it directly — Rivalry is the fight among firms already selling the same product to the same customers. Rivals do exist in this story and their quotes are what give the builder leverage, but the pressure is being applied by the customer and it is the customer who decides.
- **C. Bargaining power of buyers**, because a big share of revenue plus easy switching lets a customer force price and terms **— correct.** Buyer power grows when a few accounts make up most of your revenue, when comparison is easy, and when switching costs the customer little, and all three are present. That combination is the signature of this force, and it is why loyalty and switching costs are the usual response to it.
- **D.** Threat of new entrants, because rival suppliers are moving into business this maker used to hold on its own — New entrants are firms not competing in your industry today that could start tomorrow, held back by barriers to entry such as cost, scale, know-how, or loyalty. The two rival suppliers quoting are already in this business, so no barrier is being tested.

22. 1.5 A cinema chain's attendance falls for several periods, including in towns where it is the only theater and no new theater has opened. Over the same span, household spending on streaming subscriptions rises sharply. Which force explains the decline?

- **A.** Competitive rivalry, because the chain is losing to other companies competing for the same entertainment spending — Rivalry is pressure from firms inside your own industry selling nearly the same thing to the same customers. The scenario removes that possibility deliberately: in the towns described there is no other theater and none opened.
- **B.** Bargaining power of buyers, because moviegoers are refusing to pay the ticket prices the chain sets for them — Buyer power is customers forcing your price down or your service up while still buying from your industry, which happens when switching is cheap and comparison is easy. These households are not negotiating over ticket prices; they stopped buying tickets at all.
- **C.** Threat of new entrants, because streaming services entered a market the chain used to have to itself — A new entrant is a company that starts selling what you sell. A streaming service operates no theaters and sells no tickets to them, which is what separates an entrant from a different kind of product answering the same need.
- **D. Threat of substitutes**, because a product from outside the industry meets the same need, so demand leaves the category **— correct.** A substitute meets the same underlying need, an evening's entertainment, by a different route, so the customer never enters your industry. The tell is in the facts: the decline continues where there is no competitor at all, so nothing inside the industry can account for it.

23. 1.5 A grocery chain plans a portal where every approved grower submits a sealed quote against one consolidated weekly order, alternate growers stay qualified and visible, and contract expiry dates appear on the buying team's calendar before renewal comes due. Which value chain activity does this initiative improve?

- **A. Procurement**, because it is about finding, qualifying, and contracting the suppliers **— correct.** Procurement is deciding whom to buy from and on what terms. Quotes, qualification of alternates, and contract renewal are all parts of that decision, which is the activity an initiative like this belongs to.
- **B.** Inbound logistics, because it concerns goods arriving from suppliers before they are sellable — Inbound logistics is receiving, storing, and moving goods that have already arrived, such as a dock scanner counting a pallet as it comes off the truck. It begins after the procurement decision, and mixing the two is the most common way to file a recommendation in the wrong department.
- **C.** Operations, because better buying is what makes a stocked, priced shelf possible — Operations turns inputs into what the customer buys, such as register scans becoming restock tasks so the milk case refills before it empties. Almost anything upstream eventually helps operations, but naming the activity an initiative actually changes is what keeps the analysis usable.
- **D.** Firm infrastructure, because contracts and renewal calendars are matters of general management and planning — Firm infrastructure covers general management, finance, accounting, legal, and planning, such as a margin report by store by day. The buying team here is doing supplier selection, which the value chain names as its own activity.

24. 1.5 A retailer gives every returns desk a screen showing the customer's complete purchase history, so a refund is resolved quickly and no receipt is needed. Complaints fall, and repeat visits from customers who returned something rise. Where does this initiative sit in the value chain?

- **A.** Marketing and sales, because a smooth return is what persuades customers to choose this retailer next time — Marketing and sales is making customers aware and giving them a reason to choose you before they buy, such as a loyalty offer aimed at an item a household stopped buying. The downstream effect described is real, but the activity being improved happens after the sale.
- **B. Service**, because it keeps the customer whole after the sale — returns, complaints, and repairs **— correct.** Service is the primary activity that keeps the customer whole once the sale is done, and a returns screen showing purchase history so a refund is resolved quickly is an example of a system that adds value there.
- **C.** Outbound logistics, because it manages goods moving between the customer and the store — Outbound logistics is getting the finished product out to the customer, such as pickup and delivery batched by neighborhood into slots a driver can meet. A return travels the other way and is handled as service, not as a delivery problem.

- **D.** Technology development, because the retailer had to build a new screen and connect it to purchase records — Technology development is the support activity of improving the product and the processes themselves, such as building your own app rather than renting one every rival can rent. Building this screen involved that work, but the activity where the value reaches the customer is service.

25. 1.5 Four analysts each write one closing sentence at the end of the same study of a regional retailer. Which one is a recommendation an executive can fund and later judge, rather than a preference?

- **A.** Our biggest exposure is disruption, so we should invest in digital transformation and modernize the customer experience — This names a worry and a category of spending. Disruption is not one of the five forces, digital transformation is not something anyone can be assigned to build, and no measure is named, so a year later nobody could say whether it worked.
- **B.** We should adopt artificial intelligence this year, because leading firms in our sector already have and our costs are higher — This opens with a technology and reasons from what other firms are doing, which skips the diagnosis. The cost gap is a real observation, but nothing here names the force or the value chain activity the technology would relieve, or what would count as success.
- **C.** Buyer power is strongest — customer research shows many lapsed members shop a rival — so build one chain-wide purchase history, judged by repeat visits **— correct.** It names the force, gives the observable evidence behind that claim, states a concrete initiative rather than a category of spending, and commits to a measure — the four parts that separate a fundable proposal from a preference.
- **D.** Inventory visibility is inconsistent from store to store, which is costing us sales in seasonal categories and should be fixed soon — This is a finding, and a legitimate one that even names a consequence. It stops short of a recommendation because it proposes no initiative anyone could be assigned to build and no measure by which success could later be judged.

What to do next. Two things make this material stick. First, take one organization you already deal with — an employer, a store you shop at, a service you pay for — and name its information systems out loud: what data goes in, what information comes out, who acts on it. Second, practise the strategy method on that same organization until naming a force and an initiative feels routine. Everything after this chapter assumes both habits.